

Phil 260 Intro to Metaphysics & Epistemology

Numbered Transcript

PHIL 260 • Dr. Stephen R. C. Hicks

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Lecture 1: Origins of Reality

At a glance — Hicks introduces metaphysics and epistemology and the origins of reality.

- 1.1 Welcome to this Introduction to Metaphysics and Epistemology, where in this course, we're going to explore some of the most fundamental, deepest, important, and controversial issues that philosophers grapple with.
- 1.2 I want to start with asking you to imagine or re-imagine an experience that I'm pretty sure we have all had. That is, you go outside, and it's dark, and maybe you are alone, and you look up at the sky, and the stars have come out. But it stops you because in that moment, you realize just how many stars there are, and you start to wonder: How many are there? What exactly are they? Where did they come from? In that moment, we all become philosophers.
- 1.3 Now, if we ask: Where did they all come from? Where did everything come from? Why are we here? We have in our culture, Western culture that is increasingly becoming a global culture, two very different stories that have come down to us. I want to call them stories at the beginning, not to prejudge issues of truth and falsity— enormously influential. There are parallels to these stories in the East as well. Let's ask this question in this philosophical moment.
- 1.4 In the beginning, what happened? One of these stories we call the Big Bang Story, and it starts by saying, the best number right now, about 13.78 billion years ago—we try to wrap our minds around what that number would mean— all of the matter and all of the energy of the universe was highly compressed into what we might call a singularity, and then it went bang. That's a metaphor for saying it all suddenly expanded dramatically, and with that expansion came cooling.
- 1.5 And then very quickly, within a matter of just a few seconds, maybe seconds shading into minutes, we start to see the elaboration and creation of some of the lighter chemical elements: helium, hydrogen, beryllium, maybe a little bit of lithium, and so forth. This process stretches into minutes. Everything is expanding out, and cooling, and starting to recombine. This process goes on, maybe for the next estimate: 380,000 years or so—another big number, hard for us to wrap our minds around.
- 1.6 But then, it enters an era that they call the Dark Ages. The Dark Ages last for some hundreds of millions of years. It's called the Dark Ages because of all of the combining and recombining that's going on, there's no light, apparently, being emitted, but nonetheless, intense activity: still expansion, cooling, combining, recombining, and so on. And now, we've gone from seconds into minutes, into years, into hundreds of thousands of years, into millions of years. Maybe around 1.3 billion years into the process, we start to see proto-galaxies forming. And then, that process carries on for some other billions of years, and then we get into full galaxies. And then, as we get into

6, 7, 8, 9, 10 billion years, the current array of galaxies has come into existence, and it is formed.

- 1.7 Over the course of these billions of years, the analysis runs. Some of those stars, giant stars, have collapsed after going supernova. And then, we start to see the creation of heavier elements: iron, carbon, oxygen, and so on. So now we are maybe 9 or 10 billion years into the process, and we start to come closer to home. Our sun, a star, one of the smaller and newer stars, comes into existence, and some satellites develop that are orbiting it, so we have now a solar system.
- 1.8 Then, shortly after that, maybe 4.5 billion more years, the Earth is formed. Some billion, maybe 2 billion years go on, all of the combining and recombining, expansion, cooling, heating is going on. The earliest life forms come into existence. Another billion years goes on. There are various forms of life that come into existence. The planet is filled, mass dying off, some extinction events.
- 1.9 And then, 66 million years ago, the one that we were all fascinated with as kids: the mass extinction of the dinosaurs. Bad for the dinosaurs, but out of that, there's space made for a relatively insignificant species up to that point. The hominids can come into existence, starting maybe two million years ago, but they start to flourish more and more and evolve, and by the time we get to maybe 300,000 years ago, us, Homo sapiens, the hominoids with the very big brain capable of extraordinary capacities for thought. And now we're about 300,000 years ago, 200,000 years ago, and tens of thousands of years go by, they go by, they go by.
- 1.10 A significant event happens maybe 10,000 to 12,000 years ago in what we now think of as Mesopotamia: the discovery of, an invention of agriculture. Instead of simply harvesting and hunting what pre-existed in the environment, we take control of the plants, we take control of the animals. Agriculture becomes a game-changer for our species. Maybe 7,000 years ago, we figure out how to take some of those heavy elements and heat them up and engage in metallurgical processes working with tin and copper and making bronze and another game changer for human beings.
- 1.11 Shortly after agriculture has matured, metallurgy has matured, there's a development of very sophisticated civilizations now on all of the continents: Africa, Europe, all through Asia and into the Americas as well. Now, a mere two and a half thousand years ago, a thousand and a half years ago, Greece and Rome rise, and the story carries on, and here we are. So that's our story of where it all came from, how we got here, and where we are. It's a very interesting and compelling story.
- 1.12 But let's try another story. That's also an old story. It doesn't start with bangs; it starts with creation, the language of creation. In the beginning, there is Vishnu. Vishnu—we're now telling a Hindu story, and in the Bhagavad Gita, he says, "I am the source of all spiritual and material worlds. Everything emanates from Me." Vishnu has a

personality. He is a person, a being. He is self-describing— himself, itself, herself, the gendering we'll set that aside for another discussion.

- 1.13** But notice the language that we use here: "I am the source of spiritual worlds. I am the source of material worlds." So we have two different worlds, perhaps, that are being created, but they're coming from one source. Also, each of those is plural. Is there more than one spiritual world, more than one material world, multiple worlds, perhaps? And this word "emanation"— a very interesting word, carefully chosen in translation here. Often, when we think about emanation, we think, say, about a light bulb. We're surrounded by them. You have a light bulb that is not on; you turn the light bulb on, and light emanates from the bulb. But as it goes out, it retains its identity as light.
- 1.14** If we then extrapolate that back to Vishnu, and we say that what I am doing is me, my identity, my being, whatever I am, I am emanating, and that is what creation is, the suggestion then is going to be that really everything is the same stuff as Vishnu. Because light doesn't change as it emanates, so Vishnu, whatever his being would be, would remain the same as he has emanated out. But at the same time, we do seem to have this distinction between different things that are being created: spiritual worlds and then material worlds. So some things to be thought through and worked out.
- 1.15** Another formulation about this early creation, Vishnu says, "I am the origin of all; from Me, all things evolve." The "evolve" word is interesting because it's different from the "emanation" word. If we stick with that analogy to light, light doesn't change its nature as it emanates. It might become weaker and eventually die out, but here we have a suggestion that Vishnu, in creating the world, being the origin of all of the world, is creating a reality that is not merely emanation, but changing. It's evolving. It's going into different states, perhaps over time. If it is he who is going through this process, then he, Vishnu, is going through the process of evolution. Of course, there's a question here of whether he remains the same, creates the worlds, and the worlds evolve, but he is directing the evolution, or whether he himself is the world that is evolving. Again, some interesting things to start unpacking.
- 1.16** Now, on this evolution, there are different strains of Hinduism—six major ones that I know of anyway, just as most of the religions in the West come in major forms and sub-forms. The same thing is true of Hinduism. But Vishnu takes different forms: Brahma, Krishna, and Shiva. The description of the evolution of the world, it says it goes through cycles; evolution can be conceived of as a more open-ended process, or it can be conceived of as a more closed process of creation, and maintenance, and then destruction, and then creation starts up again, so on and so forth, on ending.
- 1.17** That seems to be what we have here. In the Brahma stage, Vishnu is in a creative phase. Then, he goes through a long period of maintenance, and then at some point, destruction takes in. It's not clear, in different strains of Hinduism, whether Vishnu is behind the scenes and these are avatars, whether all three of them are always

operative and perhaps in conflict with each other, or if it's that one comes to the fore while the others are in recess, but then another comes to the fore while the other two come. All of this, again, is to be worked out. But nonetheless, we have a form here.

- 1.18** At some point, we say that all comes from Vishnu, and so the material worlds, then, it would seem, are real, just as the spiritual worlds that he creates are real. But at the same time, all through Hinduism, we have this persistent suggestion that the material worlds are less real, less significant, less important, and it's the spiritual that is more significant, more real, and more important.
- 1.19** Purusha is the spirit of a universal consciousness; that's one state of reality that has been created. Prakriti, the material world as we think of it, is often described as illusory, and often, it comes packaged with some value judgment: it's not important, it's not to be paid attention to, one should try to deny it in various ways. That is a false or illusory reality, the material world, and if one is a philosopher seeking to come to know true reality, it's going to be the spiritual consciousness, the higher reality that one is seeking. So we have a very interesting set of stories that have been elaborated and re-elaborated in different schools of thought coming down over the course now of many thousands of years.
- 1.20** Let's try another creation story. It has some structural similarities with the Hindu story out of Bhagavad Gita, but also some important differences. If we go from India, some thousands of miles to the West, we have a different set of stories that come down to us in the Book of Genesis, the first book in the Hebrew scriptures, the first book of the Bible in the Christian tradition, significant also in Islamic tradition. So what's in the beginning? Where does it all come from?
- 1.21** Genesis 1:1, right at the very beginning, "When God began to create the heavens and the Earth, the Earth was without form and empty." Now, that is a little bit different from this formulation. "In the beginning, God created the heavens and the Earth." Let's pause and compare this because, just as in the Bhagavad Gita, there are interpretive issues right off the bat that need to be sorted out about the nature of the creation. The same thing occurs here.
- 1.22** Let's start with this one: "In the beginning, God created the heaven and the earth." So that, then, if we start to think, is to say in the beginning there was God, but nothing else, God and only God. And then, God engaged in an act of creation, bringing into existence the heaven and the earth.
- 1.23** Then, at the earliest stage, "the earth was without form and void, and darkness was upon the face of the deep." So what God brought into existence was something that didn't exist before, and it didn't yet have any form or shape or identity. We might think of it as a blob or a chunk of liquid plastic. And then God, at some point, is going to add forming to it and give it a more definite shape and definite direction. The Earth was without form and void—so God and nothing. Then something comes into existence,

heaven and earth, but it's not yet shaped. And then, the succession of acts of creation or God giving it shape.

- 1.24** Now, let's go to the other one, also very early in Genesis: "When God began to create the heavens and the earth, the earth was without form and empty." A school of interpretation here suggests that the way to read this is to say that God did not create the existence of the earth. Instead, what was already there was something that was just without form and empty, some void, plasticine, moldable. When God began to create, the raw material, so to speak, that he is working with is formless and empty, has no specific identity. So at the beginning, we have God and some formless stuff. And then, the activity of creation is God giving form to the stuff, perhaps by analogy to a sculptor. There's a chunk of clay, it's there, but it's not anything in particular. Along comes the sculptor and makes something out of that.
- 1.25** Again, we don't want to prejudge which of these interpretations is correct, but there are suggestions of both of them. And so what God then did is move the Spirit of God, move upon the waters, and "God said, 'Let there be light, and there was light.'" So the waters are there, those are more material. We know a little bit later the Earth is there, and that's material stuff. And spiritually, the Spirit of God comes along and adds form and identity to all of this stuff.
- 1.26** So what we have then is two things to think about if we're working through this tradition, this philosophical tradition. Do we mean by creation that God created something out of nothing, that it was just God and nothing else, and then suddenly this different stuff comes into existence?—that's the Latin phrase *ex nihilo*, "from nothing." Or do we say that really we should understand creation in terms of God taking some formless void-like, not particularly anything stuff and imposing structure, form, direction upon it? In which case, at the end of either process, we have God, who is described as a spiritual being and then a material creation that has resulted now with form, identity and process built into it.
- 1.27** This story carries on. There are several stages of creation. Then we get to God's greatest creation. "Then the Lord God formed man from the dust of the ground and breathed into his nostrils the breath of life, and the man became a living being." So again, we might think by analogy to a sculptor who takes some clay, the dust of the earth, makes the shape of a man, but that's just clay in the shape of a man. It's not yet a living thing, but that's the dust of the earth in the shape of a man. And then, God adds something different to it, and this is the breath metaphor—God is a spiritual being, so God, in effect, transfers some of His Spirit into the dust of the earth. So now we have material stuff with some spirit added to it; those two come together and make a human being, the being that it is right now. Something like that, right?
- 1.28** Next: woman. "The Lord God caused the man to fall into a deep sleep, and while he was sleeping, he took one of the man's ribs and then closed up the place with flesh. Then the Lord God made a woman from the rib he had taken out of the man, and he

brought her to the man." So human beings are an important, significant, interesting part of the world that we want to explain. And in this account, we have man made from dust plus breath or matter plus spirit. And then, we're going to have arguments about this one as well, a little bit later. This might be significant—the woman comes after and from the man. So the two creations are significantly similar, but also, importantly, different in some respects.

- 1.29** The next part of this story is about how we got to where we are. The serpent— there's a back story for the serpent. Let's set that aside for now. "The serpent was more crafty than any of the wild animals the Lord God had made. He said to the woman, 'Did God really say you must not eat from any tree in the garden?'" So the tempter arrives. The woman has received instruction not to eat from a particular tree, and the serpent is tempting her.
- 1.30** There's some interesting issues here about whether the serpent manipulated the woman, Eve. He's more savvy and powerful than she is. She's relatively unknowing and weak. Really, the responsibility lies with the serpent. She was putty in his hands, so to speak. Was she gullible? She knew, but the serpent manipulated her a little bit more. So there's some shared responsibility for what happens here. Or did Eve really know what she was doing? The serpent was just an excuse to do what she wanted anyway, so she bears the responsibility?
- 1.31** Again, interesting issues to start thinking through, but nonetheless, there's a sin here, disobedience of the command when God comes back after she has, in fact, stolen something that was off limits. She lies about it. Adam is also complicit in that, and then, as a result from the sin, there are punishments. Interestingly, men get two punishments. Whatever relatively easy work one had to do, maintaining the Garden of Eden, which was a lovely place. Now, they're going to have to work hard and not enjoy their work. It's going to be painful and sweaty work that they have to do, and they're going to die. The same thing holds for women, but women do get one extra punishment, and that is pain in childbirth.
- 1.32** Next generation. They have children. The boys, the most famous story here: Cain and Abel. The theme is that human beings are envious, as Cain is of Abel, envious to the point of being willing to murder. So now we have not only theft, lying, disobedience, but we add envy and murder to the roster. Over the course of the subsequent generations, we have a lot of lust and adultery and cheating and lying and so on.
- 1.33** Until partway through Genesis—this is relatively early in the story— God comes back and surveys what humans have made of his wonderful creation. "The Lord regretted that He had made human beings on the earth, and His heart was deeply troubled. So the Lord said, 'I will wipe from the face of the Earth the human race I have created, and with them the animals, the birds, and the creatures that move along the ground, for I regret that I have made them.'" A complete disaster, this first round of creation.

- 1.34** So, He does so. The flood is a vehicle. There are a few chosen human survivors. They start to repopulate the Earth. As the stories carry on, it's the same human story over and over again, dominated by sins of all kinds. Here we are. Then what happens next? Well, as we know in this set of religious traditions, there are many schisms and split-offs. The Jewish story goes in one direction, the Christian story, and the later Muslim story in another direction.
- 1.35** Now, we have the Big Bang story, and we have two creation stories, and both of them have long pedigrees. Both of them have articulate, thoughtful, well-read defenders. Both of them are saying something important to us and trying to give us an explanation of things that matter deeply to us. Built into all of those stories, the Big Bang story and the two creation stories that we looked at briefly, are some very deep philosophical commitments. I want to use the stories to tease out those philosophical issues that are going to preoccupy us for this course.
- 1.36** This is a course in metaphysics and epistemology—those are multi-syllabic words, perhaps a little bit intimidating at the beginning. I know when my first-year university students come in, their eyes get big. They have to practice saying these words. Both of them are Greek in origin for the reason that the Greeks were the first to start doing philosophy systematically. And all of the major philosophical issues that we recognize today were first identified, articulated, and the position staked out by the ancient Greeks, starting maybe 2,600 years ago. So much of the terminology that we have comes from the Greeks.
- 1.37** The core question that we ask as philosophers is: What is real? What is reality? One of the things we know is that some things don't exist; some things are merely imaginary. So we want to know the difference between what's real and what's really imaginary. We know that sometimes we can believe things that are true, some things that are false. By what standard are we going to sort out the true from the false? We need to have some reality that is going to be the source of facts that are underlying our truth statements.
- 1.38** But what is this thing, this reality that we are so interested in? The word "metaphysics" is a compound word, the meta and the physics part, or physis in the original Greek. But think of the physics. When we are doing physics, we're looking at reality. We're looking at particular things out there in reality. We might start to categorize them in various ways, and study how they act and interact, and come up with principles or laws of how those things in physics react. We might have lots of different subfields of investigation here as we are studying the physical world.
- 1.39** But if we step back and say, "I'm not interested in this particular thing or this particular category of things. I'm interested in the whole shebang,"—I just used the slang here—"the whole package." Everything that exists. So it's the reality, the physical world, but my attitude toward it is the meta step, to step back and try to take all of it in. What is true of all of it, the whole story? Or what features are generally true or universally true

of all of reality, including the question about what reality itself is, not any particular thing, but the whole story? Metaphysics.

- 1.40** Epistemology is the close follow-up question. Anytime I want to say, "This is real," and that claim is a claim of unreality, you're just making that up, or that's false, you've made a mistake, it's not true, it's only imaginary, or you're lying to me, facts are facts. All of that is to assess various claims about the nature of reality. We want to know which ones are good claims and which ones are not good claims. So, our question is, anytime someone says, "This is a fact," how do you know? Or, "This is true"—why do you think that? And that is an epistemological claim.
- 1.41** Again, it's a compound word, episteme, which is a Greek word for "knowledge," but it's a self-reflective, informed knowledge. And then, the -ology part from logos is "to give an account of something." So we're familiar with biology: giving an account of life; geology: giving an account of the Earth; epistemology: we're trying to give an account of knowledge. What counts really as knowledge? Maybe there's a distinction between truth and probability and facts and claims that are justified and claims that are speculative and things that are just wild guesses, sorting out all of that terrain.
- 1.42** So, when we're thinking about the most important things and making up our minds, what I'm really going to believe and live, I'm in a position to have a self-reflective judgment that I think what I believe is true of reality. So, what's real and how do we know? Those are the fundamental questions that govern everything, and that's what this course is going to be about.
- 1.43** Now, let's plunge into some metaphysics using that label and start to compare the stories, Big Bang stories and creation stories, as we have laid them out in terms of their metaphysical commitments. One striking contrast right from the beginning is that the Big Bang story is physicalist in its metaphysics. It says that in the beginning is matter and energy: physical matter, physical energy. It's then going to give an explanation of the development of the universe in terms of physical process. We get some chemical elements, we have some expansion, we have some cooling—that's the subject matter of physics. We have the galaxies being formed— that's astrophysics. Then, we have the formation of the solar system and the Earth, and we're into geology and all of these scientific disciplines, starting from a physicalistic basis.
- 1.44** At some point along the story, billions of years into the story, we start to get life forms. Some of those life forms become complex enough to start to have self-awareness and psychological states, and so on. But that comes after a long physical story and is dependent upon the underlying physics of the situation. The physical comes first. What we will call the psychological or the consciousness, or if we even want to start using language like spiritual, it's going to be derivative. It's a second and later part of the story.

- 1.45 In both of the creation stories, the order seems to be reversed. We start with not physical stuff, we start with spiritual stuff. God is a spirit. Vishnu is characterized in spiritual terms as an ultimate consciousness, an ultimate awareness. And what we call physical reality comes into existence after and depends upon and is derivative of consciousness. So right now, right off the bat, we have a very deep metaphysical issue: what comes first, the physical or the psychological? An opposition between those two stories.
- 1.46 We might then zero in on human beings, seeing human beings as a microcosm of the macrocosm. Both of the stories have different accounts of the human being. We have a big brain, we have a mind, we have all of these very interesting psychological traits. But it's not the case that there was matter and then some completely different psychological stuff out there that came together in the human being.
- 1.47 According to the Big Bang stories and the evolutionary stories that come out of them, physical matter became increasingly complex and psychological phenomena emerged out of that or evolved out of that, or to use a technical term, supervenes on underlying physical states. But nonetheless, they are an integrated thing. You can't just take the psychological out and still have it do what it does, and have the physical separately doing whatever it does.
- 1.48 By contrast, the other stories, the creation story, particularly the Genesis story, is quite explicit in saying the human being is a dual creature. There was material stuff that was completely unspiritual, material dust of the Earth. And then this completely different stuff, spirit, was breathed into it by God, who was himself a spiritual being. So we have, instead of an integrated being, an aggregate, this plus this.
- 1.49 It still is an aggregate because, going on to the next issue, we say that we know what happens at the beginning of life, but what happens at the end of life? We have two different stories then with respect to the human being. If it's the case that our psychological material, whatever is going on in our minds, depends on the underlying physical processes in the brain, the nervous system, the rest of the body, when the body breaks down and dies, then the prediction is that all of the psychological stops happening; it goes out of existence. But if the human being is a dual creature— it was this body, then some soul got added to it, or spirit got added to it— in principle, they can be separated again. When the body stops doing whatever the body does and dies, the soul is then freed, or it disentangles itself from the body, and it can have perhaps a further destiny, an after biological life, whatever that would be. Deep differences.
- 1.50 When we start to think about human value issues—we're going to be getting ahead of ourselves right now, but already this is built into the stories—whatever we want to think about our place in the universe as human beings, it's in the natural world. That seems to come out of the Big Bang story. On the other hand, if we came from Vishnu and we are now alienated from Vishnu and trapped in a world of Maya, an illusion, then our ultimate destiny should be, it seems, to get away from this illusory material

world and reunite with Vishnu in some ultimate state. Or, if we are, in the other story, trapped in a material world, trapped in our bodies, but ultimately God is spirit, He is the origin and the destiny, then our destiny should be to try to get away from the material world, ultimately, or welcoming getting away from the material world to get back to the ultimate source of being, the ultimate source of truth, and so on.

- 1.51 Some interesting normative issues about good and bad, about human beings in the world, have already come out of these metaphysical stories about origins. The Big Bang story says that if we start with the human beings, there were some primitive animals grubbing along, surviving not particularly well, getting eaten by predators and so forth, but then they developed a big enough brain that they could start to look after themselves more well. They were living in caves. But then, because of the power of their brains, they figured out agriculture, and life got better. They figured out metallurgy, life got better. They started to build civilizations. They built skyscrapers, magnificent. They are now developing the sciences that are enabling us to probe all the ultimate questions of the universe.
- 1.52 One way of reading that story, that way, is to say that we're going from a not-so-good state for human beings and better and better and better as we go along. It might be two steps forward, one step back, two steps forward, one step back, but it's ultimately progress from cold, hungry, not surviving very well to living now in skyscrapers, central heating, and so on.
- 1.53 The story on the other side, though, at least from the story of Genesis, seems to be a story of regression. Human beings started in a magnificent world, the Garden of Eden, but they messed it up, and as a result of messing it up, they got kicked out justifiably from the Garden of Eden. And then they continued to mess things up and make it worse and worse and worse until God says this is a do-over flood, "We do it over." But human beings, having a blank slate once again, continue to mess it up. It's a story of regression and decline rather than a story of progress.
- 1.54 It's interesting how those normative descriptions seem to be already packaged in these metaphysical stories. Often, there's a distinction between valuing the physical world and how much we should value the physical world. Obviously, from the perspective of the Big Bang, if we are human beings, we're in the physical world, we are ultimately complicated physical beings ourselves. So the physical world is it. It's our top value, and we should enjoy and survive and flourish in it.
- 1.55 In both of the creation stories, though, there seems to be a bifurcation between the material world and the spiritual world. The material world is often described more in cognitive terms as a world of illusion, and the truth of reality is up in the spiritual world, or it's described as a trap, something that sucks you into things that you should not really be concerned with too much: physical values. Really, you should be focusing only on spiritual values or as much as possible.

- 1.56 Now, human control is another issue here. If it's the case that we started not even in the caves, but maybe the caves was a move up for us, we get into the caves, and then we're getting into huts, and then we're getting into buildings, and now we're getting into skyscrapers and wonderful resort complexes and so forth, it seems like we are in charge of our destiny. We are making our world, and we are in some ways remaking the physical world in a way that's going to be better for us.
- 1.57 So there seems to be at least a glimmer of some optimistic assessment of what human beings can do in the Big Bang story—not to overstate it, but there's something there—versus the other stories, which seem to suggest sinfulness as the dominant story. The most important thing about Eve is that she sinned, and then Adam joined in on the sin. Then, we have Cain committing sins. And then, we've got the whole story of Lot and all of the people surrounding Lot and so on. There are no noble moral human beings who are emphasized in these stories doing great and wonderful things. Instead, it seems to be a story of sin. And the suggestion is that even after getting a second chance, we can't help ourselves in some sense. After Noah, after the flood, we continue to engage in sin, and it's going to take some special intervention to come along and save us from ourselves. That contrast is there.
- 1.58 Now, of course, we can take all of these contrasts and put them on a spectrum, and I'm trying to exaggerate and illustrate the pulls of the spectrum in each of these issues. Other possibilities will be to say that we don't need to go to that extreme. We can work out places in the middle. I don't want to prejudge any of that.
- 1.59 Where did it all come from—that might be the question that started us, and where are we going? Different answers to that question are built into the stories as well. In the Big Bang story, there is no script, so to speak. It's not like there is a designer or a script writer or someone overseeing the process or stage managing the process. Instead, there are simple physical elements and energy, and whatever happens, happens as a result of the expression of those laws. The universe is going. It's open-ended, as far as we know.
- 1.60 The other two stories, there does seem to be—this is a little bit metaphorical—a script. God has a plan. He created the universe for some reason. It might be known only to Him. He made it this way, and then He did this, and then He did this, and He made human beings, and He gave them some rules, and it's all going somewhere, and ultimately, there's an end of the story, and we should be working toward that end of the story, or at least accepting our role in the script writer's plot for helping the story progress. So there's a purpose. There's no purpose, so to speak, of the universe as a whole.
- 1.61 What if we go back to the beginning? That's a very interesting question, and these are ones we will be diving into some more. Suppose we say that the Big Bang story takes us back to the Big Bang 13.78 billion years ago—very interesting. So what happened

before? What happened 13.79 billion years ago? I really wanted to know. That's a hard question.

1.62 We might say that there's something wrong with the question. Maybe the "before" language. What do we mean when we say "before?" We're using time-talk. Maybe there was no time before. That time is something that happens once you've got physical stuff doing stuff, acting in the world. So that time is a derivative property of the physical universe as it's working out. In which case, we can talk about before and after as the universe is going through various processes. But "before" doesn't work, and "after" doesn't work if we try to get before the Big Bang. So maybe there's something ill-formed about the question. We just need to set it aside.

1.63 Or we could say—again, this would just be speculating—maybe the universe went through this great explosion and expansion, and it's continuing to expand. And then, eventually, the rate of expansion will slow down, slow down, stop, and then maybe it'll start to contract, contract, contract in accelerating form, back into a Big Crunch. And then out of that Big Crunch will then be another Big Bang, and so the whole universe goes in cycles of multibillion-year Big Bangs, Big Crunches, Big Bangs. Maybe. Again, we're speculating, but we don't know. We're quite happy to say we've worked out the story up to this point, and we don't know what happened yet. We're still working on it. We're going to keep an open mind.

1.64 Now, if we go to the other side of the story, suppose we try to ask the same question. Suppose we say everything came from Vishnu. At some point back in time, Vishnu—and then we say: What happened before Vishnu? And typically in this tradition, we say that's not a legitimate question. You don't ask that question. Vishnu is the beginning, and Vishnu is the end. There is no before, there's no behind, underneath Vishnu.

1.65 And then, the same thing in the more Middle Eastern and now largely Western traditions, we might say that in the beginning, there was God. Maybe there was this formless stuff that he was going to operate on, but there was God. And then, God started doing stuff. And then we say: What happened before God? Again, in this tradition, we find that we're encouraged very quickly to say that's not a legitimate question. God is the beginning. There is no behind God, under God, above God, before God. None of that makes sense. God is, in some sense, the fundamental out of which all explanation is going to come in a derivative form, and we're pretty sure about that. So, in one sense, we're willing to stop the questioning at that point. That's a cognitive difference from the other side, which is to say, Well, we've gotten to this point, and we were going to keep asking questions and see if we can push it further back.

1.66 Now, let's do some hard work and massaging on two stories to try to tease out a lot of metaphysical issues. What if we switch over to the cognitive issues or the issues about knowledge, the epistemological questions that seem to be built into the two stories as well?

- 1.67** So the Big Bang story. We listen to the stories, fascinating, very cool. It gets your juices flowing. You're thinking cosmological scales, and wow. But how do we know any of this? Why should I think any of this? The story is to say, anybody can know this, but you have to start with your own eyes, just like we started looking at the stars. Maybe start getting some measuring instruments and keeping track of the stars. Do some physics, do some chemistry, do some biology, put it all together, build up a big, complicated, logical, mathematical theory. Then, on the basis of that, you can become increasingly confident that you know that this story is true.
- 1.68** If we're just amateurs, we might say, that's an interesting story. I can understand it, and I have some evidence in my degree of knowledge. That means I should believe, perhaps, this is probable. But if I really want to know, then I have to do a lot of observation, a lot of science to increase my justification in believing in this big, complicated story. But in principle, you can do it. Anybody can do it, and this is the method that you would use.
- 1.69** By contrast, if we were to ask what makes the story about Vishnu and the cycles of Brahma, Krishna, and Shiva, true? It's a very fascinating story. Interesting stuff in there. But why should I believe that? What's my source of information? We're not initially directed to go and look at the world and start doing scientific procedures. We are directed to a scripture. "Read this book. Take this book as authoritative. Internal to the book are all of the answers that you need."
- 1.70** Now, where does the book come from? And then, typically, the answer will be—and this is similar in the Western scriptural traditions and in the Eastern scriptural tradition — that this was communicated from Vishnu or from God to certain special individuals, that they received information from God. They received—sometimes we use visual metaphors; they received a vision. But they were passive, perhaps going about their business, and God spoke to them, and revealed Himself and laid some serious truth on them. They then wrote it down—that's the scripture part, and that is the source. So the ultimate source is God speaking to these special individuals. That is, in many such traditions, a claim of revelation or mysticism.
- 1.71** Mysticism is slightly broader. The idea of mysticism is perhaps everybody, or perhaps, again, only certain special individuals, if they train themselves properly, typically by trying to separate their minds from their body as much as possible through meditation processes or prayer or whatever, to try to train their minds, leave the illusory material world and ascend to the higher spiritual world and see the truth for themselves. That's another mystical claim. But that makes the human being the active seeker instead of the more passive one who is revealed to.
- 1.72** Different traditions will make different emphasis here. So, we have revelation claims, mystic claims, some combination of those, but typically, they have to get written down in an authoritative book. Some religions, of course, we're not looking at them right now, have only oral traditions, but scripture becomes absolutely important.

- 1.73** And then, the important question is, for those of us who are not recipients of the revelations or we are not ones who have had the mystical experience, I want to know and believe what is true, and I am leaning toward this religious tradition, then the words that we typically here emphasize are "accepting the tradition," the traditional view, the received view, that has come down more authoritative human beings in the past, or the word "faith," which is a cognitive commitment to a doctrine: I accept this book on faith.
- 1.74** What faith means is another big, complicated set of issues. We'll be coming back to that one. But it does not seem to mean that you go out and you do experiments and you do science and mathematics. It is a matter of committing to and accepting a certain tradition that is recorded in scripture or by some cognitive means. So we have very different epistemologies that are urged by these two stories.
- 1.75** Another difference—this one I think comes more in degrees and is more amenable to it — but when you talk to people about the state of their belief, you talk to people who are leaning toward the Big Bang theory, they're more likely to say things like, "We know a lot of stuff, but also there's a lot of important stuff that we don't know." There's an agnostic claim that is built into their self-assessment of how much they do know. "There's so much more to learn," and "I think we can make progress. Right now, we have a lot of hypotheses that we are considering. Some of them are just possibilities, some of them are more probable. Some of them I'm pretty reasonably certain about." But it's very hard to get people in this tradition to say, "I know it all for certain. I'm absolutely sure that I've got the whole story worked out."
- 1.76** That cognitive difference, again, comes in degrees, is typically not found quite so much on the other side of the divide. Instead, what we find there is, "This scripture is the authoritative truth. I believe it wholeheartedly," and a resistance in many cases to any suggestion that there is anything in it to doubt. The individuals involved will often go through some cycles of doubt, but then their faith will be re-energized. So there's a cyclical process. But the idea here is that one should be making a wholehearted commitment. If one is going to have the faith, it's a total lifestyle as well as a total cognitive commitment to a set of beliefs that one holds to be absolutely true.
- 1.77** And then, in those traditions, if one has advocates of them who are making claims of revelation or claims of mysticism, they will almost always assert those claims to be infallible claims. You cannot question the mystic. You cannot question the person to whom the revelation has occurred, and that person has no obligation to explain to you if you are not one of the persons who has received a revelation.
- 1.78** So again, we can array this set of cognitive attitudes on a spectrum from very strong faith-held certainty to skepticism and agnosticism at the other end of the spectrum. But there tends to be a waiting of the Big Bang story toward this part of the spectrum and a waiting toward more certainty and perhaps dogmatism in some cases on the other side of the story.

- 1.79** And then, what are our sources of information? In both cases, where should I go to look for more information? We will typically then have one side—we'll start over here this time—saying, "Here is the book, the scripture. Read this book." This will contain all, or if not all, pretty much all that you need to know that is true and important about the universe and our place in it. And the other side, if you want to know the whole story, they'll give you a large stack of textbooks and a lab and also point to nature and say, "Go get a whole lot of work done." And again, that's a very different epistemological orientation.
- 1.80** So where are we? We have this big question: What is reality? One question is: What is the sub-question? What's the stuff of reality? Sometimes we use the word ontology, in this case, as a sub-branch. Can we identify the furniture of the universe, so to speak, or how many different kinds of things make up the universe?
- 1.81** We know that there are conscious, psychological, spiritual, whatever we call that, that's real. There's also material stuff. They seem to be different. Do we divide all of reality into those two categories: spirit and matter? There's two different kinds of stuff that make up the world. Or is there really just one stuff, physical, material, reality, and the spiritual is a byproduct of underlying material processes? Of course, we can go the other way. We can say, really, only the spirit is real, and matter is just an illusion. Just from those two candidates, in our ontology, we've got at least three possible metaphysical positions.
- 1.82** Causation—there's the stuff, the stuff does stuff, and as a result of the stuff doing stuff, other stuff comes into being or the original stuff is in a different state; there's been change, and we start to talk about causal processes.
- 1.83** Cause and effect—what's the nature of cause and effect? Is there one unifying causal force? Are there multiple causal forces out there in the world? Is there a direction? Is it the case that we started up here and we're all just going down, down, down, down, down, pessimistically? Or we started down here, but the direction of causation is progressive; it's getting better and better? Is it cyclical? Maybe we get a little bit better, but then it gets worse, and then it gets better and gets worse, but then, over time, ultimately, it's a wavy but nonetheless flat line. What is the nature or the direction of the causal process of the stuff of the universe?
- 1.84** Any of the claims that we make about any of those things, what are the facts? We have to develop an apparatus for distinguishing what's a factual claim from what is an illusion. Now, we have this grand possible claim that all of the material world is an illusion. We might not want to go that far, but nonetheless, every epistemology has a recognition that illusions do exist. Sometimes I think something is there, but it's not there, or I misidentify what seems to be there, or it's only apparently there, but not really there. What's the distinction between facts and illusions?

- 1.85** We have this concept of truth and a contrast concept of the false. Sometimes we come to this from assessing our own stock of knowledge. There are things that I thought I knew to be true, and then I just go through a process, and I realize that is not true. So I form this concept of falsehood. But in principle, what's the difference between the true and the false? What processes reliably keep us on track for the truth? What processes are fallibilistic or unreliable processes that almost always take us in the direction of falsity? Can we identify those things?
- 1.86** When we self-assess much of our knowledge, we might want to have certainty. It'd be very good to know that the things that I believe are absolutely true with certainty. Then with confidence, I can believe them, I can act on them in the world, and I know I'm going to be successful. But chances are good, the world is very complicated. Sometimes my knowledge is partial.
- 1.87** So what I need to do, when I'm self-assessing my own cognitive process, is to be aware that I'm not in a position to be certain. I might have a lot of data, and I might have thought it through, but this is only, say, a high probability or maybe a low probability, or maybe it's just a bare possibility. Training my mind and my self-assessment of my mind to know where I am on that cognitive line is very important. Philosophers spend a lot of time sorting that one out. Can we identify consistent methods of using our senses, using our memory, using our grammar, using logic, using mathematics, and so forth to yield what we want to call the norm?
- 1.88** Now, we've told two types of stories so far. I want to say that's already really complicated stuff, but it's more complicated. I want to throw out that maybe there's a third type of story that we should consider. Both of those stories are characterizable as narratives, and narratives try to put everything in a package and put a nice, pretty bow on top of it, so to speak, metaphorically. But maybe that itself is an illusion. While we have the creation stories and the Big Bang stories, and there are very different stories in lots and lots of different ways, but they are both stories, and they are both just so stories or made up stories that we tell ourselves because we would like there to be a nice, tidy story with a bow on top of it, but the reality is there is no story.
- 1.89** Maybe reality is not knowable to us. Maybe reality in itself is just a big steaming mess—that's a metaphor. It's not even got its act together enough to be a steaming mess. It's chaos, arbitrary. But nonetheless, we are making up categories and stories and imposing them in a nice tidy way on a reality that itself is not there. The ultimate story is reality is a mess. It's unknowable. And at the end result of our philosophical investigations will put us there. We are into the philosophy now. When we come back, we'll start to plunge into the details of some of those sub-stories. I'll see you then.
- 1.90** You mentioned Muhammad and Vishnu, Krishna, all those things, and Jesus. Those are all people who said, "I heard something, follow me. This is the way." The big difference with Christianity and Jesus, He said, "I am God." How does that—how would you fit that into an accurate form, given what you're trying to say here?

- 1.91** I would say that's an important sub-question inside the mystical tradition. So the first group of people you mentioned, a lot of them, "I heard something," or "I received a vision." In many cases, it's perceptual metaphors that are being used. So, "I heard something," but the claim there is not that I am the author of what I am reporting to you. I am a vehicle of transmission. The second part of that, "follow me," then is a suggestion to the person that you should trust me and my claim that my vision did come from where I said the vision came from, and you can't get that vision yourself, so you have to follow me in order to be on the right path and so forth. Good.
- 1.92** Now, that's to say the metaphysics is more complicated because you've got three parties: the source, the mystic, and then the followers. You're right in the claim of Jesus; if Jesus says, "I am God in a certain form," He's in effect cutting out the middleman. "I am speaking directly to you. Follow me." Now, in that case, He does have two claims that need to be justified. One is the claim that what He said is true, and then His justification for that is not going to be like the other people say, "Because I got it from this guy." He has his follow-up claim to say, "I am this guy. I am the God." So then the next question will be: Why should we believe that claim?
- 1.93** I think the conversation would carry on to say: What's the justificatory story going to be for that "I am God claim?" If that justification story account can be made good on, then that He is God stands good, what God says, Jesus, God/Jesus says, is reliable, then the people will follow on the basis of that. So I think that's the difference.
- 1.94** If I'm putting it correctly, generally speaking, the Big Bang story is, "Ask and keep asking." There's that element, right? And the creation stories is, "This is the answer. Don't keep asking questions past that." However, in Judeo-Christian tradition, there's also free will. So honestly, I'm curious in the Vishnu story or in Islam, I'm currently reading the Quran, but I'm not that far along, do they have the concept of free will? And how do they—
- 1.95** Some of them do, and some don't. In many cases, the exact same philosophical issues that we grapple with in all of the Western traditions are elaborated in the sophisticated Eastern traditions, certainly in Hinduism and in Buddhism. Some of the other ones, I am not expert enough in to say, but my judgment is that Hinduism and Buddhism are the most philosophically sophisticated of them. Some of the others have some sophisticated philosophical positions that lead them to say we should not be doing philosophy—that's an interesting exit from the whole process, but that's not quite what you are asking.
- 1.96** So if we come back to the Western traditions, and I don't want to oversell it right now because in the Jewish, Christian, and Islamic traditions, those are now, for some of the sub-schools, very sophisticated philosophical traditions as well. And there, all of these metaphysical and epistemological issues need to be engaged with, not on grounds of, "We're just asserting this." Many of them are willing to say, "We're going to have theories of language and interpretation and hermeneutics and so forth. We're going to

make arguments for and against, and we think that those arguments can be based on observational data." But that's to say there are sub-philosophical schools within the broad traditions.

- 1.97 In the broad traditions, this starts to become more of a sociological claim about whether the majority of people in those broad traditions are doing the hard theological philosophical work versus accepting things on faith, accepting things on authority, accepting things as certain or not, and then an internal debate within those traditions about which philosophical approach is truest to the tradition. I don't want to prejudge any of those issues.

Lecture 2: Essential Questions

At a glance — Essential questions, including the Ship of Theseus and identity.

- 2.1 A classical metaphysical question comes down to us. It's always fun to ponder. It's called the Ship of Theseus. Theseus, one of the ancient heroes of the Greeks, went off, had many adventures. Imagine if, after all of his various adventures, we were able to retain and look after his ship. But of course, it would need repairs over the course of the centuries and so on.
- 2.2 But suppose we stayed even within Theseus's lifetime. So he buys a ship or has it built, and he goes off, and after the first voyage, a storm hits, and the sail gets ripped, so the sail gets replaced. Then, they get into a battle, and many of the oars on one side get replaced. And then, from wear and tear, some of the planks on the main deck have to be replaced as well. Then, at one point, they run into some rocks, so some of the underlying construction of the keel gets replaced.
- 2.3 Then, over the course of many years, say 80% of the original pieces of material that made up the ship of Theseus have been replaced. The question is: Is it still the ship of Theseus, or is it now a different ship? If you say that no, it's 80%, but what if it was 90%? Or 99%? What makes the ship of Theseus, the ship of Theseus? What's its identity? What can change? What has to stay the same?
- 2.4 It's often said that philosophy begins with wonder. We wonder where it all came from. I like to think that all of us become philosophers by the age of four when three-year-olds, and especially four-year-olds, have learned to say, as they are in their exponential language-building process, "What is this? What is this? What is that? What is that?" You find yourself as a parent answering all kinds of questions, and you find your vocabulary expands not as rapidly as the kid's vocabulary, but nonetheless. "What are things?" We want to categorize them, assign labels to them, and so forth.
- 2.5 Then, the kids fairly quickly ask when things change, "Why did that happen?" Then we give an explanation for that. Then, they realize they can ask why again, "What about that thing? Why did that happen?" Or if it's something that's this big, "What is it?" We break it down to underlying constituents or processes. If we say, "What's under that?" Then, it's "what" all the way down, or it's "why" all the way down. We keep asking why until we get to the end of the story, supposing there's an end of the story.
- 2.6 Suppose we start there, at the beginning, and we start asking, "What are we talking about?" We start naively, and I start observing the world, and I seem to say that the first thing is that there are things or there are beings. There's this being, that thing. It is, it is, it is. There are all these individual beings out there, or things, or existence. That seems to be a starting point. What if we took all of those individual beings and we started to say that they all be, so to speak, or they all are. This thing exists, that thing exists, they all exist. If we try to say, what about not particular beings, but all of the

beings? We talk about being as such because all of the individual beings share something in common: they are, they exist, they're real.

- 2.7** What about reality as a whole, being as a whole, existence as a whole? What can I say about it? So the question of being. Individual beings are. What is this being that they all share in common? They all exist, so they have this— is it a property of existence? Is it an attribute of existence? And existence as a whole, what is that that they all share?
- 2.8** Now, we could look at the beings and then note that they all are, but also that lots of them are different. So this being seems to be something, but it's in some ways different from this being, different from this being, and different from this being. So they all be, they all are, but they have an identity or a set of attributes or a nature or something like that. Sometimes their identities are similar, sometimes they are different. So what is this identity? What makes this thing the thing that it is? And what differently makes this thing the thing that it is? What's the difference of their identities or the similarities of their natures?
- 2.9** Things are, but we also notice that they change. I might say this thing was like this, and it has a certain identity, certain attributes, certain properties, whatever language we develop there. Then I follow it, and as it changes, it no longer has all of the same attributes or all of the same properties that it did. It has changed. So we have things. They are. They are what they are, but they're also changing. What makes this change? What makes change possible? What makes it necessary? What is change?
- 2.10** I might drill down into some of those changes and say, one of the changes is, say I'm looking at myself and I notice my hand here, and then my hand goes over to here, and then my hand goes back over to there. That's the same hand, but something is different. I might develop the word space. Nothing has changed except the here-ness and the here-ness. Or my other arm is changing. Now it's changing. Space seems to be real. I see you. I see you, but something is different. So it's space. So now we have: things are, they are what they are; they seem to change, and one of those fundamental things of change is spatial change, if we're developing the concept.
- 2.11** Now, I might also then say, I'm looking at my hand again. My hand moves over here. My hand moves over here and over there. The hand is. It's the same hand in each case. It's the same space change in both cases, but something is different again. This space change, that space change. Time is different. I have my arm here, I raise my arm, I lower my arm. Arm is what it is. It's an arm. Change of space is the same, but the time concept is different.
- 2.12** So, a basic investigation—we've now got four or five basic categories, or concepts, or realities, or features that we are aware of. Then, as I become more sophisticated, I might seem to say there seems to be also something constant in all of these things. There's an I, aware of these beings. There's an I or a self or a consciousness that is aware of the time change, the time difference, the space change. I might even say

there's a self-awareness: I am aware that I am aware. That also seems to be a reality of some sort. Maybe we have 6 features, aspects of reality.

- 2.13** Now, let's go back through each of them and see what we can do in a more sophisticated fashion, because each of these is posited as a basic metaphysical candidate, basic metaphysical category, basic metaphysical feature. But about them, as everything is controversial in philosophy, there is controversy. Along the way, I want to say what we can say, what's controversial about it, and name-check the important names in the history of philosophy on each side of the controversy. This will be an introduction. The idea is we'll get the basic material out for further reflection.
- 2.14** What can we say about being? One of the things we immediately run into is a language issue, even internal to our own language, where different philosophers will describe the same phenomenon with different words. Being, existence, things are, things exist—those are synonymous—reality, what is, actuality, factuality. What all of these words as concepts are trying to capture is out there, reality, something exists, et cetera. Using all of those for now as interchangeable terms, some philosophers will, of course, insist on subtle differences, but for now, let's keep our language flexible and say that we're all trying to focus on the same phenomenon, whatever it is that we are talking about.
- 2.15** What can we say about existence in general, or all of being, or reality as such, whatever verbal formulation that we are trying to get to, reality? We started looking at individual beings, but now we're trying to generalize to reality as such, being as such, all of being. Sometimes we will use the capital letter B to indicate that we are talking about the big picture, big perspective on beings as such, not particular things that might happen to be beings in general. What can we say, think, conceptualize, when we're operating at that level?
- 2.16** One thing we might do is just to say, "It is. It is. It is. It is." I affirm existence. That's where everything starts, an affirmation that existence exists. Reality is real. What is, is. One entire school of metaphysical philosophy will say that is the right answer. That's where you start: accepting, affirming reality as existing. Another school will, though, say that we can always ask that why question: Why does reality exist? Why do beings be?—using the ungrammatical form just for fun. Why does existence exist? Yes, all of these beings are collectively or integrally. Reality is real. But why is reality real?
- 2.17** The first big controversy in this unit is to say that one of those two positions has to be false because they're saying different things. Aristotle is on the side of the affirmationists: the fundamental axiom is that reality is real. Everything that we want to do on the basis of that, we want to say anything else about reality has to start by saying that what is, is. And of course, what is not, is not. And there we have an is, is not, reality, not real. All of our statements then are going to be true or false, depending on whether they are speaking of a reality that is or is not. So we have a metaphysical axiom of existence or being or reality. We try to get beyond that. Instead, we go on taking reality as it is and trying to say things about particular things inside reality, and

those might be true or false. "What is" makes statements true or false. That has to be the first thing that we say.

2.18 Now, the other side wants to say that at least grammatically, we can form the question. I can say that yes, reality is real, being is, existence exists. We know, as a rule of grammar, we can always ask why. Why don't I just take that proposition and ask why? Why does existence exist? Why is reality real? Or there's a distinction between what's real and what's unreal, between things that exist and nothingness, there's a difference there. So what explains the existence of existence?

2.19 We might then go to Stephen Hawking, the very philosophical physicist of the 20th century. We'll read this through how this comes to this very intelligent gentleman. "Even if there is only one possible unified theory, it is just a set of rules and equations, what is it that breathes fire into the equations and makes a universe for them to describe?" So we have the universe, and we have all of these equations now. As physicists, we're developing to describe the universe, but the equations are just words, formulas, and so forth. What then—he uses the metaphor—"breathes fire into it?" What makes them describe the world the way it is? That is to say, why is the world the way it is?

2.20 "The usual approach of science of constructing a mathematical model cannot answer the questions of why there should be a universe for the model to describe. Why does the universe go to all the bother of existing?" It's a fun formulation. He's anthropomorphizing a little: I'm the universe. Should I exist or not? I guess I will make the effort. I'll bother to exist.

2.21 It could be that there's nothing, but there isn't nothing. There is existence. Why is there existence and not the nothing? To him, that seems like a legitimate question. So we're not going to start by affirming existence. We're going to try to get under existence or behind existence or behind reality in some way to explain why reality is in the first place. There's a deep divide amongst metaphysicians over exactly this question.

2.22 Now, the other side wants to immediately say that that is an illegitimate question to ask. Why does existence exist? Grammatically, it meets the conditions of our language, but our language has to be governed by true facts about reality and ultimate axioms about reality, and there is no way to get behind, underneath, or beyond existence. When we use the word "why," what are we trying to do? We're trying to say there's a cause and effect. But what's cause and effect? Cause and effect don't make sense unless you're talking about existing things. The cause exists, the effect exists. So "why" questions presuppose cause and effect. Cause and effect presupposes existence. So you can't then take cause and effect and "why" questions that presuppose existence and apply them to their precondition. That's an illogical move. So grammatically, it's fine, but logically, it's not fine. It's the counterargument here.

- 2.23** Or if we say, our point is we started with all these individual beings, and we've generalized to all of being, all of existence, if we're going to try to explain all of existence, where would we have to be in order to provide an explanation? It sounds like we'd have to step outside of existence. Here we are outside of existence, and here we find something that explains the coming into being of existence. Stepping outside of existence, what does that mean? Well, if we have said all existence just means everything that exists, you can't step outside of everything that exists. Even the metaphors we're trying to use here—"step outside"—it doesn't make sense.
- 2.24** What's outside of existence? Notice we're using a spatial metaphor, outside of existence. Well, what would be outside of existence if there were such a place, a state, or whatever? If existence is everything that is, then we would have to be stepping into what is not, into nothingness. Somehow, the question is asking us to imagine ourselves or project ourselves into the nothing. But if you're in the nothing, what can you do with nothing? Nothing, right? You can't make existence or explain existence from nothing. You've got a big bunch of nothing. That's all you're going to have. It's just a big bunch of nothing.
- 2.25** The argument, there's a counterargument here, to say it's an illegitimate question. The fact that you reach these absurd results, illogical results or irrational results show that that's not the question to ask in the first place. You have to start simply by affirming reality, existence, being.
- 2.26** Now, the other side wants to say that we don't like that, of course. Most famous, more recent exponent of this is Martin Heidegger, German philosopher of the 20th century, who's very dismissive of that counterargument. What he then says is, "Thinking begins only when we have come to know that reason, glorified for centuries, is the most stiff-necked adversary of thought." So this question we say: Why is reality real? The other side wants to say, that's an irrational question. It's an illogical question. You can't reason about that reason presupposes existence and so forth. He wants to dismiss that entire approach to reason as the enemy. Somehow we have to do this in an admittedly non-rational way. One of the things Heidegger will then do is start to explore the questions more emotionally, less rationally, less logically. There's an epistemological set of issues that will come up later and so on.
- 2.27** All right, that's the controversy over being: Do we start with existence exists, reality is real, or do we try to get underneath it or behind it and explain it some way or other? Things are, but they also are what they are, and they share certain common features. But they also are different; everything has its own identity, its own uniqueness, as well as being, in many cases, similar to other things. What can we say about identity? Things are, but they also are what they are.
- 2.28** We might formulate this question in a lot of different ways. Why is reality the way it is? Maybe things could be different. What are things? Yes, they are, but what? So it is, but

what is it? And whatever the way things are, why are they that way? Why not in various other ways?

- 2.29** Or we might use essence language. We might say things are a particular set of attributes, but there's something essential that makes them the things that they are. So I'm a human being. I'm made up of a whole bunch of material stuff, say, all of the foods that I ate, the oxygen that I took in. But there's something that is essentially different from me, say, from another animal that might also be eating the same kinds of foods and taking oxygen in and so on. We have different essences. We are different kinds of beings. So all of those are the kinds of questions we want to grapple with.
- 2.30** Now, Aristotle, again, wants to say there's a tight connection between identity and existence. And so we start by saying what is, is, and that to be something in particular is built into the nature of things or the reality of things. And so the way he formulates it in his metaphysics is by means of a principle of non-contradiction. We'll just read it through. Where would we start most fundamentally? "Such a principle is the most certain of all; which principle this is, let us proceed to say. It is that the same attribute cannot at the same time belong and not belong to the same subject and in the same respect."
- 2.31** So everything that it is, whatever attributes we talk about, either this thing, it has it or doesn't have it. This other attribute has it or doesn't have it. Has it or doesn't have it. And the entity, the being, is the collection, the integration, or the expression of all of those particular attributes. This is then fundamental to logic, because if things exist and they are what they are, then, when we are thinking more complicatedly, we have to identify things as they are, but in a non-contradictory way. It's impossible for us to say this thing has this feature and it doesn't have this feature. Either it does, or it doesn't.
- 2.32** A follow-up question, a close corollary: things are what they are; is it intrinsic to them, or is it acquired? There's a subtle issue that has divided so many philosophers on this over the century. On the one side, the formulations we'll see is reality has a nature, or existence is identity, to be is to be something. So there's a tight connection between being and identity.
- 2.33** Other traditions will say that existence and identity are separable, and their conjunction is something that we have to explain. We might say, initially, something is, but it's not anything in particular. It doesn't have any form, or structure, or identity. Something comes along to take that formless thing that doesn't have any identity in particular and gives it an identity, makes it the thing that it is, or identity is imposed upon things, or essences are added to things.
- 2.34** These are highly abstract formulations. Let's get down to a little bit more nitty-gritty. Thomas Aquinas is probably the most important name in sophisticated thinking in the entire tradition about why there is this issue about existence and identity in the first

place. He wrote a book in the 1200s called *On Being and Essence*; chapter 4 is the one you will want to focus on for this clarification.

- 2.35** Mostly, Thomas Aquinas is in the Aristotelian tradition. His major project was to try to reconcile Christianity with Aristotelian philosophy. On this issue, though, I wanted to suggest that he is more Platonic and less Aristotelian. Nonetheless, he is the big name-check on your list of future reading for this issue.
- 2.36** If you think of the Christian or the Judeo-Christian account of creation, one of the things we saw was that it says, initially, the Earth was without form and void or empty and not anything in particular. So that then is to say it existed, but it had no identity yet. What gave it its identity was an external act by God of imposing an essence or adding an essence to various things. In principle, identity and existence are separable, and we have to explain their conjunction, perhaps by means of an external agent like a God. The other side of the argument wants to say no such external agent is possible. The identity of things is built into their existence, and that's the distinction.
- 2.37** We see this coming down to us, and why I say that Aquinas is more Platonic than Aristotelian on this tradition. Of course, an ancient philosophy would be a very useful complement to what we are doing right now. Here's a quotation from Plato. What Plato wanted to argue was that all of the things that exist in this material world are real, but they're only a lower order of reality, or in many cases, semi-real, shading in many cases into unreality. So they are, but what really is is the essential natures of things in a higher reality.
- 2.38** So we might say, for example, that I have three drawings of a triangle, and because I'm not a very good draftsman, my triangles are not drawn perfectly. But we'd all recognize that they are triangles. But if we were to measure, say, the triangles, triangles are supposed to be straight lines, three lines enclosed, add up to 180 degrees. We say that the lines Hicks drew are not very straight. It's a little bit wavy. Or the angles, if we measure them carefully, only add up to 179. So, this is sort of a triangle. It almost has the identity of a triangle. It exists, but as an imperfect semi-real triangle.
- 2.39** Nonetheless, we have this idea of perfect triangularity, the essence of what it is to be a triangle: three sides, all straight, interior angles, 180 degrees. If we don't find perfect exemplars of those things in this world, there must be a higher world that we tap into to come to realize those essences, the nature of triangularity in its pure abstract form.
- 2.40** "But of the heaven which is above the heavens, this is the region whereof I must now speak. For thither, the truly existent essence has found the intangible essence visible only to the mind, the pilot of the soul." So there's a tradition here that says this reality is sort of reality, it exists, but it is only sort of what it is. Things that are really, essentially, truly what they should be, essential things, are in some other reality. And there's an epistemological correlate to this: since our senses are tied to this physical world, we must have another capacity, our mind's eye, or a soul that's able to tap into

this higher reality. So again, is it the case that existence and essence, our being and identity, are related or not?

- 2.41** Name-checking another important person coming into the 20th century. We might say the Judeo-Christian tradition or the Platonic tradition says, first there are all of these essences up in the mind of God or up in the world of Forms, and then things that are in reality exist without any—lower reality exists without any particular identity. Then they are formed or shaped or essences are added to them. In that case, essence really comes first, and then existing as a particular thing comes second.
- 2.42** Jean-Paul Sartre, part of his metaphysics, is to deny the existence of God. So there can't be any essence-imposing being on reality. It just must be the case that things exist at the beginning as nothing in particular, including the human being. We don't have a human nature, so to speak. We just exist. We come into the world, and somehow our essence or what it is that we are going to be is something that we create ourselves. It's not created by God; we do it ourselves. But the metaphysical formulation is: existence comes first, not essence. It's the opposite of the tradition that says that essences exist first, embodied in, and existence comes second.
- 2.43** The Aristotelian tradition, I'm going to say, carries on in the 20th century in Rand, to name-check another important name here. Notice this formulation as we read it through. Existence first or essence first? We have those two traditions. This position here is going to be that's a false alternative. Existence is identity. To be is to be something, and we're right back to Aristotle.
- 2.44** "To exist is to be something, as distinguished from the nothing of non-existence. It is to be an entity of a specific nature made of specific attributes. Centuries ago, the man who was—no matter what his errors—the greatest of your philosophers,"— she's referencing Aristotle right there—"has stated the formula of defining the concept of existence and the rule of all knowledge: A is A. A thing is itself." It's not that the thing is, but it's not anything in particular. We need to add something to make it what it is. These things that make things what they are can be in some nonexistent state in some other realm.
- 2.45** "You have never grasped the meaning of his statement. I'm here to complete it: Existence is identity." We have three positions. To be is to be something. Existence is identity. Versus the two that want to separate them and say: this one comes first, this one's secondary, or this one comes first, this one's secondary. On the basis of this, Aristotle, going back to Aristotle, then again, wants to say we have, if things are and they are what they are, the basis for a rational, logical approach to epistemology. If you want to deny this principle, then you have to do something non-rational, non-logical in your epistemology.
- 2.46** "To hold equally to the opinions and seemings of those disputing against one another is foolish, for clearly one of them must be mistaken." It's an either-or. So either existence

exists and existence is identity, or essences are independent of existence, or beings are without anything, any identity in particular.

- 2.47** Change—suppose we've worked out our minds, what we think about things existing. Do we affirm or try to get behind? Things are what they are. Did it come that way intrinsically, or was it added? Things are changing. How do we process change? We could talk about difference, variance, process, different philosophers, and trying to capture this aspect of reality, using different languages. Let's keep them all more or less synonymous for now.
- 2.48** Number of sub-questions—things, when they change, do they change in a causal way so that there's order in causality? Maybe that order is something that we can then capture in laws or principles when we do our thinking, perhaps our science and so forth. Or is the case that things are, and they are what they are, but the reality of all of that is just that it's a big chaos, and there is no cause and effect structure or order in the world?
- 2.49** This question, now closer to the Ship of Theseus issue: Is it that things change in a flux capacity, and so things lose their identity as a result of the change, or is it a case that things can change but retain their identity? We can say that a caterpillar forms a cocoon and then transforms itself rather dramatically into a butterfly. That's a big process of change. There might be an identity process that works through, but is the now flying butterfly the same being as the caterpillar, or is it a different thing?
- 2.50** There are laws in principles—the language is a little bit important here, and sometimes subtle. If we want to say there is causality, there's an order, there's a structure to various things. When we do our thinking about it, we say here's a law of biology or a law of physics, or a law of chemistry. When we use the "law language," often we are implicitly saying that there must be a law-giver that created the law and therefore imposed it on reality. In many cases, the "law language," we're trying to start to describe the laws of cause and effect, comes bundled with the assumption that God or a divine-like creature is the ultimate law-giver.
- 2.51** Sometimes a tip off to the other metaphysical tradition will be to say that what we are doing is not identifying laws out there in nature, but rather we are capturing causal processes in principles, and that the principle does not assume that there is a principle maker or principle giver in the first place. So just be attuned to that language for now.
- 2.52** When we're talking about causality, is it the case that there is such a thing that is real out there, a cause and effect structure, or is it that what we call causality is something that we have come up with and imposed on reality? We know it is possible for us to take a whole bunch of data and impose a pattern on it. So there's a question: Is that what we always are doing? We are never actually capturing real patterns that are out there in reality. The question of causality, coming out of the question of change, as we're trying to understand, raises a whole bunch of other subordinate questions.

- 2.53** Now, very early in Greek philosophy, there came to be a divide between philosophers. They're called the pre-Socratic philosophers, because they were sophisticated philosophers before Socrates. Socrates was then the teacher of Plato, Plato was then the teacher of Aristotle, so we were generations earlier. One generation before Socrates was Parmenides, who wanted to argue that things exist, so we affirm reality. Things are what they are, so we affirm identity. Things that are not, are not, so we're still affirming identity. But we cannot make sense of change. There's something paradoxical about the notion of change.
- 2.54** His argument was captured here: "What is, is, and what is not, is not. And what is not can neither be nor be thought of." That's hard to wrap one's mind around, but let's dive into it. Here's a thing that is, and suppose we say it has property A. For this thing to change, property A, for example, would have to go out of existence. And then, some other property that does not currently exist would have to come into existence. So A, whatever that property is, that is, has to unexist itself, whatever that means. And something that is nothing, does not exist, would somehow have to existify itself—we're making up words here to show the ridiculousness of the process.
- 2.55** If you've got nothing, you can't get something out of nothing. And things that are, can't go out of existence. So for him, it was a puzzle that could not be resolved. The conclusion is, if we say things are and they are what they are, and we're going to be logical and affirm those principles, we have to say that change is an illusion. Our reason tells us that this is correct. Our senses tell us that things change and go in and out of existence, and so forth. What this means is we have to trust our reason, but not our senses.
- 2.56** We're going to go in a different epistemological direction as well. In this sense, Parmenides is the first to use anachronistic language, what we would call a rationalist. A rationalist is someone who is in favor of reason, but reason divorced from or not reliant upon empirical sensory information. The contrast will be to empiricists who will say, we are eventually going to get to reason, but we start with our sensory information, and we build up to our reason. So this is the first anti-empiricist position or the first rationalist position. Again, these are anachronistic labels at this point.
- 2.57** Now, immediately in the next generation, rather, his student Zeno is famous for developing some paradoxes about change, also to show that change is impossible. The argument is: suppose we start a race and we have two people—it could be two human beings, sometimes it's a rabbit and a turtle, a hare and a tortoise, or sometimes it's Achilles shooting an arrow. The idea here is if you start with two beings, suppose you have a hare here or a fast rabbit and a tortoise here, and you say, "Let's have a race," and the rabbit says, "Oh, man, you're having me race against a turtle. This is insulting. I'm going to give the turtle a headstart." The turtle starts off, and the turtle goes halfway down the field.

- 2.58** Zeno then goes on to argue that it's mathematically impossible for the rabbit to catch up to the tortoise and overtake it because, for the rabbit to catch up to the tortoise, first it has to get to where the tortoise is now. It's going to take a certain amount of time to get to here, but by then, the tortoise has advanced. Now the rabbit has to get to where the tortoise is now, and that's going to take it a certain amount of time. But by the time it gets there, the tortoise will have advanced again—less amount, but still advanced. Then once again, the rabbit has to get to here, but then the tortoise will advance. We can keep going this process mathematically on into infinity, which proves, logically, mathematically, that it's impossible ever for the rabbit to catch up to the tortoise once it's had a headstart. That's a mathematical proof. It's very logical, it's very rational.
- 2.59** Now, our senses, when they are interacting with the material worlds, "Oh, come on, we know that the rabbit is going to beat the tortoise." But that just then shows that our sensory understanding of the way the world really works is illusory. We go with reason, we go with mathematics. So this is one side of an early divide over the issue of change, and it wants to rule out the existence of change. Ultimately, the universe is one unified, unchanging, eternal.
- 2.60** Now, there might be a connection to many of the religious traditions, because if we ask: What is the nature of ultimate reality for many of those religious traditions? It is God. God is everything. God is ultimately eternal. He is, that is to say, out of time, which is to say He's not changing. As a perfect being, He does not need to change. If you're already a perfect being, any change would be in the direction of an imperfection. So it's impossible for God to change. So whatever considerations lead to that religious conception of ultimate reality being an unchanging reality, these philosophers are getting to it by a different analytical route.
- 2.61** Now, on the other side of the divide, the argument comes from Heraclitus, who wants to say that obviously, the world changes. If your logic, and mathematics, and reason are incapable of grasping this basic fact of reality, so much for them; we go with reality as we see it. But there is still something paradoxical about change, that what change proves to us is that identity is impossible. The famous formulation is to say everything flows, that is to say everything changes. We use lots of water, liquid metaphors.
- 2.62** His most famous aphorism that comes down to us is the one that says, "No one steps into the same river twice." So if you say, what is our concept of river? We have this idea of one word describing one feature in reality: this thing we call the river, the Rio Grande River. But what is the Rio Grande River? Well, it is these boundaries. It is this water. It's not this land over here. It's not that water over there. It is defined this way. If that's what we mean by the Rio Grande River, that's what it is, that's what its identity is, then every split second, it's different. I may step into the river now I step out, it changes. It's different water. Some of the sand or soil at the side of the river has also

washed away. The border has changed, the water has changed. When I step into it, it's a different river. It's not the same.

- 2.63** So that then is to say, if we affirm that change exists and is real, then we have to say all of our concepts, language, that try to freeze reality and fix reality and say things are what they are and they are the same, that if I grew up by the Rio Grande River and I went off to college and had a career and came back 40 years later and said, "Oh, it's good to be back to—" Well, it's not the same river. You can never go home in this sense, metaphorically speaking.
- 2.64** So change is real; therefore, identity does not exist—that's the Heraclitus position. Identity exists, things are what they are; therefore, change is impossible—that's the Parmenides position. The Aristotelian position is going to affirm causality as a function of identity—things exist, they are what they are, and so what things do who or how they change is built into their identity.
- 2.65** Now, he has some sophisticated language that he develops about potentials and actualities, but things are what they are right now. But part of their identity is built into them a potential that they can act, they can change in various ways. The caterpillar has the butterfly in it, so to speak, as part of its potentiality. So there is an existing thing with a certain identity. It goes a certain causal process, and we can identify this without paradox. We have three positions then with respect to the nature of change.
- 2.66** Another fun one: Albert Einstein. Here, he's speaking a little bit tongue-in-cheek on change: "There's only two ways to live your life. One is as though nothing is a miracle. The other is as though everything is a miracle." And that's fun. But I'm using this because I want to get to this miracle point. How are we to understand the concept of miracles and claims about reality of miracles? Because suppose we say, as a result of our analysis, we come down either on the Heraclitus side or the Aristotle side and say change is real as a result of things being what they are and doing what they do. And so there's order, cause and effect structure in the world. Does that leave any room for miracles? Or do we want to say that that analysis would lead us to conclude that miracles, as traditionally conceived, could not exist?
- 2.67** If we say, for example, there are laws of cause and effect or principles of cause and effect, is it possible for those to be overridden? Because typically, as we understand a miracle, it is defined as a law, or a regularity or a principle of order being overridden. We might then say that our understanding of the existence of reality, if we're going to try to affirm miracles, things are what they are. Causality is built into it in a certain way, but there's an external being or agent, a God, that can override those causal principles and change them in perhaps a temporary way or perhaps in a more grand way, changing entirely the laws of physics, the laws of mathematics, the laws of chemistry, whatever they are.

- 2.68** So is it the case that existence has its identity, it has its then principles of causality, and those are strict and built into reality? Or do we say that, in some sense, there is also an outside being that is able to override all of those things? That's another change position.
- 2.69** Space and time, let's move on to them, consider them. Again, hard to wrap our minds around a lot of these things, fundamental philosophers, physicists, and others. Let's plunge in. Do we think of these as absolute features— this is the language of this sub-debate—as relative or as subjective? We're going to get three of them.
- 2.70** On that dimension, do we think of them as finite or infinite? We all run into the question of, as I imagine myself, looking up at the star, if I were just to keep going and keep going and keep going, would I ever reach the end of the universe, whatever that means, the edge of the universe? So there's only so much, and then it stops, or is space infinite in that direction, and I could keep, in principle, going? That's finite or infinite.
- 2.71** On the time issue, if we go back in time, is there such a thing as the beginning of time? If we go into the future, will there be an end of time such that time did not exist at some point? It does now exist, but it has a beginning and an end. Or is it the case that we could, in principle, with our investigations, just keep on going back and back and back in time without end? The same thing is going to happen in the future; it will carry on and carry on and carry on without end. How do we conceptualize these two features of the world? Because we know things are in space and time, seems to be the case.
- 2.72** The most famous representative of the absolute position is Isaac Newton. In his Principia Mathematica, a landmark work of physics in the early modern world, he argued very explicitly for space and time as absolutes. There's space and time and then everything that exists and the things that are what they are and are doing whatever they're doing that the science is describing exist inside this absolute space and time. Things are subject to space and time as a primary fundamental.
- 2.73** The quotation here: "Absolute, true, and mathematical time of itself"— it's an independent existent or an independent feature of reality—"and from its own nature flows equably without regard to anything external." So that is to say space is not dependent upon anything else. The relative position is going to say that space is relative to or depends on something else. But he's going to argue space is primary. That was time, sorry. "Absolute space in its own nature, without regard to anything external, remains always similar and immovable." So these are axiomatic and fundamental.
- 2.74** Perhaps metaphorically, you can think of space and time as a giant container, like a big box, a big spatial box with an internal clock of some sort. So the entire universe then is this big box of stuff regulated by some intrinsically universal clock that is what it is and goes, whatever. It's going to be regular and uniform across all of space.
- 2.75** Now, the contraposition, to wrap our minds around, is most famously associated with Albert Einstein, starting a little over a century ago, and he argues the reverse position,

that rather than space and time being primary and fundamental, what is primary and fundamental are existing things doing whatever it is that existing things do, and space and time are dependent or relative or functional outcomes of whatever it is that existing things are doing.

- 2.76** So light is moving in one of his famous experiments. So he asks you to imagine yourself, say, on the front of a spaceship. And so suppose you are going and you shine a flashlight ahead of you. Suppose the light goes at a certain speed. Suppose your spaceship is traveling at one one-hundredth of the speed of light. What would time seem like to you? The argument is it would be different if you were traveling at one-tenth of the speed of light. So you have these entities moving and changing of their own accord, and our measurement of time is going to be a function of us relative to light.
- 2.77** Now, the light is a real thing. The spaceship we are on is a real thing. We are real things doing the measuring and so on. But nonetheless, time is going to be different depending on whether you are going $1/100$ th, $1/10$ th, or approaching the speed of light. It's the relative function.
- 2.78** Probably the same thing with respect to space in a moment, but here are the quotations from Einstein that are directly relevant on this. For him, the relativity does not undermine the objectivity of space and time. Objects are real; their motion is real. Space and time as relational phenomena are dependent upon those underlying objective phenomena. "The actual objects of physical reality, physical objects are not in space, but these objects are spatially extended." So the space is a function of the object, not the object a function of a position in space. That's the objectivity part; everything comes down to those things.
- 2.79** The relativity, though, "Every description of events in space involves the use of a rigid body to which such events have to be referred." The rigid body, in this case, a little bit metaphorically here, might be, I use a rod, and I will take this rod, and if I'm measuring space, that rod is a real object, and I'm relating that real object to some other real object. So it's a relationship between these two objects. That's what we mean by space. The objects are what they are.
- 2.80** Let's try another physical example. Imagine, again, a spaceship. We got a person on a spaceship at three different times, so to speak. Now, imagine that person standing on the spaceship— you actually would do this inside the spaceship, but the graphic doesn't work quite as well if you try to put the person in there— has a ball in his hand, and as the spaceship is traveling, he's just bouncing the ball up and down off the floor. So the ball is moving, and you ask him to describe the motion of the ball. What he will say is that the ball is moving straight up and down. So I would draw a line, then I would bounce it again. When I was in the middle position, straight up and down, over here, straight up and down. And that is an objective relational statement about the movement of the ball.

- 2.81** Now, the spaceship is at the same time moving, and we bring in another relative observer. So, as you are outside of the spaceship and you're another person, but you are observing this guy in the spaceship going by, and he is bouncing the ball up and down. What, though, will the motion of the ball look like to you? Well, at the beginning, the ball was here, but then he threw it down, and the spaceship moved forward, so the ball was describing a wavy line. So, what you would say is: "This is what I saw; I saw the ball moving like this." Now, that is not a straight line up and down, but that was an objective assessment for the person on the spaceship, and this wavy line in a horizontal direction is also an objective description for you. So we are objective, talking about real features of moving objects. Nonetheless, our space and time characterizations of what's going on in these changes is relative.
- 2.82** Time—"When you're courting a nice girl, an hour seems a second. When you sit on a red-hot cinder, a second seems like an hour. That's relativity." Einstein having fun here, but he's also making a point about the psychological dimension of time. There's a great deal of discussion about our awareness of time in addition to our objective relative measurements of time. This is a real phenomenon as well.
- 2.83** Some philosophers will want to argue that the other two accounts, there's absolute time and absolute space or relative time, relative space being objective, point us in the direction of space and time being understood properly as really subjective, that they are not objective features or intrinsic features of reality in any way, either Newtonian or Einsteinian, but merely functions of how we try to perceive the world. These are philosophers and physicists, some of them, who see time thoroughly as subjective.
- 2.84** The big names, when we're doing our name-checking here, let's start with David Hume. David Hume, on all of these metaphysical issues, is very skeptical, arguing he can't make sense of the notion of being, of entity, of identity, of causality, and so on. He had skeptical arguments, dismissing all of those as pointless, and the result of the Humian approach to philosophy is basically to say we should not be trying to do metaphysics or trying to do philosophy at all. These all end up being meaningless concepts.
- 2.85** But on this issue of time and space being subjective, the more important point was his influence on the next generation philosopher, Immanuel Kant, who had been a Newtonian physics person as a younger philosopher trying to keep up on the physics of the 18th century. But he read Hume and realized that Hume's skeptical arguments had enormous power and, to some extent, were unanswerable.
- 2.86** The famous quotation here is when Kant says, "I openly confess, the suggestion of David Hume was the very thing which many years ago first interrupted my dogmatic slumber,"—the idea here is I'm sleepwalking through my philosophy career, following in the tradition of various philosophers who think reason can objectively come to no reality and we can figure out physics and how the world really works—"and gave my investigations in the field of speculative philosophy quite a new direction."

- 2.87** That new direction then was announced in his first and most important book, *The Critique of Pure Reason*, where, in his Kantian language, he makes the point and argues for space and time as having no intrinsic or objective relative notion at all. Rather, we have a mind, and our mind has a certain structure that we take data in from reality, and we filter and construct it and manipulate it in various ways and then present that to our mind's eye. By the time it gets to our mind's eye inside our head, so to speak, it is spatially and temporally constructed. But that's only because our mind has a space and time Lego block, or Meccano block, or Robo blocks set of filters—using those metaphors—that it constructs our subjective reality.
- 2.88** The first famous line here is, "Space and time are only forms of sensible intuition, and so only conditions of the existence of things as appearances." They're only apparent; they're not real. They are, the way he puts it here, "forms of sensible intuition." Our capacities of sensing, of intuiting the world, have these formal structures built into them. So whatever comes in basically gets spatial structure added to it, temporal structure added to it, and then it gets fed to our apparent mind. What's really out there? Not space and time. We don't know; we are stuck inside our heads. What we then have is three views on the nature of space and time.
- 2.89** This is going very quickly over deep material. I just wanted to have a sideways recommendation. We have a whole course on modern philosophy with a deeper dive into Hume, Kant, their environment, if you want to know more about those metaphysical and epistemological issues.
- 2.90** Now, awareness of all of this stuff—we've been talking a lot about stuff, and we're trying to use examples of hands and arms and so forth. What is this awareness thing that is certainly real? It seems real. Whether it's giving us real knowledge or whether it is not giving us real knowledge—another set of questions here. When we turn to epistemological issues, we will come back and take that feature up and subject it to, hopefully, equally deep examination. But let's pause there for now.
- 2.91** You went a lot on Aristotle, the Ship of Theseus, and humans are beings, therefore we exist. Yet we exist with different, not the same cells, different cells than we were born with. We change. Yet the elements are all the same since time eternal: carbon, oxygen, water. And at the same time, our souls don't change. How can we explain that one?
- 2.92** Okay, well, that's not one. You weren't asking about one.
- 2.93** How can we explain those?
- 2.94** Yes, exactly right. So, the issue of soul or not, I want to save that for our later lecture, when we talk about the human being in particular. So do we say the soul is something that is an entity, that is an unchanging entity of some sort, or do we see it as a function of the brain and nervous system and so forth, all of which are subject to change? That's an important issue, but let's put that one to the side.

- 2.95** The first network of questions you were raising, or the Ship of Theseus questions, applied to the human being. Yes, we might say—I guess this will be the question—every time I go and visit my mom and she says, "Oh, there's my little boy," well, I'm not her little boy anymore, but I am her little boy, and I'm the same little boy. So is she making a metaphysical error in saying that I'm the same kid that is now an adult, and of a certain number of years?
- 2.96** I think that's an interesting question, but I think we also want to say that identity questions—this is now me speaking and not just commenting—I think identity questions are often contextual. So one context might be: Do we mean the identity of someone over the course of a lifespan? If someone over the course of a lifespan, then we would say part of the identity of living beings, like human beings, is that they do go through this process of change. Cells die, new cells are added, and so on. And so there is identity across change, and how we work that out is one thing.
- 2.97** At the same time, we want to say that sometimes our context is different because I might very well say that I'm not the same person I was in high school. Sometimes that might be the real deal. When I was in high school, I wasn't interested in philosophy. I was just interested in girls and sports. Now I'm interested in philosophy and girls in sports, but more of my interests are changing in various ways. Maybe I believed certain things about politics and religion in my teen years, and I don't believe those things anymore. I had these social network friends at that point, but I don't have those anymore. So in that context, it's also appropriate to say that no, I'm not the same person anymore. So, I think you do have to specify the context within which you are asking the identity question in the first place.
- 2.98** To take another simple example, the ship of Theseus can be more complicated, but a simple example would be, suppose you have an heirloom that is your grandfather's hatchet. He used to chop wood for whatever, and so when he died, he made sure that it went to your dad. But before your dad passed it on to you, he used it for a while, but the handle broke. So he went to the hardware store, he took the old handle out and put in a new handle. And then, when you got it from your dad, you realized that some of the nails or the screws in the top that attach the handle to the head had rusted out. So you went to the hardware store, and you put some new screws and so forth and wedged it in a different way. Is it still your grandfather's axe?
- 2.99** Here, I think the way to answer that is, again, by specifying the context. The context is—the important thing is that something was passed from grandfather to son and on to son, and I have a part of that, and I am also part of that chain of transmission of being, including all of the psychological and sentimental value, however we capture that. So this is my grandfather's axe that I have here. If we are dropping all of the sentimental stuff, if we're dropping legal status stuff, and was it in the will or not, and we are just focusing on material constituents, then I would say your grandfather's axe was characterized by these three constituent elements. Now it only has one of those three

constituent elements, so it's one-third your grandfather's axe and two-thirds not your grandfather's axe. I think both of those are perfectly good answers to the same question as long as you're specifying context.

2.100 Now, to get to human beings who are obviously a lot more complicated, there is partly an empirical question because I don't think every cell in our body changes. My understanding is—don't quote me on this—but that there are some cells in our brain that do not change. I don't know whether that amounts to 3%, 5%, 8% or 1%. Then, the same thing about some of the major organs, whether a lot of the cells in the heart rejuvenate, is it all of them or not all of them? Again, the contextualizing of the question is the first step.

Lecture 3: Nature & Super-Nature

At a glance — Nature and super-nature — the natural and the supernatural.

- 3.1 Let's plunge in by these slightly narrower concepts and issues of the natural and the supernatural. There are long traditions in the intellectual life of associating natural with natural philosophy, out of which the sciences grew, and the supernatural with religions, out of which many contemporary versions still remain. We sometimes make a division between philosophy and religion.
- 3.2 The Venn diagram, so to speak, is complicated because some philosophies are religious, and some religions are philosophical. Some philosophies are not religious, and some religions are not particularly philosophical. So I'm more interested in the intersection concepts and the place of the natural and the supernatural in all of these. Let's start from the religion side of the equation, because so far we've spent two sessions looking at everything from a strictly philosophical perspective, although we've had some religious philosophical stories that have come down to us as well.
- 3.3 The question of what religion is itself, of course, fraught with complexity, difference, and controversy. What I've done is, I've got four definitions I would like to offer to you, two of them coming from people who are advocates of religion and two of them who are not advocates of religion. They're just anthropologists who are taking a sociological study or a historical study of religion.
- 3.4 So the idea is, we sample the territory, and we see what emerges from them. So here's Karl Rahner, 20th-century philosophical theologian. What is religion? Religion in humans is "the event of God's absolute self-communication. This capacity for the infinite, which is always given to us, is what we call the supernatural existential." So we have the existent word, existence, being, reality, and now the adjective supernatural, and I've highlighted that one. That's not in the original.
- 3.5 Another 20th-century statement. This is from an anthropologist. "Religion is a set of rituals, rationalized by myth, which mobilizes supernatural powers for the purposes of achieving or preventing transformations of state in man or nature." We have nature, man as part of nature or at least one giant foot in nature, power mobilizing all the language of change, built in there. And then, I've highlighted again: supernatural.
- 3.6 Going back to the 1800s, anthropologist James Frazer, "By religion, then, I understand a propitiation or a conciliation of powers superior to man, which are believed to direct and control the course of nature and of human life." So in this case, I've highlighted superior. So we have man; with whatever level of powers we have, there is a superior power. Then all of nature, there is a power superior to nature, super to nature. That's what we mean by religion in this quotation.

- 3.7** Then once more, this is from a religious theologian, philosophical one of the 1800s. "Religion is the belief in an ever-living God, that is, a divine mind and will, ruling the universe and holding moral relations with mankind." So here again, we have the universe and then a being that rules the universe. So that means its powers are above, beyond, superior to, greater than those of the universe. And the being is a mind, and a will has some personality to it, or personality features, and is ever-living, eternal. So a time dimension is added in that respect.
- 3.8** In this case, the organizing common concept, the one that is emphasized, is this concept of the super: supernatural, superior to nature, superior to the universe. This comes out of the Latin, "over," "above," "beyond," "on top of," whatever. So now what we have then is built into religions. If we take these four definitions as representative, accurate, arguable, that there is a metaphysics of there being a natural reality and a supernatural reality, and that the supernatural reality is the one that governs, has more power, is over, beyond, and so forth.
- 3.9** Now, surveying the various religions that are out there, that superior or the supernatural issue seems to come in two different flavors. One, somewhat metaphorically, I think of as conceptualizing the relationship in a vertical direction. So we have nature down here, so to speak, and when we talk about the supernatural, we point up there. It is superior in height, but that's, of course, still a spatial measure. Therefore, we don't necessarily mean that the supernatural being or beings are spatially extended, but that metaphor, nonetheless, we use it.
- 3.10** So we have nature below and then superior to that, in an above position. This is not necessarily a normative claim yet, that the superior is morally superior. Sometimes that's explicitly built into, but that would be additional to just saying it's more powerful. It governs, it rules, it creates, it causes various things to happen. We might add that normative dimension as well.
- 3.11** Others don't have as hard and fast a distinction between the natural realm and the supernatural realm. Instead, they will say it is on a continuum. Things are more natural and more supernatural, but there might be a lot of fuzziness and overlap in the two concepts. So, if we think, for example, of someone who is a shaman in a religion, the shaman might be a human being, but he's not an ordinary human being.
- 3.12** An ordinary human being might be a purely naturalistic being. But the shaman is someone who sometimes becomes partly supernatural because he— usually it's a he, but sometimes it's a she—has certain powers that inhere in the shaman that are not part of the natural world. As a special being, he is a leg into the supernatural world and a leg into the natural world. And there's not a hard and fast boundary line between those two aspects of the shaman.
- 3.13** And then, further into the supernatural being would be beings that are not at all natural, but rather purely supernatural in their traits. So we then are taking it as a

spectrum of more or less, rather than natural ends here, supernatural begins, and those are two completely separate kinds of things. So that's a subordinate question within the question of—I suppose we've decided, rather, the question that there is a supernatural dimension, and now we're asking: What is it like in relation to the natural? And we have two immediate positions that emerge there.

- 3.14** Now, cutting across those are some other dimensions as well. If we say there is a supernatural dimension, how do we characterize that supernatural dimension? We know that some emphasize the personality traits of the supernatural. God or the gods have a mind. They have a personality. They have a will. They have desires. They have goals. We might give them proper names: Zeus or Brahma or whatever we call that person.
- 3.15** Others will say, when we think about the supernatural, we should think of it not in personal terms, but rather in impersonal terms. We sometimes might somewhat metaphorically use the concept of energy, or Being with a capital B, but not presuppose that the being, the higher being, has a will or a mind. It is just a state of being. Or it is spirit without mind, but it's an energy that directs the universe or creates the universe or emanates into the universe in various sorts. And again, we can array this along a spectrum from understandings of the supernatural as more impersonal and more personal.
- 3.16** How many beings are in the supernatural realm? If we do know, some emphasize there's one, only one. Others are, there are many. There are maybe many supernatural beings of varying gradations of power, sometimes zones of influence and zones of power. We think of the Greek gods, for example. There's a lot of them, and goddesses as well. The natural world is divided up into zones of power and influence, sometimes overlapping. Those would be many, but personal. Sometimes we might have the idea they're all rather impersonal, but they are conflicting energy forces that we find their manifestation in the world.
- 3.17** Or there's one, one God, one ultimate being, one ultimate reality, whatever we conceive of it, and we are resistant to the idea of multiplication. We can then, again, array religions, I think, along this spectrum. Some are very fierce in their monotheism. Sometimes we'll start talking about trinities—three aspects or three beings—and whether that still is monotheistic or whether one is now saying there are three beings. Are there angels? Are there demons? Are there various other kinds of beings in the supernatural realm as well, to outright thousands and perhaps more gods even being created as the universe goes on? So that's an important sub-dimension, rather, of debate and discussion within the religions.
- 3.18** Always superior and always having power—that was a common feature in all of the definitions that we looked at. Whether the amount of power that the supernatural dimension has relative to us and the natural dimension is another important dimension. Some will argue that it is entirely the powerful being, not very powerful, but

if we start saying omnipotent, or all-powerful, the assumption of that is that the supernatural being has all of the power. If it has all of the power, we can't have any of the power; there can't be any powers that we have.

- 3.19** It might be, though, that we say it has all of the power and always uses the power, and everything that happens on Earth is totally controlled by God or the gods. Or we might say that sometimes this supernatural being delegates some power to us. Maybe it chooses some special individuals and delegates some supernatural powers to them, that if you pray before one of the specially delegated individuals, he or she can wipe your moral slate clean by saying a certain prayer. That's a delegated power from the supernatural, for example.
- 3.20** Or it might be that some human beings have special powers that, by their own efforts, they can tap into the supernatural and use some of that power for their own benefit. And all of the religious traditions that have shamans and magicians are typically at that end of the spectrum. They will say, if I say the right words, abracadabra—to use the comic book version of this story here—then that word, and it might be in the context of saying a whole lot of other words, but that necessarily means I can make the supernatural do something that I want to do, and it can't override me.
- 3.21** Or if I take the right mixture of ingredients, if we think of the stereotypical witches, I get my recipe, and I can mix things, and then usually I have to say the correct words as well, but I can summon supernatural forces and make those supernatural forces do my will. So this issue of degree of power and the dispersal of power is another important sub-dimension within all of the religions as well. As a result of that, we know, of course, people can take different positions along all these dimensions, and then we have a multiplicity of religions that result as a result of that.
- 3.22** Now, that then is to develop a possible taxonomy of different kinds of religion. That descriptive taxonomy is very valuable because, if we want to say, "I should believe in a God or the gods, or I should not believe in the God or the gods," we're going to proceed to have discussions and arguments about that. It's important to know first what it is that we are talking about. Are we talking about an impersonal being that has all of the power? Are we talking about multiple beings with dispersed power? Are we talking about one being that is omnipotent and all-knowing, and we have no power, et cetera, et cetera? So that we know, when we are having a discussion or just trying to make up our own minds, which specific one we are arguing for. All of that by way of background.
- 3.23** I want now to plunge into the next important step in the philosophy of religion, because philosophers almost uniformly want to say, we should be able to reason, argue, do some logic about this, and make up our minds with respect to the supernatural or not, by those tools, philosophical tools, rather than, say, just believing because I'm part of a tradition or whatever it is.

- 3.24** So, I want to start looking at some of the philosophical arguments for the existence of God or gods—we can leave that one open for now, which one is going to pop out—and spend some time, particularly on two of the most important arguments. Now, there's a distinction in the literature between theology or approaches to religion that emphasize argument and those that don't emphasize argument.
- 3.25** So one of the divisions then is to say, we're going to do theology, that is to say, giving an account of the gods or of God, theos from the Greek, -ology, by means of nature. And this is the one that the philosophers like the most, where they find the most overlap between philosophy and religion. Because what this approach says is, what we need to do is do what philosophers do, when they start by looking at reality.
- 3.26** They start by looking at nature. They say, well, there are things, and these things seem to be what they are, and they move around, and we're trying to explain the way the world is. The world that we're trying to explain is the natural world that we start with. And we think that maybe the best explanation for whatever it is we're interested in the world is to appeal to a supernatural dimension, a God or the gods, and that then is going to be our best philosophy.
- 3.27** Versus those philosophers who will then say—and these are a minority of philosophers, but a majority of people who approach the issues committed to a religion already—who will say, the right way to approach is through revealed theology. That then is to say, we presuppose that revelation has already occurred. God or the gods have spoken to the prophets; they've spoken to the mystics. The word of God or the gods has been revealed, it has been captured, and that is our starting point in the scripture, and we work with revelation. That is not as necessarily going to be amenable to making arguments about existence and causality and being and the nature of time and so on.
- 3.28** So there is a fork in the road. Again, the Venn diagrams are complicated because some people who start with religion will do natural theology. Some people who start with doing naturalistic philosophy will end up adopting a revelation kind of approach. But we're going to be working now with starting in the natural theology tradition.
- 3.29** Suppose we want to say, I'm aware of beings. I see this being, I see that being, there's this thing, all of these things are. Then I make that move where I expand out, and I say, "Wow, the whole thing, all of being, all of reality, all of existence, why is it here? Where did it come from?" And one move that I would have heard, when I talk with other smart people, is to say, "All of it came from a God. God created the universe." And we say, "Why do we make that move? Why don't we just say the world exists or being exists? Why do we have this extra move to a God?"
- 3.30** And we might then say, "Well, existence exists, but it couldn't just exist for no reason, or it couldn't just exist for no cause. Whatever exists, there has to be a reason why it exists. There has to be a cause for whatever exists." So if we're asking the question

about all of existence, then it has to be that the reason is outside of the realm of being, and there has to be a God-like creature that created reality.

- 3.31** Now, the usual immediate follow-up move then is to say, so now we are saying that God is a being who exists, and this God created the universe, and that is our answer to our question. We made that move because we said whatever exists can't just exist for no reason. Whatever exists must be the cause—sorry, the effect of something that brought it into existence.
- 3.32** So if God is existent, then wouldn't we have to ask that same question about God? Why does God exist? And here we face a fork in the road. We can say that anything that exists, there has to be a cause of its existence. Therefore, there has to be a cause of God's existence. And that would then have to be like a superior God who is superior to nature. But then, of course, we would then say that a superior God exists, and whatever exists has to be the effect of a prior cause. There must be a previous God, and nobody likes this direction of investigation. Because then we end up saying, well, there's God, and then there's a super God, and then there's a super duper God, and then a super duper duper God, and we're not really getting anywhere.
- 3.33** So that fork in the road, we want to close it off. So we go back, and then we say that no, there are beings, there's the universe. It comes into existence as the effect of God's creation. When we ask, "Why does God exist?", we say that's not the right question. There is no explanation for the existence of God. God just is. He is an exception to the principle that everything that exists has to exist as a result of the effect of some previous cause. So if universe, it needs an explanation, that explanation is God, but God himself needs no explanation.
- 3.34** But then, of course, the response then is going to be that if you're going to make exceptions, why go to this extra step? Why not say, well, maybe every particular being in the universe, we have to be able to give an explanation for its existence—it exists, and there are some prior thing that caused it to exist—but the universe as a whole, it just exists. We have a choice. We can either say existence exists, and then we go on to try to explain all the particular things that are going on inside the universe, but we don't ask why the universe as a whole exists; it just does. Or we say, existence exists as a result of the effect of God's existence, and we stop there. And we don't ask questions why God exists. He is the beginning. And that's a hard choice on this first existence issue.
- 3.35** All right, suppose we say, there in that case, we started by talking about being as a whole, and we were led down certain paths, and we have to make a choice. What if we don't start there? One of the always interesting questions about philosophy is: Where does one start?
- 3.36** Let's try a different starting point. Instead of asking about being as a whole, suppose we ask about a particular being. Here's a particular being. We say, why does this being

exist? We say that we have cause and effect. It exists, it is what it is, and we have cause and effect. This being exists because there were some other beings before in time that existed, and they did some stuff, and this thing came into existence. So it might be, I exist. Why do I exist? Well, because my mom and dad got together in the past and out came me.

3.37 Then we say, OK, so that makes sense. I exist. I have my identity. I'm the result of a certain causal process. This is very interesting. I want to go further back. Why does my mom exist? Now, she's another particular being. We supply the same principle. We say that she must be the effect of some previous causes. Then we say, it's because her mom and dad got together earlier in time, did some things, and as a result of that causal process, my mom came into existence, and then I came into existence, and so on.

3.38 This is great. Then we say, now we go back and say, what about my mom's dad? Then we're just going to do the same process. We're just going to keep going back. Now our question is going to be: How far back can we continue to do this process? This is the effect of this cause, which is the effect of this cause, which is the effect of this cause, which is the effect of this cause, and we, at a certain point, are going to reach a choice. We're going to say that this process just goes on forever.

3.39 You'll notice I've used a time concept—forever. It does not stop in time. You can always go further back in time and find a previous cause that gave rise to whichever effect in time you are interested in right now. But it's an open-ended process going infinitely back in time. That's one option.

3.40 Or we might say that no, you can't do that. It has to stop. You can't just keep going back forever in time. There has to be a first cause. That's the termination of the process back in time. Here's the first cause. It got the whole shebang going, and then all of these other effects occur. Then we want to say, OK, there has to be a first cause. What kind of being would it have to be to be able to cause everything that happens in the universe, but itself not be caused by something previous? That's got to be a pretty special being. That's got to be a God-like being.

3.41 So we have this choice. We can say things go back and back and back and back forever, or we can say things go back in time and, in effect, time stops. There was no earlier time because before then there was just God, and God is out of time. He's eternal.

3.42 OK, so now our brains are hurting a little bit, trying to keep all of this in mind. But notice what we are doing. We're talking about cause and effect, and we are now into the issue of time. Because one side wants to say, time can stretch back however we understand it, naturalistically, without end. There can be potentially indefinite or infinite time in the past. The other side wants to say, no, that's not how time works. Time has to have a beginning. There's a very first thing that happened first. There was no before that, except for God, but God is not a being subject to time.

- 3.43 What I would suggest then is, we now have a very interesting argument about the existence of God or not, but really, this is an argument about time, and we need to sort out our thinking about time, as we started doing in the previous lecture, in order to come back to this issue. All right, so that's two kinds of considerations.
- 3.44 Let's move on to another possible starting point. Suppose we're not interested in saying: Why does being as a whole exist? That's not our starting point. We do some reasoning based on that. Suppose our starting point is not we start with this particular being and we ask what causal process, going back in time, brought that one into existence? That's not where we are starting. So as we start saying, OK, the world exists, beings are, and things have come into an existence through whatever process going back in time, wherever that terminates, what we are interested in is the beings that exist right now and how they interact with each other, the causal processes that are currently existing in the world.
- 3.45 One of the things that we notice is stuff like this. This is a picture of a human cell, one cell. We know there are billions of them in our bodies, and we look at this, and we've done our science, a lot of very impressive science to get to be able to come up with this diagram, and we go, "Wow, that is amazing." So, a lot of science and a lot of philosophy is a whole lot of wow. So the universe exists and is very causal, but it's extraordinarily complex. How is it possible that such a complex universe exists?
- 3.46 It's everywhere that we look. We find complexity all over. When we go into the very small, we scale out to the very big. It's just amazing. Why is the world so complexly ordered? What keeps all of those cells working together in the way that they need to work together, so that the larger complexity, the human being, can continue to function? Why doesn't the whole thing just fall apart on a regular basis? What makes cause and effect hang together and give rise to all of this amazingly stupendous order? So yes, things exist. Yes, they are what they are. Yes, there is causality, but we're interested in now complexity, complex order, complex cause and effect, that seems to be a part of the world.
- 3.47 Now, a traditional argument here is called the teleological argument, another Greek phrase from telos, things are end-directed. So we're trying to give an end. All of this order seems to be for a direction. In this case, we would say everything inside this cell is for a purpose or for a telos or for an end. So this cell can perform certain functions as part of an organ, as part of a human being. And the human being will have teloses, purposes, functions, and so on in life. So things are part of an overall order for a purpose.
- 3.48 The casual name for this argument, the teleological argument, is like the universe is designed or like the cell was designed in order to fulfill that function in a larger and larger ordered system. Now, this is a natural theology argument because it's perhaps the most famous and widely discussed argument for the existence of God, with respect to the prominence of this dimension, particularly as we become more scientific in our

thinking. In many cases, philosophers who are friendly to religion and advocates of religion who are friendly to philosophy and science and so forth love this argument the most.

- 3.49** So, it starts from an observation. We are looking at the natural world. We're not looking at scripture. We're not seeking mystical insights. We're not just adopting things on faith. We're looking at the world. Everybody who looks at the world and starts to do their science can see the natural world is cause and effect, and that cause and effect gives rise to various forms of complex order.
- 3.50** Now, then we say, where does this complex order come from? Why is it so complex? Why does it all hang together? Why doesn't it, so to speak, just fall apart? And we make the assertion that it can't just happen, and often the language here is "by chance," or "by randomness," or "by luck." It has to be, since it is so complex, the case that natural universe can't have given rise to it. It has to be something that imposed that order or complexity on the natural universe.
- 3.51** So, a formal way of putting that then is to say, yes, the natural world is orderly. Complex order, though, cannot arise from within the natural universe itself. If we go back to this cell, we might want to say, we can't just ask all of the constituent elements of the cell to somehow self-organize themselves into this amazingly complex structure. It's too much to ask of nature.
- 3.52** Now, if those two premises are true, then we can make an inference. So, if the universe is complexly ordered, and complex order can't arise within the natural universe itself, there must be an external orderer who imposed order on the natural world, an external-to-the-universe orderer. Now what we have is, the world is, it's various stuff, but there's an external identity imposer that we are appealing to. Now, we are using this very neutral phrase: external orderer. What do we mean by that? Let's go to work on it a little bit.
- 3.53** We would just say that if you're going to design something like a cell, how smart do you have to be? You're going to design a whole galaxy and make it all hang together. How powerful do you have to be? The claim is going to be that whatever this external order is has to be extraordinarily powerful and it has to be extraordinarily intelligent. Those two traits we can assign. And it follows then that there must be a super-intelligent, super-powerful being, external to the universe, that has imposed this complex order on the universe and maybe continues to maintain it and so forth.
- 3.54** Now, a terminological thing, super-intelligent and powerful external orderer of the universe, that's a lot of syllables. Why don't we just say God? God is what we mean by super-intelligent, powerful orderer of the universe outside the natural universe. So we've got a proof for the existence of God. There must be a God. That is our conclusion. Now, this teleological argument, or design argument, has some rhetorical

power, has some logical power, and everything is controversial in philosophy. So let's turn to the controversies about this particular argument.

- 3.55** As these arguments get more complicated, of course, a certain amount of it we can do in our heads, but I do recommend, methodologically, start to write stuff down and start to write notes on the side in order to keep your thoughts organized, because so far what we have is a seven-step argument. Maybe we can hold seven steps in our head, but we can't hold seven steps in our head plus seven responses and seven counter-responses. Geniuses even can't get up to that level, so we do need devices. What I recommend in this case is a chart. What I've got here is the initial argument. Just put it on a chart, and I'm just going to then, as a methodological point, say, well, what are the kind of criticisms, questions, that we can raise about each step of the argument? Enter into the debate that way.
- 3.56** The very first premise is to say the universe is orderly. We put our critical thinking hats on, and we say: Is that true? Do we think the evidence supports the claim that the universe is orderly? This is not particularly controversial in the literature, because most people want to say, they are science-friendly people, there's lots of order in the universe, or we are religion-friendly people trying to explain the orderly universe that we think God has created. So we're not going to have an argument about order in the universe. Yes, there are these complicated cells. There are the seasons when mother and father elephant get together; they create baby elephants, not banana milkshakes and so forth. There's causality everywhere that we look. We're not particularly interested in that argument.
- 3.57** Complex order cannot arise within the universe itself. This is the big one. This is the one that generates the most controversy because the other side wants to ask, why not? Why must it be that causality has to be imposed on the universe? Why can't causality be a function of the identity of the things that exist in the universe? Do we need to explain order from the top down? God up here, imposing order on stuff below, or is it the case that order can emerge metaphorically, bottom-up, simple ingredients interacting with each other, generating complexity?
- 3.58** Do we need to think of the identity of things as imposed from an external source, or is identity built into what it is to exist? Things are what they are, and as they act, causality falls out of that. If you have different things acting causally and they interact, then you're going to get more complexity. This then, of course, is the big debate between evolutionary theory and design theory.
- 3.59** This stage of the argument then says we can certainly understand the religious advocates who offer the design argument and their conception of God. What we at this point need to do is plunge into evolutionary theory and make sure we are conversant with that and see which explanation ends up being better.

- 3.60** This terminology of bottom-up versus top-down, partly the idea here is imagining, again, God as a kind of sculptor looking over a workbench, say, on which there is raw materials. He might be making a clock, or he might be making a cell, or he might be making a galaxy. But he is, from the top, on the stuff down there, imposing the order on it.
- 3.61** The bottom-up one is—let's try an example. Suppose we went to the top of a mountain and we had a bag, just a big bag of, a canister, say, of hydrogen and another canister of oxygen. We have two relatively simple chemical elements separated. Suppose we had a pretty sturdy bag and we released a certain amount of hydrogen and a certain amount of oxygen into the bag. And then we took the bag, now we're at the top of the mountain, and just threw it off the mountain, and it goes tumbling down hundreds and hundreds of feet, and then thousands and thousands of feet until it gets to the bottom. This is a thought experiment that philosophy comes up with.
- 3.62** Then we say, at the bottom, when we go down, and we open up the bag, what would we expect to find in there? Well, we'll have some hydrogen, we'll have some oxygen, but is it not likely that we'd also have some H₂O? That, as all these billions of hydrogen atoms and billions of oxygen atoms are banging around and interacting with each other, at some point, it's two H's and one O, they will bang into each other and bond. They will bond because that's what oxygen and hydrogen do when they bang into each other. Now we have got two things that were initially simple, but we now have something more complex, some water molecules that have come.
- 3.63** If we were to take that back up to the top of the mountain, add some other chemical in it, throw it down again, we just kept doing this year after year after year, would we not build up increasingly complex molecules? Now, that's the thought experiment that says we can do it from the bottom up. Simplicity can generate complexity if you have enough time. The other model wants to say, no, that's not possible. It has to be top-down if it is complex. This really is the premise worth thinking about most deeply.
- 3.64** Third step of the argument. This one is typically not controversial. If it's the case that the natural universe is orderly, and it must be that order comes from outside the universe, then there has to be an outside-the-universe order. So the logic is fine. No controversy over that step in the argument.
- 3.65** The next step is more controversial. Some controversies here. This is, say, the orderer must be super-intelligent and super-powerful. How did we get from needing some sort of external ordering to there being one external orderer? Is that a legal step, so to speak, or a logically legal step to make? Why couldn't it be that there are two or three or a committee of external beings that are ordering the universe and managing the complex order?
- 3.66** In many cases, people who are singular theists or monotheists will then have an additional argument to say the order of the universe shows, say, such a cohesive design

that it must be the plan of a singular mind. Whereas the polytheists, the many-theists, will typically argue, actually, if you look at nature, there is a lot of order; but there's a lot of disorder, and there's also, it seems to be, competing orders.

- 3.67** We might say, the plants go to seed in the fall and they're dormant during the winter. Then in the spring, the rains come, and the earth softens, and so the earth gives life. But what happens partway through is, there's a huge storm that goes through and wipes out the entire crop. The polytheist explanation is going to say something, "Mother Earth is giving birth to life." We can understand that. "She has been inseminated by the rain god, who is usually masculine, but there's some other god who's angry with them. Even though they have created this new life, it just wants to destroy that life." They will say that's a better explanation for the combination of order and competing orders and disorder in the universe.
- 3.68** So, at this point, logically, are we entitled to say there is a single orderer, or are we open to the idea that it must be a possible number of orderers that are engaged in explaining the complex order of the world?
- 3.69** Step five of the argument has a minor controversy attached to it. We use the present tense. Therefore, a super powerful orderer exists. Now, the countervailing hypothesis to think about is to say, well, maybe we agree with the argument up to this point, but there was a super-intelligent and powerful godlike being that ordered the universe, but then, having ordered the universe, basically set it off and then retired, or went out of existence. This being's one metaphysical function was to impose order on the universe and get it going; but having done its job, the universe goes of itself.
- 3.70** Or do we need to say, no, the universe was ordered by this God, and there is ongoing maintenance by the God of the order. One of the sub-debates here then is typically between what we call in the literature the theists and the deists. The deists will say, theists are assuming that God still exists, but if you really believe in an all-powerful, all-intelligent God, then you can believe that he did his job so well, when he imposed order on the universe, that it doesn't need him actively managing it, so to speak. If you believe that God made the world and then has to step in to keep it going, why didn't he make it better in the first place, so he doesn't need to keep stepping in to keep it going? That sub-debate grants everything so far, but makes that suggestion there.
- 3.71** Terminological point in the sixth step of the argument, let's call the being God. You would say that's fine, as long as you are not importing particularities about your version of God at this point. Our usual test question here is then to say, suppose we get to this step in the argument, would you be comfortable with saying, "Let's call the super intelligent and powerful being Allah. Let's call this being Yahweh. Let's call this being Gitchi Manitou." If your hackles are going up or you're resistant at any point, "No, I don't want to use that name. I want to use God," make sure that you are not, by God, meaning something more than a neutral, intelligent and powerful being, and not importing lots of other stuff from your particular religion.

- 3.72** Strict logic, the argument here is. Maybe all of those other particular features will become part of the package, but you're going to need a separate argument to add those additional features. All we have is an order that's intelligent enough and powerful enough to explain the order that exists in the universe right now.
- 3.73** So, God exists. Say, well, if there must be a super-intelligent and powerful orderer, and we've agreed that it's fine to call this orderer God, then the conclusion follows, logically, that God exists. So the argument then is fine.
- 3.74** But we now have an array of questions, criticisms, that we need to sort out, make sure we understand both sides of this column, think them through further, and come down on whether we think the case has been made or not been made.
- 3.75** There's one other interesting point that is made. We say, look, we've now reached the conclusion that God exists, but earlier we had this proposition. There must be a God because complex order that does exist can't just come from the universe itself. It has to be from some external source that added the complex order on the universe. So, complex order requires some being outside of itself to impose that order. That's our premise.
- 3.76** Now we are saying, as a result of that, we've concluded that there must be a God. What if we ask that same question about the God? We say that God is super-intelligent and powerful, and he exists. Is God complexly ordered? Here we have another fork in the road. Because suppose we say, we believe that God exists, and you ask us this question. Does God have a complex order within him? It seems like we have a yes or no answer to that question.
- 3.77** What happens if we say no? No, God is not complexly ordered. Then we have a problem because then we have a being that is presumably very simple, but it's not the kind of being that's going to be able to impose this extraordinarily complex order on the universe and keep it going. So we can't say no.
- 3.78** So then we say, God must be complexly ordered. And that makes sense because what do we mean by intelligence? Well, by intelligence, we mean some sort of consciousness or mind that's really got its act together, all of this stuff, and knows how everything fits together and so forth, a really complex mind. What do we mean by being very powerful? Well, being able to impose whatever kind of complicated causal structure you want on something as giant as the universe. So God must be an extraordinarily complex being, with all of this power and knowledge.
- 3.79** But then we have this premise two, that we use. It says, complex order cannot arise from within the thing itself. If we apply that to God, then it seems like the logical conclusion is, we have to say, where did God get his complex order from? We can't just say, well, he just generated it himself. It came from an outside being that imposed complex order on him, and he in turn imposed the complex order on the universe. Then we've got a problem, because then we say, what about that being? Is it complexly

ordered or not? We have to say yes. Then there has to be a super duper, extra-complex being that made this extra-complex being, that made God, that then made the complex—and nobody likes that direction either.

3.80 Then what we say is, the universe is complex. There has to be an external orderer, God, and he is an exception. He is self-ordering. His complexity, he generates it from within himself or by himself. Then again, we are back to saying that if we're going to start making exceptions, why not make the exception in the first place? We say the universe exists. It is what it is. Things do what they do, and complex order emerges. And so the universe evolves its own complex order. Why this extra step of saying, it can't do that? Then we have this other complex self-ordering being.

3.81 OK, fast run-through, very complicated argument. I want, again, to underscore the method that we are using. It's a philosophical method. This one is most associated and developed to a high art with Thomas Aquinas, in my judgment, one of the two most important religious philosophers of all time.

3.82 One of the things that makes him one of the most important is that he makes a point of saying, we're going to argue there's this premise, this conclusion, and we're systematically going to lay the arguments out, and we are going to fair-mindedly consider every possible criticism and objection to this argument, and then mount counter-objections and so forth. It's going to be a natural theology based on evidence and reason, and we think that, whatever the arguments come out, that's what we are going to believe.

3.83 All right, let's try another argument that's in this natural theology tradition. But in fair-mindedness, we've been looking at arguments to prove the existence of God. Let's do the same thing with an argument that tries to disprove the existence of God. The most famous argument in this tradition, I imagine we are all familiar with it, colloquially, is called the argument from evil. What we're trying to do in this case is explain or justify the rightness of God's creation, supposing that we, by hypothesis, allow that there is a God who created the world.

3.84 The way this argument works is by means of a negative method. It grants its opponents' assumptions and shows that, if we accept those as assumptions, nonetheless, we reach an absurd conclusion or an impossible conclusion. Therefore, those assumptions have to be false.

3.85 Now, in this case, the assumption of the philosophers on the religion side is to take a standard Western—there are some Eastern analogs to this— conception of a singular God that is infinite in all relevant dimensions, infinitely powerful, infinitely knowledgeable, infinitely good. God is a perfect being, and so, in all of these dimensions, he has to be perfect, and the perfection is an expression of his infinity.

3.86 So, we expose and we will allow, for the sake of argument, that there is an all-powerful God, an all-knowing God, an all-good God who is a creator of the universe. That's a

supposition hypothesis. Now we break it down. The next premise would be to say, suppose this God is all-powerful, and we imagine him before he has created the universe. "I'm going to make the universe." The logical thing is to say, well, he could make any kind of universe he wanted, right? Multiple possibilities, but he could create the best possible universe, the most perfect universe. If he's a perfect being, he can create a perfect universe.

- 3.87** Second premise drills down on the knowledge claim: God knows what he is doing. As a perfectly knowledgeable being, he knows all of the options; he knows all the consequences of all of the options. And so, he can select knowingly and accurately exactly the kind of universe he wants to create.
- 3.88** Next premise drills down on the benevolence point or the goodness point of God. If God is all-good, well, what's built into being good? Well, you want to make the best that you can make. You want to make things as perfect as you can make. If you want to make things a little bit bad, you're not perfect. If you want to make things or you try to make things, but they don't end up perfect, you are not perfect. If you make something and there's a mistake and you say, "Oops, I made a mistake," then your knowledge must not be perfect.
- 3.89** So, flipping those all to the positive, if God's all-knowing, all-good and all-powerful, then the universe that he made would have to be a perfect universe. Then, this is where the natural theology part comes in. We look at the universe, we look at nature, we look at the world as it is, and not too controversial, not perfect, has problems—bad things, sometimes evil things. The conclusion then is that God does not exist. There is no infinitely powerful, benevolent, and knowledgeable God.
- 3.90** OK, again, an argument, we lay it out. We end up with a seven-proposition structure. It has some rhetorical force. It has some logical force. We can all feel the weight of it, but let's do the method. Look for the controversies, structure the criticisms, break the argument down into its component parts so that we can consider each thing in isolation without trying to hold the whole shebang in our minds at one given time.
- 3.91** So, one standard response is the free-will response, to say, yes, in fact, God did make a perfect world. And if we go inside the Judeo-Christian tradition, he made the Garden of Eden. He made everything else around it. It was fine; it was no problem. He put human beings. Where did the badness and the evil in the world come from? It came from the free will, the bad choices that the human beings made. Therefore, the responsibility for the evils and bad things that exist in the world do not lie with God, they lie with human beings. So God is off the hook, so to speak. This is probably the first and most common response to the problem of evil.
- 3.92** Now, the other side then wants to jump in and say it seems like there are a lot of "evils" in the world.—I put those in quotation marks. There are a lot of bad things in the world that don't seem to be a result of human free will. So, hurricanes that kill a lot of people

and destroy a lot of stuff—that's not a free will issue. Diseases are not often a result of human free will. There are various things. Why do those things exist in the world? Or my go-to example, mosquitoes. Why on earth are there mosquitoes? Those guys are evil little monsters, but they are not a part of human free will. Actually, they're opposed to human free will.

- 3.93** The question then is, it's a reasonable question to say, we want to say, some of the evils in the world can be explained by human free will and bad choices, but not all of them. We still have a residual of bad things and evil things that need to be explained by some other channels.
- 3.94** Another interesting example is to say, sometimes we think of God in a paternal form, and here we're getting into good and bad, right and wrong, some moral issues as well. We might say that God is functioning like parents act with respect to their children. You create a child, the child has free will, and you let the child make mistakes, and you don't step in and stop the child from making every possible mistake.
- 3.95** So bad things are going to happen, and it's a normal part of children growing to adulthood. But even so, the argument is going to make, that parents are going to stop children from doing some things. They're going to override the free will choices that some human beings make, or little human beings make. So if we're using this more paternal understanding of God, that he gave us free will and he's going to leave us to make mistakes, there still is going to be the question of why he doesn't step in when some of those mistakes get really big and really bad? Just as a good parent would step in, respect the child's free will in some cases, but override it in other cases.
- 3.96** So the big examples here, the go-to examples, would be Hitler and the Holocaust. We might say, well, Hitler was exercising his free will, but still, if God has more power than Hitler, you would keep the abuses of power there. Or all of the serial killers that are out there. When you say the serial killers are evil, they're exercising their free will. But nonetheless, just as good police officers would stop and override the free will of the serial killers, if God is a good being, there's a reasonable supposition that he should override some of the worst excesses of free will. So, how strong is the free will response?
- 3.97** Now, another kind of response doesn't rely on free will, per se, but rather to say, God is a good being, and he allows evil and bad things to exist for a good purpose. And that good purpose is to teach us the difference between good and evil. So the good purpose is, human beings come to be morally knowledgeable. How do we come to be morally knowledgeable? And the argument here is that God is what we, philosophically, would call an empiricist. He wants human beings to learn by experience. And so you learn the difference between good and evil by experiencing good and experiencing evil. And that's the only way to get to learn the difference between the two.

- 3.98** So, that's a good purpose. Standard response: Could that purpose be served by some other method? Could God have given us a conscience that automatically tells us the difference between good and evil? Could he have given us an instinct that sorts out the difference between, so we're coming into the world knowing already the difference between good and evil? And that way, we don't have to experience good and evil in order to know the difference between?
- 3.99** Another standard response here worth thinking about is: How much evil is necessary to teach this lesson? So, particularly in the last century, the Holocaust is a good example. Suppose we say we have to learn that murder is evil. Maybe there has to be a few murders, and we become aware of the murders, and we say, "Oh, murder is really bad because now I've experienced its existence, and then I'm resolved that I'm not going to become a murderer." That's how I become a better person, as a result of that.
- 3.100** Well, how many murders do I need to know about before I get the point and I get the lesson? I've seen a couple of murders, not sure yet. I've seen a dozen murders. It seems like that's pretty bad. Do I need 100 murders? Do I need 100,000 murders? It seems like most people would be able to understand the difference between murdering and not murdering—not murdering, good; murdering, bad—after very few murders, if at all. We don't need 6 million. We don't need 12 million.
- 3.101** So all of that seems really overkill. And if it is overkill, why does it exist? You might let your child get bitten again by a mosquito or two, so the child experiences the discomfort that comes with that. But you're not going to send your child into a hive—I'm not sure if it's a hive—a hive of mosquitoes. You'd have a thousand bites, or one or two is going to serve enough to make the point. So, another interesting back-and-forth set of issues to think about.
- 3.102** Sometimes the argument is on utilitarian grounds, that God allows a certain amount of evils or bad things to occur in order that future good things come into existence. We know this, that sometimes we have to bear costs, suffer pains, and so on in order to get to a better state. "No pain, no gain," we might say, for example. This is to adopt a utilitarian moral principle, but some controversy here.
- 3.103** First thing is, if we're going to say this, we do need to make good on what the greater good claim is. If you want to say, "This bad thing happened, it must be for the greater good," we have to be able to identify: What is the greater good that we think is going to come out of this? If we don't have an answer to that question, then it starts to sound like we're just saying something in order to avoid the problem, and that's not a good state to be in.
- 3.104** It becomes hard when we try to do that. It's quite fine if you are taking your child to the dentist, and the dentist inflicts pain on you, on the child, and the child says, "Why are you making me inflict this pain?" Well, we can explain to the child, "This pain is necessary, or the discomfort is necessary, because your teeth are very important, and

they're going to be with you for the rest of your life." So we can make good on that claim. Some pain and discomfort is necessary for a greater good down the road and so forth.

- 3.105** But some examples, it seems very hard to do so. One example I always go back to is when I was in graduate school, I was passing through an airport, and there was a newspaper. I just picked it up, and I was reading the newspaper. Apparently, what had happened the night before in this city was, a little girl had been shopping with her mom, I think in a convenience store or drugstore, and she had picked up a tube of toothpaste off the shelf. You know, the way little kids are always picking stuff up, saying, "What is this? Can we buy this, Mommy?"
- 3.106** And it had exploded in her hand, and she ended up losing the limb and becoming blind in one side and disfigured on this side. And the story turned out to be that some guys had decided it would be a lot of fun to take a toothpaste carton, open it up one end very carefully, put a pipe bomb inside the toothpaste tube, smuggle it back into the store and put it on the shelf, knowing someone eventually is going to pick it up. It's got a motion-activated detonator. Big fun, from this sick perspective. It was a little girl.
- 3.107** But what was interesting about this newspaper account was that the mother who was interviewed for the story was raising exactly the theodicy problem. And her question was, "Why would God let this happen to my perfect little angel?" Those words are emblazoned in my memory. Exact quotation. Why would God let that sick person do that? And presumably he knew about it, if he's all-knowing. Presumably, if he's all-powerful, he could do something about it. But if he let it slide, then either that doesn't speak well of him, or it means that he does not exist. That's the question here.
- 3.108** Now, we've got an answer. We say, "Well, there's a greater good." But what is the greater good in this case? This girl is going to go through the rest of her life disfigured and blind. What is that? You might say, "Well, maybe the greater good is that people like me will read about this in the newspaper and be redoubled in our belief that we're not going to do bad things to little kids." I was already there, but maybe there's lots of other people who needed the lesson.
- 3.109** Maybe she will develop strength of character and become the kind of person who goes off in her adult life and does amazing things that she otherwise would not have done if she had not. Maybe, but do we know that? Do we know that there's a greater good, and so on? So, tough questions still about this stage of the argument.
- 3.110** Another variation, and the response side worth paying attention to, is the issue of retribution. Most religions in contemporary times have a conception of the afterlife, and in many cases, an afterlife is a moral sorting, a moral justice, a place where people who do bad things get their due results; people who have suffered bad things get compensation; and people who do good things get their rewards.

- 3.111** So we might then say, "Fine, bad things happen, but God's goodness and his power is an ultimate kind of goodness and power. He keeps track of everything, and in the afterlife, people who have suffered get compensation, and the people who have done things pay their rewards." There's an ultimate, divine justice cause-and-effect principle operating here in the afterlife.
- 3.112** Now, the tricky bit here, the way the argument goes back and forth on this, is to say that maybe it gets made up for in the afterlife, but it still doesn't change the fact that it was evil when it happened in this life. It doesn't seem like it had to happen.
- 3.113** It might be, to take a silly example, if I just walked up and kicked somebody and caused you some pain, and you had to go to the hospital or whatever, and I said afterwards, "Oh, I'm sorry. Here's a thousand bucks for your pain," so I'm making retribution, but it doesn't change the fact that I shouldn't have kicked you in the first place, and that was just a bad and wrong thing to do. If there is oversight on the part of the powers that be that are watching and they're still letting things happen, that seems negligent. So, afterlife, how much of a solution is that?
- 3.114** A stronger claim, and this is going to take us closer to tomorrow's material, starts to focus on cognitive. We are saying that, in this world, there are all of these bad things and evil things. That is to say, we know the difference between the good and the bad. We have a standard and we're applying that standard to these particular things.
- 3.115** But maybe the point is that we, from our human perspective, don't have the right perspective on good and evil. Maybe we are applying the wrong standard or only a human standard. From the proper divine standard, what we think of as evil things are, in fact, not evil things. They are, in fact, good things. And that is a kind of claim. It's not trying to justify the bad things. It's trying to say that, in fact, those bad things are not bad things; you are just looking at bad things the wrong way.
- 3.116** Now, that's an interesting question, and it raises some side discussions about whether we are becoming skeptics at this point, because often in our religions, we want to say we actually know what we are talking about. Our philosophy and our religion give us a guide to knowing the difference between good and evil, and so we should apply those standards in our lives. If, to avoid the problems of the problem of evil is that we say, we don't know what the difference is between good and evil, we have to be agnostic and say it's only a human perspective, then that's undercutting our claim to any sort of moral authority inside our religion.
- 3.117** But again, the more serious question would then be, if we're just looking at things the wrong way, can you give us a better way of looking at this thing? It sure seems like it's an evil thing to me. Just point out how I should adjust my perception.
- 3.118** So, here's my grandfather who's suffering from Alzheimer's, and it's just devastating to him and to everybody who knows him. That seems like a really bad thing. Now, if you come to me and you say, "Oh, you're just looking at it the wrong way," well, what's the

right way to look at Alzheimer's? "Oh, yeah, this guy raped 20 women. That's evil." "No, we don't really know." Come on. We do know that's evil. I'm just "not looking at the rape from the right perspective?" Come on. It seems like it may be an argument-avoidance issue that's going on here.

- 3.119** Now, one final subcategory, since there's a huge literature on this, is to say, look, maybe this argument, the theodicy argument, works against a very robust understanding of God. We've got God, who is infinite in all dimensions. He's all-powerful, all-knowing, all-benevolent, and so on. Maybe the argument shows that that package cannot coexist with an imperfect world. Rather than those saying that we're just going to get rid of the concept of God altogether, maybe what we need to do is modify our concept of God and weaken the concept of God in some respect.
- 3.120** So, what if we said, for example, that God is maybe not infinitely powerful, but just very powerful, but all-knowing and all-good? We back off on the infinite power claim a little bit to make room for another kind of divine being that is also quite powerful and quite knowing, but evil. And so we adjust our metaphysics or our cosmology, in this sense of saying, really, what's going on in the world is not one divine being with his plan for the universe, but rather, there are at least two of them. And it's a great battle between the force of good and the force of evil, and they are contending it out. What happens in the real world is, sometimes the divine evil force wins, and sometimes the divine good force wins.
- 3.121** That avoids the problem of evil, because now we have a good explanation for evil, and we're not caught in any sort of a logical contradiction. What if we then perhaps said, God is all-powerful and all-good, but he's in a process of cognitive development, say. In the beginning, he has before him an infinite number of universes that he could create, and he chose one that is pretty good, but it wasn't a perfect one. And he is in a process then, over time, of trying to improve upon his creation as he goes on. So he's learning as he goes.
- 3.122** Now, there, we say, he's got a lot of power, maybe infinite power. He's good, all-good. But we're backing off a little bit on the omniscience claim. He's just very smart. That would avoid the problem and make this one go away, and we can still believe in a kind of God. Now, advocates of the argument for evil, they'll say, "Oh, that's all fine. Our argument is only directed against the idea of an infinitely perfect God." So, if you want to have one of these modified versions of God, we still have an argument in order to go ahead.
- 3.123** Now, what has occurred, though, is, in the subsequent history of this debate—and we're just getting our feet wet on this very fascinating issue—is that those who will say, well, the argument works against most of the major world religions that really want to have a perfect God, so we can set at least those religions aside. Those who, though, want to say, all of these heresies, as we're now going to call them, that back away from the idea of God being perfect in all dimensions, that God does not have all of the

power, or God does not have all of the knowledge, or God is sometimes not good, sometimes he's a little bit angry and loses control of his emotions or whatever—all of those variations that try to preserve as much as possible are heretical.

- 3.124** We're going to stick to our guns and say God is a perfect being. This is a set of problems. Maybe we don't have logical explanations for the problems, but we're going to dismiss the heresies as heresies. To the extent that we can't answer, logically, say, the theodicy problem, we're going to reassert faith as the proper method of coming to. "It's a matter of faith. God works in mysterious ways," that kind of language.
- 3.125** The point of all that, then, is to say that we now are getting to epistemology. We've been proceeding on the assumption of observation, reasoning, argument, logic. If we reach a conclusion, for example, that that doesn't work, for some of the things we want to believe, we might start considering alternative epistemologies. Of course, if we think those are the right ways of doing everything, we still have to justify them—observation, logic, reasoning. We'll come back next time for those epistemological issues.
- 3.126** So, just a possible observation question based on it. You introduced Hume that begot Kant, right? Kant. They seem to be idealists, meaning reality is just basically perception. It's not real, it's not objectivist. If I look at all the other ones that you brought forth, from the classics and the pre-Aristotle, pre-Socrates, is this the first time that this enters the chat, is the question?
- 3.127** Nice contemporary formulation. Yes and no. So, as a philosopher, we always have to make a distinction. So, first, Hume is not an idealist. He ends up as a skeptic. And so, what he wants to do is end up saying, "If you want to tell me the world is material, you want to tell me it's spiritual, I'm not playing that game. I'm hands-off." So the idealists, though, are the ones who want to say—they take Hume's skeptical arguments. They show, say, that material, physicalist understandings of the universe can't work, that we are trapped in a subjective idealist state, and that's the direction that Kant takes. So Kant is a kind of idealist, but then once we plunge into Kant, there's further distinctions that we need to make.
- 3.128** This is not, though, the first time that idealism enters the chat. This is now a kind of idealism that is driven by epistemological concerns. We might then say, this is epistemological idealism for various reasons—this is where we'll be going tomorrow. We don't think we are observing the world. All we are aware of is these ideas inside our head, by our mind's eye. We might think that there's a world out there, but we don't know what it is. But all we know, for epistemological reasons, is this ideal realm or the realm of ideas.
- 3.129** That is the epistemological idealism, because metaphysical idealism has a long history. Actually, many of the classic thinkers we've looking at so far end up being metaphysical idealists. If you take, for example, a standard view of many, but not all, major religions, if you ask the in-the-beginning question, "What exists?" "It's God." "Well, what is God?"

"God is a giant mind," so he is a mental being, but what constitutes his being is all of his ideas. Then, as a result of God having certain ideas in his mind, that comes first. Eventually, reality is created and comes into existence.

3.130 That then is to say, ideas have priority. It's a primacy of mind, primacy of consciousness, primacy of spirit, a type of position. That's a metaphysical idealism. Everything that exists is a product of an idea, God's idea.

Lecture 4: First Perceptions

At a glance — First perceptions — sense perception and the mind.

- 4.1 The fascinating and glorious things about human beings is the power of our minds. Whether our minds are a function of underlying biological complexity, whether they have an independent existence, is a matter of controversy, of course; that's a metaphysical issue. But when we turn to issues of epistemology, as we are going to in the next few sessions, reflecting on just what we are capable of with our minds is an astounding place to begin.
- 4.2 We can focus on all sorts of things in our visual field, our auditory field. We can explore things with our tactile senses and smell them in detail, and develop instruments that enable us to go into astounding, smaller and smaller detail all the way down. We can scale out to galaxies and beyond. We can think about our present realities and project the future. We can go far into the past, and all of these things matter to us. We can acquire, not yet about things in the future, but knowledge about the things of the present and the past. We get jokes, and that is another sign of intelligence, and we enjoy humor. We employ metaphors and allegories and figures of speech and elevate all of that into poetic language; we enjoy that process as well. As well, we do the, sometimes what we call the hard-core sciences—physics, mathematics— in order to extend our knowledge.
- 4.3 Now, one measure of how we appreciate that, not only about ourselves as individuals — we want to be the best that we can be, physically and mentally—but we also celebrate it in other people who extend our knowledge of the universe, but also illustrate to us just what is possible for human beings in general.
- 4.4 The one sign of that is that we all recognize this individual. We can put a name to him: Albert Einstein. Even if we don't know the math, we don't know the physics, we know who Albert Einstein is, and he stands for something: an ultimate possibility for human beings. If we know our history of mathematics as well as our history of physics, we might know Ramanujan, a largely self-taught mathematician from India, an astounding genius in that field who eventually made his way to the West as well and collaborated with some of the great mathematicians in Western Europe.
- 4.5 We might also, to the extent that we use our minds not only for super realistic, important things of figuring out reality, we like to use them as a form of exercise and play, and we like puzzles. We might recognize Marilyn vos Savant, who apparently scored the highest anybody has ever scored on IQ tests, another recognizable genius, and had a long-time column in which she took up difficult mathematical, statistical, logical puzzles, and explained things in a way that other intelligent people who are into this thing could enjoy. So we celebrate the human accomplishment cognitively in a variety of directions.

- 4.6 Now, this is complicated. We know the universe is complicated. We know that our capacity to come to believe things and hopefully know things about the universe is a complex set of capacities. Part of what we need to do is figure out not only the way the universe works, but the way our cognitive capacities work, so we can say we have, in fact, figured out the way the universe is, and that requires a certain amount of self-reflection on what's going on inside our minds.
- 4.7 As we know, everything in philosophy is controversial. One of the things that's controversial in philosophy is: Where do you actually start properly to do philosophy? Often, by the time we start self-consciously doing philosophy, we are already adults or close to being adults, and we have already used our capacities in certain ways, acquired certain beliefs, and we have certain habits of thought. How do we sort through all of that and figure out where we can appropriately begin this whole enterprise that we call philosophy?
- 4.8 What we have tried to do in other earlier sessions is to say, let's start with this particular thing, this being and this existent, and try to see where it came from. Then maybe we go back to the previous set of existents and say those ones caused this one to come into existence, and we try to figure out a beginning by using that particular route. But we might also, as we did earlier, say, let's start by scaling out and not looking at things in their particularity, but think about being in general or existence in general, and see if we can figure out some general truths or realities and start our philosophical enterprise from that.
- 4.9 For this, I would like to start by asking us to observe another being during the process of its cognitive development. So one of the things we know is that adult cognizing is a long developmental process. It's fascinating in the case of human beings that when we give birth, the children do not come out of the womb, so to speak, ready to take on the world all by themselves. They are dependent physically. They're largely dependent cognitively. It takes a long time for them to develop physically, and cognitively, and emotionally until they are ready to take on the world by themselves.
- 4.10 That we share with lots of other animal species, but if we look at most other animal species, it might be a matter of a season. The young are born in the spring, but by the time winter comes, they are adults, and they are ready. They know what they need to do and how to behave in the world. Sometimes it might be a matter of a whole calendar year or two or three years until the young animal is fully adult. But if we think about our two-, three-, four-year-olds, even our five-year-olds, they're just getting started. Human beings seem to be unusually long in the period of their physical development, but even longer in terms of the maturity of their cognitive development.
- 4.11 Not only is it complicated, it is long-term, and probably it's long-term because it is so complicated, and so many moving parts need to be learned and then figured out how to work those together and how to operationalize those in our bodies and our actions before we're ready to take on the world.

- 4.12 So maybe a way to start on this issue of knowledge, now that all of us are adults, is to go back to the beginning and observe what happens when children start off on their cognitive development. So, we imagine a child was just born, no language yet, no sophisticated theories about the world. Instead, the child comes out, born normal, healthy, has five senses, and very quickly starts to take in information, as we might say, or absorb energy of different forms from its five sensory receptors. Its eyes start to track things, its ears start to pick up sounds, and we can sometimes see the child trying to orient itself toward the sound. It picks things up and manipulates them with its hands, and everything goes in the mouth, often with children, so taste is happening.
- 4.13 This happens largely automatically. Of course, we as parents might try to stimulate, and guide and direct by putting things with strong perceptual features in the child's visual field to emphasize certain kinds of sensory development. But what happens fairly quickly seems to be that these disconnected senses—so we might have to imagine that we see a visual field and we can take our eyes out of focus now as adults and just see it as, say, a bunch of moving color blobs, so to speak. But those do automatically get integrated to the point where we're aware of percepts, objects that persist across time.
- 4.14 For a child, for example, even a one-day-old child, the mother's face comes into the visual field. It might initially just be a blob of colors and undifferentiated from the background and what's in the foreground. But fairly quickly, the child learns how to integrate all of the particular sensations and recognize the mother's face as integrated; these things all hang together. If the mother then moves across the visual field, the child's eyes track the mother's face as an integrated perceptual entity separate from the background and so on.
- 4.15 So, the child is moving from sensation to perception, as we say. Sensations: individual color, blobs, sounds here and there; but those, integrated into units, we call the perceptual state of organization. This goes on. The child becomes more sensory, perceptually sophisticated with eyes, ears, touch, and so forth.
- 4.16 Then, at a certain point, kids start to typically imitate parents. They start to vocalize, they start to use words, and they use those words in a way that seems to be picking out categories of things in the world. So they will, say, in the house have a cat or a kitten, and the child will start learning the word kitty because the parents perhaps have pointed to the cat frequently and said "cat," or "kitten," or "kitty," or whatever it is, and then the child starts to do so.
- 4.17 What's interesting is that the word "kitten" very quickly gets transferred to other beings that are the same. So, if you go visit the neighbors, for example, or a family member, and they also have a cat, maybe they also have a dog, the child will say "cat" to the cat over there, but it will not call the dog cat. So, it seems to have a recognition that there are individual things: cats, dogs, mom, dad, brother, sister, my blanket, spoon, and so on. But also categories of things: this cat, this individual cat, is like other cats in the world. And so it has a general category. Other things are human beings, and

there are dogs, and we drive in the country, and we see cows, and they're different from horses, and there are trees, et cetera. So we have all kinds of individual words for general categories of things.

- 4.18** As far as we know, some animals might be able to form some concepts, a limited number of concepts—this is ongoing research, and we don't want to prejudge any of that—but it's something that definitely human beings do, and it seems obvious that they very quickly outstrip any other animals in terms of the number of concepts they have, such that by the time they are a year or two or three years old, they have hundreds, if not thousands of words that they can use with some facility to pick out particular things in the world, including mathematical concepts for counting, concepts of the past, concepts of the future. Then, at a certain point, the child is also aware sensorially and perceptually of things going on inside his or her body and inside his or her head, self-awareness and concepts of internal processes that are going on.
- 4.19** Fairly quickly, children start to have the capacity to take individual concepts and put them together into larger units that we start to call sentences. So it's not just "cat," but "cat eating," or "cat's bowl," "cat walking over to bowl in order to eat," and the child can then take all of those individual concepts: "cat," "walking," "bowl," concepts of movement, and put them together into a new unit, a sentence, "The cat is walking across the room in order to eat." Or we're out in the woods, "The dog saw the fox and ran after the fox quickly. But the fox escaped it through the woods, and the dog looked sad and came back to us." That's a whole lot of words. We all understand it, obviously, but three-year-olds can get it as well.
- 4.20** So the capacity then of taking sensations and integrating them into percepts, putting our percepts of individual things into general categories that we call concepts, taking those concepts, integrating them into sentences—extraordinary. Already by the time kids are three or four, they're able to do this.
- 4.21** If we are good parents, we read to our children. We start off with simple pictorial stories, helping them develop their sensory capacities, their perceptual capacities, learn all of the particular words. We start to have very short sentences. The sentences start to get longer and longer. And then, at a certain point, we start to have sentences that have actual stories in them. It might be a 20-page picture book. It's got pictures and words and sentences and so on.
- 4.22** But the child, over the course of, say, the five or six or seven minutes that it takes for you to read through the story with him or her, gets that the story has a beginning, and then there was a development, and then maybe a plot twist, and then there was an ending, and all of that hangs together in the child's mind as a narrative, as a story, and the child is able to get it. So it's not just this sentence, and this sentence, and this sentence. It's this sentence integrated with this sentence, integrated with this sentence, integrated with that sentence, adds up to—we start to use languages like plot, where a plot is an integration. But now we're talking about perhaps 20 pages, and

each of them has a sentence or two on them, and we might be up to 80 or 100 concepts. So it's already very sophisticated stuff that is going on.

- 4.23 Also, children learn to make the distinction between stories that are often just made up—it's not real, but nonetheless, it's intelligible, and I can understand it—versus stories that are meant to be real, little theories about the way the world works. As I can remember, I would have been seven years old in second grade. But then a couple of years later, my sister went through the same exercise and my brother a year later. Then, 20 years later, when I was in the parenting mode, kids coming home from second grade were still doing this same assignment.
- 4.24 The assignment—maybe you have this one as well—is you do a diagram of the weather cycle where kids in second grade are learning about the weather. There's the sun, there's the water, there's rain, there's precipitation, temperature up, temperature down, and so forth. After they have studied this in a second-grade science class, all of these very complicated things going on in the world—what, water can become gas? Water can become solid? So, there's some sophisticated physics already that's going on in there. But the children then do a diagram of this whole thing.
- 4.25 Typically, what they will do is they will draw—the seven-year-old art skills—a picture of the sun, a picture of a body of water, a pond. This will be labeled "the sun," this will be labeled "the pond." Then, there will be wavy lines that go up, and that will be labeled "evaporation." Then, at the top there, another word "condensation" appears, and there are pictures of clouds, and then there's pictures of some watery drops coming down, precipitation or rain that falls into a stream, and the stream runs into the pond, and the whole cycle starts over again in an unending fashion.
- 4.26 That's very sophisticated. This is not just a story. The claim here is that this is a reality. This is the way the world works, but we're at the level of being able to understand theories. This is open-ended. Already, we're only seven years old, but we become better at using our senses, better at perceiving, developing more and more concepts, learning to be able to put concepts together in increasingly sophisticated sentences to capture subtleties and very complex thoughts, to start developing our imaginative capacities—how can I recombine these things to make stories—to make narratives, and also open-ended in terms of our understanding of reality and all of its complexities and so on, until, hopefully, by the time we are adults, 18 years or so, we're ready to take on the world cognitively on our own. We get our diploma and off we go. But still, we continue to learn.
- 4.27 Now, taking all of that, one thing is to talk about the capacities: I have a capacity for senses, I have a capacity for forming concepts, I have a capacity for imagining stories, I have a capacity for—all of those capacities are exercised and practiced until we become good at exercising those. We have developed then names for the exercise of those capacities well, such that they do the function that they are supposed to do.

- 4.28 Observation is a skill. So we want to learn and practice how to be able to observe the world well, to notice things that need to be noticed, to notice similarities, to notice differences on an ongoing basis. Maybe at some point we start taking music lessons that increase our capacity auditorially. Or we will take drawing lessons, which is both eyes and hands integrated. Or as adults, we might get into tasting wines and developing our palate so that we can increasingly notice subtle differences and similarities and so forth.
- 4.29 The capacity for forming abstract concepts and saying, "This is the category of things that I am talking about, and it is different from this category of things over here," that is the process of definition. What we want to be able to do is form not only abstract categories, but notice the relationships among all of our abstract categories. And so very often we start developing taxonomies of things. We might say there are kittens and cats and dogs and puppies, and there are cows and calves and horses and baby horses— I can't remember right now what baby horses are called—colts or fillies and so forth.
- 4.30 So all of these categories are here, but then we might put them all into a broader category of mammals because we look at all of those and we say they have similarities, but they're different from other things like these plants that are over here. And the plants are different from these things over here that are just rocks. So we then say the mammals and all of the plants are in a broader category of living things. So we're developing, by means of the definitions, what makes these things similar, but what differentiates them from the other categories, these complex structures. Then, of course, we can take dogs and divide them into sub-breeds, and so the process is open-ended that way.
- 4.31 We all carry around, by the time we are adults, very sophisticated taxonomical structures in our heads for all of the words. And an important part of learning, anytime we come across a new word, is to try to figure out what class of things does it capture and how does it fit into this complicated taxonomical scheme that I am carrying around in my head.
- 4.32 When we start forming sentences, the science of grammar is absolutely important here. We're talking about objects and objects moving, but sometimes we're talking about an aspect of the object or we're focusing on a feature of the movement. We develop the language of nouns and verbs and adjectives and adverbs, and then all kinds of other relational concepts and helper concepts that keep us organized, as we're constructing increasingly complicated sentences, where the punctuation goes, so we know what things get grouped together and what things don't get grouped together. So grammar becomes then a skill and a tool that we use in organizing all of our concepts.
- 4.33 When we start talking about very sophisticated integrations of all of that stuff, we're making narratives. How do I tell a story that's going to be a 200-page story as an adult and have it all organized such that everything is where it needs to be, and it flows, and

it's compelling, and people are actually interested in this? All of the skills that go into rhetoric are relevant here.

- 4.34** When we start doing theories, we want to figure out the way the world actually works, but we want to keep our very abstract claims about the way the world works, mapping onto it. How do we make sure that, logically, whatever observational basis we started with, we have derived the concepts correctly, we're putting the sentences together correctly, we're making appropriate inferences? Sometimes, we're using mathematical methods. What gets those mathematical methods off the ground? How do we integrate those with the rest of our logic so that by the time we get to some incredibly complicated, very abstract general set of claims about the universe, we can be pretty confident that, in fact, we're still connected to reality?
- 4.35** So, when we do epistemology as a general science, here we want to figure out what really is knowledge, and to give an account of knowledge. This is all very complicated. We know it can go wrong. We can misdefine things. We can assign the wrong labels. We can make grammatical errors that confuse our thinking. We can make mistakes in math. We can make logical fallacy inference moves and so on. How do we sort out the way that we are using our minds in all of these very sophisticated ways such that we actually end up with knowledge, not just something that's a belief, not something that's a fantasy? That's what the science of epistemology is about.
- 4.36** All of these capacities, exercised well, will yield knowledge. That's the hope, that's the dream, that's the plan. We seem to be very good at it, although we know we can get it horribly wrong as well. As philosophers, we want to step back and figure out what the process is.
- 4.37** Let's start talking, at the beginning of the process, with sense perception. How does it work? We say things like we have senses, hopefully we have five senses, and the idea here is that they are in contact with reality. Energy impinges upon our senses: chemical energy in our noses, tactile structures with respect to our touch, light energy, sound energy. On the basis of that, we do stuff with all of that information and we've come to be aware of the world. This is philosophy. Everything is going to be controversial, including the status of sense perception. Does it, in fact, put us in contact with reality? If so, is that contact direct? Is it accurate?
- 4.38** The first and most common story is a position we call direct realism. As the name suggests, our senses are directly contacting reality, and they are our first point of contact with reality. When we are sensing the world, we are actually sensing the world directly, reality. Now, we might then, of course, go to take some of our sensations and put them into memory for later reflection. We might take this sensation and that sensation and combine it in various ways. So there's other sophisticated things that start to happen. But the moment of sensation and senses being integrated into perceptual objects is a direct phenomenon. We've got five of them, different modes of

energy out there impinging upon us, and we have developed capacities to respond to different forms of energy.

- 4.39 We also know that other animal species have sense organs; some of them the same as ours, some of them different from ours. We know, for example, that bats also hear things. We can sort of tell where things are, but bats apparently have a very strong capacity for echolocation, so they can be 20 feet away from things emitting their sounds, and as the sound patterns come back, they can tell if it's a round object or a cubical object that is 20 feet away. We can be aware that there might be something over there. So the sophistication of the receptors and the capacity for responding to subtle or not, structures in the energy that's hitting them, varies from species to species.
- 4.40 Dogs, famously, with their noses, are able to respond to chemical structures that are typically in the air. We are fairly sophisticated with our noses, but dogs are exponentially more so. One anecdote I always remember is I had a dog once and we had a handyman coming to fix something. The dog, she was a good dog, we said, "Sit here." The doorbell rang. She goes crazy for a moment, but sit. And the handyman comes in and he stands in the foyer for a moment. He's not quite sure where he's supposed to go. The dog is sitting very nicely and quietly, but she is intensely curious.
- 4.41 And then, I say, "Yeah, come on back this way." I take him toward the back of the house where the thing needs to be fixed. He starts over here, and then he realizes that I've gone over there. So he takes a step over here and says, "Do you mind if I use the restroom?" I say, "Yeah, no problem. It's down there." So he went into the hallway, the restroom, came out, and then I was waiting for him. I took him, and he had to go around a sofa out into the back. So he does his thing, and then 30 minutes later, he leaves.
- 4.42 And then, we say to the dog, "Okay, you're free to resume your business." The first thing that happens immediately is her nose goes to the ground where he had been standing, and she traced exactly the path that he had gone through the house: here, over here, over there, bathroom break, out of the bathroom, avoid that sofa, out into the backyard. I don't think I can do that. Probably none of us can do this, but we know that dogs can do that.
- 4.43 She was an ordinary dog, and some dogs are even more capable of doing so. What she is aware of then we want to say is there are still chemical structures floating around, close to the floor, and her nose has the receptor cells that are able to respond to those energy structures in a way that's informative to her and reveals knowledge about where this guy, in fact, went. That's one story about the nose. We have similar stories about all of the other senses as well.
- 4.44 This position then wants to say that reality is real; it is out there. There's all of these objects, these objects are doing things, and they are, in some cases, made up of energy

or they're emitting energy, where the energy takes a lot of different forms. What has happened with human beings is we happen to have developed five organs, as we call them, that are sensitive to or receptive of different energy structures that are out there in the environment.

- 4.45 So, when a certain pattern of energy impinges upon the organ, the organ responds. Its response is what we call sensation. To use a technical language, a sensation or a bunch of sensations that are integrated into a percept is a relational state. What happens is that the organism as a whole is now relating to reality, or an aspect of reality, such that its sense organ has come into an altered state as a result of this energy directly impinging upon that. That is sensory awareness or perceptual awareness.
- 4.46 It is direct; there's no gap, so to speak, between our noses and reality, or the chemicals that are entering our nostrils and the cells that are in a certain structure, reacting a certain way. It's a direct relational state. It's possible that we will remember the smells, that we might then try to conceptualize the smells or whatever pattern of energy we are talking about, but that is a later phenomenon. So, it's a direct relational response to energy structures from the environment impinging upon us.
- 4.47 I'll start name-checking some of the big philosophers and psychologists who are important in this tradition. Middle part of the modern era, Thomas Reid is known as the philosopher of common sense. He titled one of his books, *An Inquiry into the Human Mind and the Principles of Common Sense*. Partly, he was starting to react against what he saw as a lot of bizarre and weird skeptical philosophies, particularly with respect to sensation and perception. Here is just one statement: "Every man feels that perception gives him an invincible belief of the existence of that which he perceives, and that this belief is not the effect of reasoning, but the immediate consequence of perception." Our perception is immediately putting us in contact with reality. We believe that when we reflect on that process. We don't reason about it; we are aware of it, and we have an immediately formed belief based on our perceptions.
- 4.48 Later, in the 20th century, probably the most famous direct realist associated with this was a psychologist. He was at Cornell University: J. J. Gibson. One of the things he always emphasized was that we talk about sensation and perception in the organs, but we have to see them as part of an integrated system. Our awareness of the world is an embodied phenomenon. We have to not think about the mind as some soul or spirit or some disembodied awareness state. That perception is, as he puts it here, you "see the environment not with the eyes, but with the eyes-in-the-head"—so the eyes are connected to the head; there's a whole state that happens there—"the-head-on-the-body"—it's not something separate from the body—"resting-on-the-ground." It's a totally embodied, integrated experience as we respond directly to energy, in this case, impinging upon our eyes.
- 4.49 Rand, also in this tradition from the philosophical side, where she makes this claim, "Man's senses are his only direct cognitive contact with reality." She is a direct sensory

realist on that point. Then, she goes on to argue cognitively for a hierarchical. It's a foundationalism, as we call it. "And therefore, his only source of information. Without sensory evidence, there can be no concepts; without concepts, there can be no language; without language, there can be no knowledge and no science." That's how we build things up. But from this perspective, however sophisticated our later cognitive states end up being, we always have to be able to trace it back to the original sensory information.

4.50 So science, no matter how theoretical it is, has to go back to the language that it's using, the concepts that are built into that language, the perceptual basis that gave rise to those concepts, the sensory information. That's why then in more directed forms, the scientists ultimately have to do the experiments and show the observational data for whatever the high theoretical concepts are. That is the direct realist position. Everything is controversial. Many philosophers and psychologists will label direct realism as a naive position. They want to argue for various reasons, that we're now going to turn to, that that cannot be the correct story about reality.

4.51 A standard example will be to say, here we have reality on the left— we've all likely had this experience, when we were a little younger and daredevil, perhaps. We stand in the middle of the railroad tracks, hopefully not for very long, and we look into the future. When we look down to our sides, we can see that the railroad tracks are in fact, parallel. But then, when we look off into the distance, it looks like the parallel lines are coming to a point.

4.52 Now, we know that that's impossible; parallel lines are not supposed to meet. We know that in reality, these two lines do not meet, but my senses, it seems, are telling me that they do meet. Therefore, we have what we might call an illusion. What our senses are giving us is not an awareness of reality. It's a subjective state. We might start using language like, "That's not reality; that's only how it appears to us." There's a difference between reality and appearance. We're only aware of the appearance, but what's the connection of the appearance to reality? That's an issue.

4.53 There's another fun one: you take a glass of water, you take a pencil, you know the pencil is straight, you can feel it, you can look at it, you look at the pencil after you've put it in the water, and suddenly it looks like it is fractured or it might look like it is bent. We might be able to, of course, talk about the theory of light refraction that we've learned from lots of observations, and do the math and explain this phenomenon. But this criticism of direct realism will say, what you are directly aware of is a broken pencil. But the pencil is not really broken, so you cannot be aware of the real pencil. It must be something else that you are aware of, something in the subject.

4.54 Another interesting example: this is obviously a two-dimensional rendering of a three-dimensional cube. Try this. You may have already tried this, though, but is it the case that the corner is facing you? Choose one of those corners. I'm choosing not the one at the far upper right, but if you follow almost a 45-degree angle to the left, choose that

corner. Is that the thing that is closest to you and the cube is receding from it? Make yourself do it. Or, say, can I make that go away, and I can choose another vortex, as the one that is closest to me, and suddenly that one pops into the background? Can I manipulate this cube? Now, the cube is what it is. It doesn't change, but nonetheless, my perception of the cube and its orientation changes. Therefore, it can't be that I am directly aware of the cube as it is. I'm only aware of some subjective phenomenal state.

4.55 Another famous one: the duck-rabbit in this tradition. We look at this drawing, and it is possible for us to see the things going off to the left as the beak of a duck, and there's the duck's eyeball more or less to the right, and then its head goes around. But again, we can manipulate our perspective and say, I want to see this as a rabbit, and say, don't look at those as the beak of the duck; look at those as the ears of the rabbit coming out from the back of its head—there's the eye, and what was formerly the back of the duck's head now becomes the mouth and nose area for the rabbit. It is what it is. The reality is real, but I can look at it and say: "duck." Or then, I can just arbitrarily choose I'm going to look at it and see "a rabbit; there it is." So it isn't the case that reality is telling us what it is. It's something more subjective.

4.56 This position then wants to say that yes, we do come to have beliefs about reality by means of our sense perception, but our sense perception itself is not direct of reality. Instead, what we are aware of is something going on inside the subject, inside the mind, and there is an indirect inference to an external reality. What happens is there's reality out there, some representation—it's not a presentation of reality; it literally is a re-presentation of reality inside my mind. Then cognitively, I infer something about an external world.

4.57 Now, to illustrate this, I would like you to admire my artistic skills and introduce you to Mr. Head. There he is. Now, hopefully, you can recognize this as a side profile, schematic of a human head. We'll give him an eye, even with eye lashes, so very realistic eye. Mr. Head is out in the environment, and there are some trees out there, and that's Mr. Head's dog. Obviously, it's a dog, right? Yes. So, he is watching his dog going for a walk in the trees.

4.58 Now, we know something about the physics here, so we need a source of light. This is the sun, and the sun emits rays that then hit the dog, hit the trees. They bounce off and they go into Mr. Head's eyeball. And then, inside Mr. Head's head, there is an image, a representation of reality out there. So little images of trees and the dog appear inside his head. So something like this model.

4.59 Now, all we have is these images floating around in his head. We still have to have him become aware of those images. So we might say, zooming in, there's a little Mr. Head, Mr. Head's mind, or Mr. Head's internal self, then is aware of the images or the representations inside his head of an external reality. But the point is that he's not aware of reality. He's aware of images inside his head. That's what the awareness is. So it has to be in some sense indirect. And what he might then do—and of course, once

we start doing this automatically, it seems natural to us—is infer that the images are representing the real deal out there in reality. But that's an inferential move.

4.60 Some problems with this. We might say this is a little bit unfair because we have that head. We don't really mean that there's a head inside there. We still have this issue of how this head, if there's a head, comes to be aware of the images. We have the images, but somehow there has to be a method for them to come to be aware in the head inside Mr. Head. We haven't explained that yet. Do we want to have images inside Mr. Head's head inside his head? Then, there's another little head in there that becomes aware of it. Then, we would have an infinite regress, so we don't want to do that.

4.61 Maybe more fairly, instead, we'll maybe talk a little metaphorically, and say that it's not a head; it's his mind's eye. We do use that language fairly frequently, "the mind's eye," which, of course, is going to raise the question: Do we also have the mind's nose, the mind's ears, the mind's taste buds, and so on? But let's just stick with the mind's eye right now. The point going is going to be if that's Mr. Head's mind or his self or his consciousness or his awareness, and it's something going on inside his head, what it is directly aware of, is these images inside his head, and he then has to engage in an additional move, an inference, that something else is outside there.

4.62 He might make the inference that I have these images of trees in my dog, but how do I know that? That's a puzzle because I can't, on this model of understanding, jump outside of my head and say, "Oh, here are the images inside my head, and here's real reality directly." And I compare the two, and I say, "Oh, yes, this is an accurate representation of real reality out there." I can't do that. I am trapped inside my head. All I have is the images here, and if I make an inference that they are representing external reality, that they are accurate representations, that they are true representations, that's a challengeable inferential move, and it's not obvious.

4.63 One of the issues that indirect realists or representationalists, as we sometimes call, have to grapple with, and they grapple seriously with, is: Are we going to end up being stuck inside our heads, or can we make good on this claim that we can go from these images or these representations and show that they do give us information that leads us to make legitimate inferences about the nature of reality out there? That's a bit of a challenge. But right now, we have to put a question mark out there until we can make good on that.

4.64 An analogy that might be worth thinking about is to imagine oneself as a person inside a submarine. So, you are inside the submarine. The submarine, in effect, then becomes like Mr. Head's head, and you are Mr. Head's self inside Mr. Head's head, or you are the conscious being inside the submarine. And then, all of the submarine's detection instruments—its sonar, maybe it's got some cameras mounted here, there, and the other thing—are contacting whatever is outside of the submarine, and then along various cables and methods, sending signals that are displayed on the screen, these various images. Then, you, inside the submarine, look at those images, and you make

inferences about what is out there, presumably in the ocean. Something like that is the representationalist or indirect realist model of how sense perception works.

- 4.65 Here's a quotation, not from a philosopher, but from a very philosophical scientist making exactly that point and raising the problem. "The belief in an external world independent of a perceiving subject is the basis of all natural science. Since, however, sense perception only gives information of this external world or a physical reality indirectly, we can only grasp the latter by speculative means." What we are aware of is images, sense perceptions, and those are a subjective state inside the brain. So, it's a speculation that there is a material, physical reality outside in the first place. That's the first speculation. Then the second speculation is that science is the right series of inferences about what's actually going on out there.
- 4.66 So, we have a problem here, and this problem opens the door to the third typical theory with respect to sense perception, and that is the idealist position. In a nutshell, what idealism will say is that what I am aware of is these mental representations inside my head, my mind, and various ideas kicking around there, but all of those things are not physical. My mind's eye, myself, my eye, my ideas, these images inside my brain, all of those are mental or they are ideal. We cannot make good on this speculative inference that there is a physical, scientific reality outside of our heads. All we know are things that are ideal, so therefore idealism is the truth.
- 4.67 Now, the root to this in modern philosophy always goes through Descartes. Sometimes he is cited as the founder or the father of modern philosophy. It's another controversial claim. Francis Bacon, who is an empiricist, is much more of a direct realist, but Descartes was an indirect realist on perceptual issues and started a trend that led to idealism, for him, unintentionally.
- 4.68 He agreed with the arguments to say that if we take perceptual illusions seriously— the railroad tracks really are parallel, but they appear to me to be converging; that is an illusion, and there are lots of other illusions that we reflect on—we must be aware only of images in our minds. So, how can I know that at any point, my senses are giving me accurate information about reality? I'm a little bit skeptical about that.
- 4.69 He also has a famous argument called the dream argument, where he says that we know that sometimes we're awake, sometimes we are sleeping. But what goes on when we are sleeping? Well, we subjectively start to make little realities inside of our head. We have all of these dreams, and while we are sleeping, we think that those dreams are reality. And then, sometimes we wake up and then we say, "Oh, that was just a dream. This is reality."
- 4.70 But Descartes then posed the question. He said, how do you know right now that you're not in a dream state, but you think you're awake? And you can't, by inspecting all the things that are going on in your dream, figure out some clue that's going to tell you whether it's a dream reality or real reality? So, it might be right now that you are

dreaming and everything that you think is an external physical object is a subjective creation of your own mind.

- 4.71** Descartes has some other skeptical arguments in this vein, and he thinks that he can address them. So he does not want to be a skeptic. He does think there is an external physical world, but he's taking these as serious challenges to be overcome, and he gives it a go. The problem is that most philosophers after Descartes don't think that his solutions to those problems work. So, what we are left with is this idea that maybe I have a whole bunch of illusory perceptual states, and maybe what I think is real really is just my subjective self generating a whole lot of things. There's no way to know for sure that I am anything more than a mind generating its own subjective contents.
- 4.72** So, unintendedly, Descartes puts modern philosophy or some strands of modern philosophy on the road to a position that we call solipsism. All that I know exists is myself and the contents of myself's mind, and I can't know that anything else exists. And one form of that is idealism. The most famous idealist after Descartes is a guy named Bishop Berkeley, and he was well aware of the empirical traditions and the rationalist traditions, the Baconians, Locke and Descartes, and so forth.
- 4.73** He took the bull by the horns and said that we ultimately then have to say that all that we can know for sure exists is ourselves and the content of ourselves, our minds and the content of our minds. That is all idea stuff. We might call it spirit stuff if we want to take it in a religious direction. Or we might just call it consciousness stuff, and it is self-generating. If I am not aware of something, then it goes out of existence.
- 4.74** The famous phrase here is, "To be is to be perceived." It's not the case that things exist, and they may or may not be perceived, but their existence is independent of our awareness of them. That position we typically call the primacy of being or the primacy of existence. Things are what they are. Consciousness is, in that sense, the dependent variable. It has to come to be aware of things, but things don't depend on consciousness for their existence. This ontology, from this epistemological route that we've been tracing, says we can't verify existing independent entities. All we can verify is our minds and their contents. So to be is to be perceived by some perceiver.
- 4.75** Bishop was his official title. So one of the standard problems, people will come back to idealism and say, "Are you saying then that I'm looking at all of you right now and all of the apparatus surrounding us right now, so it all exists, and I can know all of that exists, and I exist, and I exist now aware of all of these things. But if I turn around for a moment, suddenly I'm not perceiving any of you. But if to exist is to be perceived, and I'm not perceiving you right now, is it the case that you don't exist anymore?" It would seem that I have to say, "Yes, because you're not perceived, so for me, you don't exist. And then, when I turn around, oh, you come back into existence." That sounds really weird.

- 4.76 Berkeley's solution to that was to say that it is only weird from the perspective of your individual consciousness. Now, I am a bishop in the church, and my view is, this answer to that problem is to say there is a consciousness that is always perceiving reality and keeping it in existence, and that being is God. So God is a universal, omnipresent, always existing consciousness, and it is His perceiving of reality that is constant and keeps it into existence. And then, our individual consciousnesses just tap into various things. So they go in and out of existence, so to speak, for me, but not really, because God is ultimately sustaining all of reality.
- 4.77 Now, I want to mention Kant as an inheritor of these skeptical traditions in the next generation. On the basis of all of this, one of the things he argued is that philosophy needs to have a Copernican revolution. We're familiar with Copernicus from the history of science. We go from the idea that the Earth is at the center of the universe and all of the planets and the Sun and the heavens rotate around the Earth, to the idea, it's a revolutionary idea, that the Sun is at the center of the system and we are a satellite, among other satellites, going around the Sun.
- 4.78 What Kant argues is that by the time we get to the latter part of the 1700s, well into modern philosophy, the indirect realists and the idealists have trounced the naive direct realists. What we then need to do is abandon objectivity as our lodestone. Previously, what philosophers have been trying to do is to say that our minds can come to know reality. We are aware of reality, and then we go on to do some manipulations of reality cognitively, but we can possibly stay in contact with reality, and that's what objectivity is all about.
- 4.79 But now what we know, having developed our more sophisticated theories—indirect realism—is that what we are aware of is subjective images that are presented to our mind's eye—whatever that is; we haven't talked about that one yet. We can't really make an inference that there something out there that's physical or anything in particular. So maybe we should stop talking about that, and that's a revolution. Abandon objectivity as our lodestone. Abandon coming to know external reality as our objective. We retreat to subjectivity and work within these subjectivist constraints.
- 4.80 "I had, therefore,"—that's the other famous line—"to deny knowledge." There's no way we can come to know reality as it really is. We've only looked at the perceptual issues here. "In order to make room for faith"—let's save the faith quotation for a little bit. But we're denying the possibility of objective knowledge across the board. So, we have, by analyzing perception and sense perception, talked ourselves into a fairly dark place.
- 4.81 I want to say—sideways advertisement—we have a whole course on modern philosophy, where we deep-dive into Descartes, and Bacon, and Hume, and Kant and the others, if you want to know the nitty-gritty on this.

- 4.82 Now, Thomas Reid, of the same generation, recognizes—he's a direct realist—what, from his perspective, these ridiculous indirect realists and idealists are up to. To be is to be perceived? Oh, come on. "There's nothing so absurd which some philosophers have not maintained." The chief absurdity is that when I think I'm directly aware of things in reality, we have Mr. Head's self becoming too aware of subjective images and blah, blah, blah. He says, all of that is the sign of an absurd conclusion if you end up denying something so patently obvious.
- 4.83 But as philosophers, we still need to go back and figure out: Where's the mistake? Because there is a logical progression. There are some real issues about illusions, and so forth, that need to be dealt with. Is a way then the direct realists can respond to all of this in a way that doesn't set themselves up for the absurdities down the road?
- 4.84 Direct realists do then carry on. They're aware. What they try to do is tease out all of the logical steps and claims that the indirect realists and the idealists are making, and have a response with a more sophisticated direct realist theory that can handle all of those things.
- 4.85 The first thing that they want to do is to say that we have to abandon the idea of a homunculus. That is a term of art where we want to think that somehow what our consciousness is, or our mind is, like a little miniature Mr. Head or a little self of some sort inside your head that comes to be aware of various things that are going on. To try to, as much as possible, when we are doing sensing and perception, get ourselves to think of ourselves as our sense organs in direct response with reality. So this idea that somehow my sense organs are alien to me or external to me, and I am just this internal subjective homunculus or this internal subjective self that is waiting somewhere back in my head for the senses to deliver all of these images and make presentations to me, that's not the right model.
- 4.86 If we go back to examples of touch. For example, right now, I am standing here. I am aware of some tactile things going on at the bottoms of my feet—and we can all do this experiment. The wrong way to think about touch is to say that what happens here is that there are some things happening at the bottom of my feet, and then those are communicated up through various neural things. It goes up my spinal cord, it goes into my brain, and then there are some touch images that are then presented inside my mind, and there's something in the back of my mind that then comes to be aware of those touch images. Then, inferred, "Oh, there's something happening down at the feet there."
- 4.87 The better way to put it is to say that at this point, you are your feet. The whole system here is an integrated response mechanism. And yes, some things are happening at the feet. That's integrated with what's happening all through the system up to the brain. But the whole body, this whole embodied experience, is what we call the touch state.

- 4.88 If we try it with the nose, I have sensory cells that are waiting, and if certain chemical structures come by and impinge directly upon those cells, those cells transform, and they get into a state. That's the sensation. The sensation is in your nose, and it's your nose, your nose's cells directly responding to structures in the environment. It's not the case that we say that there's the chemical structures, they impinge on the nose, and it goes back through various things to the end, and then there's some smell images, and then your mind's nose smells those mind images—that's the wrong model. We are embodied direct contact.
- 4.89 None of this is to set aside the issue that we also remember various things, that we can fantasize about various things. What we do with all of this direct sensory information is a whole other set of issues that we have to deal with. But become your body, become your sense organs at the beginning and train your vocabulary when you're talking these issues to reflect that direct embodied state.
- 4.90 Another analogy I like to think of is to say that if you think of a guitar string. So here's a guitar, and the guitar is a string of usually a metal of some sort, and it's at a certain tension, and it is there. If a finger usually impinges on it, it adds some energy to it, the guitar string starts to vibrate, and the guitar string gets into a certain state. Think of yourself as a guitar string; that you are the guitar string. It's not the case that there is inside the guitar a little guitar head or a guitar awareness that becomes aware of the guitar state's responded.
- 4.91 So, the idea here is that our senses are not things that take information and process it and structure it in various ways to present it to a mind, but rather they are the response, the direct response to energy patterns, and you are them, or they are an aspect of who they are.
- 4.92 Now, to come back to Gibson on this point, he starts off with a realistic point. He says that when we look at reality, reality has structure, lots of structure all over the place, however we are interacting with it. But that is a realist claim.
- 4.93 For example, suppose you are in a room, and you have a light source above. Then you have walls in three different directions, and you have some piece of furniture in the middle. That light comes out. It's not all uniform. The light gets structured as it bounces around in the environment. So by the time it ends up in your eye, it has light energy structured in various ways. And then, our eyes have cells that are themselves structured that respond to the structures of light that are impinging upon it. So, each of these points of light reflection and refraction in this simplified direction are different from the others. So it hits us in an array of energy pattern.
- 4.94 Here's a long description of this. "When we are observing, it is not the abstract reflection of a single ray at an ideal mirror for the surfaces indicated are imperfectly smooth, and a beam is therefore scattered in various directions depending on the microstructure of the surface"—that's all on the objective side or out there in reality

— "that is on the arrangement of a large number of 'micromirrors,' each being a tiny facet of the surface. This scattered or diffuse reflection is characteristic of environmental surfaces." So reality is complicated. It's structured in various ways. Energy, as it moves around, that comes to us in structured form.

- 4.95** Now, on the subject side: "The purpose of a dual ocular system is to register this structure or, more exactly, the invariants of its changing structure." So rather than saying that what our sense organs do is take undifferentiated information and impose a subjective structure on them and present that to the mind, what is happening is they are responding to or registering the structure. The structure is there, and they get into a structured state as a response to that, rather than that we don't know what the structure is, and we're imposing structure and presenting that to the mind. So that's the important flip.
- 4.96** Vision also has to be not only embodied but seen as in time, and most of our sensory experiences are in time as well, not just momentary things. So one of the big criticisms the direct realists will use is that the critics will take pictures of individual things and take them out of time and typically strip away as much structure as they possibly can to see and to try to draw various inferences, and that's the wrong way to do it as well. So not only is it lots of structured information and receptor structure, but it's in time.
- 4.97** "Vision is not a series of snapshots taken by a stationary observer. The ambient optic array should not be treated as if it were frozen in time and as if the point of observation were motionless." So the things themselves in the environment are moving, often slightly. We ourselves are often moving slightly. We have two eyes, and each eye gets a slightly different perspective. We're often shifting our eyes and shifting our head. So, it's an embodied across-time reaction to structured energy that is hitting us.
- 4.98** So if we take this one, for example, what we have done is we have taken a three-dimensional object and put it into a two-dimensional surface. So we're taking away a huge amount of information. And we've also stripped away any possible light sources. As we know, in reality, a light source would be hitting all of those surfaces differently. And so by the time the light comes to our eyes, it would be different. So we've taken away all of that structure, and we've frozen it in a moment in time.
- 4.99** Then, we're supposed to, on the basis of this, say that it's a big perceptual illusion. Ha-ha, therefore, we're not aware of reality. We're just aware of subjective states that are going on inside. It's, from this perspective, a totally unrealistic example to use to draw any general conclusions.
- 4.100** What would really happen is I can move my head this way or that way. If this were a real three-dimensional object out there, immediately, it would be apparent what corner is closest to me. Even if I didn't move my head, if there were a light source, and there would in reality be a light source, the light source would tell me what areas are

shaded, which ones are not shaded. The information would arrive structured in a way that tells me really what the orientation of the object is.

- 4.101** Then, one more example came across, you might have seen this one on various social media. Is that a raven bird with the eyeball and the beak pointing off to the right? Or is it a cat that has twisted its head to look up at you, and what you're seeing is the eyeball looking up at you and the ear poking? It's interesting and fun, but there's no rule in reality that says crows in a certain position and cats in a certain position can't look pretty similar in a moment of time. But any ambiguity would be immediately resolved if we put this in a split second of real time, because the raven, if it's a raven, would move. In fact, this is a cat. The cat would move, you would move a little bit, and you would know the real, uh— "Hello, kitty." And with that, we will stop.
- 4.102** My question is an epistemological question, but it may bleed into the ethical domain. It has to do with transhumanism, the qualia, our experience, and some new findings in brain plasticity and sensory addition. Professor David Eagleman, a PA professor, talks about how we could add senses. He uses the example of the space belt. The Russian scientists discovered this space belt that can orient you to the north. What's happening is these people who have this new sensory modality have a new qualia, and they can only talk about it with other people who have had that same qualia.
- 4.103** So, given this new kid on the block with sensory additions that we're discovering in neuroscience, how would the natural realists, and perhaps the other two schools of philosophical thought, address this change in qualia? And is it bad? Is it good? How would we think about it?
- 4.104** I would say from my perspective as a philosopher, all of this has to be speculative. The people you're referring to, the professors who know neuroscience, would be the first on the ground. But as you are pointing out, I think quite rightly, that the neuroscientists do have to be sensitive to the philosophical language they're using to describe these phenomena, including concepts like qualia, which sometimes is a charged term. But I set that aside because that was not the heart and soul of your question. It was the transhumanist aspect.
- 4.105** I think we want to keep ourselves open to the idea that we have five senses, and those five senses have evolved to be responsive to five different kinds of energy structures in the world. But those are not the only five kinds of energy structures out in the world. It's certainly possible that other organisms have evolved different sensory modalities to respond to those different kinds of energy structures out in the world. I suppose, given enough time, we also could do so, or perhaps it could be augmented by technologies that are doing so.
- 4.106** Now, the example that comes to mind in my case is thinking about birds and how, in many cases, most birds are able to navigate many thousands of miles. It doesn't seem like they have a cognitive map. They don't consult their Garmin or their Google Maps or

whatever it is. They haven't developed that technology. It's not even the case that they necessarily recognize landmarks: "Oh, there I see this, that, or the other," so they develop a visual map.

4.107 The theory I have heard is that they are sensitive to the Earth's magnetic field. So that is another source of energy array. Magnetism is a real phenomenon. The Earth is a magnetic entity, and so the magnetic—I'm not sure what the right language here is, but the waves of magnetism vary across one's location on the globe in a north-south direction. And so the hypothesis here is that birds have developed a sense organ that is receptive to changes in the magnetic structure. And so they know how to orient themselves according to that.

4.108 So then, I think your question is, well, if the birds can do it, why can't we do it in principle? That would be a transhumanist direction. So I would say I'll keep an open mind and see how it goes.

Lecture 5: Then Conceptions

At a glance — Then conceptions — moving from perception to concepts.

- 5.1 I want to move up the cognitive complexity ladder, so to speak, from sense perception to concepts. When we are perceiving the world, we are aware of particulars, particular entities, particular actions, particular smells. When we are conceptualizing the world, we are generalizing an abstraction. We are considering things as members of groups, and then we have a word, typically, or sometimes a phrase, that we use to describe or label that group of things considered as a member of this abstract general group.
- 5.2 For example, I might say, we have some people here, but I've already used an abstract word, and I put you in a group when I was gesturing with my hand. I'm aware, perceptually, of all of you as individuals. I'm aware of all of you, perceptually, as individuals, and you each have unique traits, sensorially, different shades and colors and distances. If you were to make sounds, different sounds and so on, all of those particulars about you are completely different.
- 5.3 Yet, you're all members of the human group. While on all of your particular dimensions, you are not the same, you're all the same in the sense of being human beings. So there is an abstract trait—humanity or being a human—and I have categorized you into a general group. And all of these other things that are over are too different, so they don't get in the group. We leave them as uncategorized individuals, or if I start noticing things about them, I might put those into groups.
- 5.4 I've used this "humanity" or "human" or "person" label. What is it that I am seeing or smelling or tasting or whatever it is, that you all have that? Because I can say you have a certain color, you have a certain sound that you make. If I touched you, you might have a certain texture. But what is the humanity that I am touching or smelling or hearing or whatever it is? So, there's an interesting set of issues about this cognitive move that we make from particulars to abstractions, why we do the groupings that we do.
- 5.5 Now, that we do, and that it adds enormous power to our cognitive range, is apparent. One of the famous examples in the philosophy is called the speckled hen example. We, and any animal that has a perceptual sense of vision, can look at a speckled hen and see that it is a speckled hen. So, it's a hen, but it has a whole lot of speckles on it, and we can be perceptually aware of the hen.
- 5.6 But if I ask you the question right now: How many speckles are on the hen? You can't see the number of speckles. It has a definite number of speckles. We could do a cognitive operation that we call counting, developing 1, 2, 3, 4, 5, and so on, an open-ended numbering system. We could then engage in operations and come to know what number of speckles the hen has. But we have to do that by operationalizing concepts. We can't just see it with our eyes, so to speak.

- 5.7 Another parallel example, here's a zebra. When we say it's a zebra, it's a striped zebra, and it's got a bunch of black stripes and white stripes. We see all of them. There's a definite number of them. There's the big zebra and the little zebra, and each are particular entities. We put them in this category. But again, you can't perceive the number of stripes on the zebra. Is it 19? Is it 22? Is it 31? You don't know right now, even though you're seeing the zebra perfectly clearly. We have to deploy mathematical concepts or arithmetic concepts and do counting as an operation. The world doesn't come to us precategorized, and there's no little sign that pops up and says, "This is a 31-striped zebra, not a 29-striped zebra." That's not what happens here.
- 5.8 These are learned. One of the things that's interesting also is that, when we operate on the sensory perceptual level, those are automated processes. So, the energy impinges upon our eyes or our ears, and our bodies react automatically. Much of the integration of sensory information into awareness of objects that persist across time is also automatic. All of the evidence points to our concepts being learned. In many cases, we can time this developmentally, when the child first starts to use words, instead of just pointing at things, to use words because that means the child has some new mental unit or new way of looking at the world that he or she wants to operationalize.
- 5.9 So, an example of this, some anthropologists have studied lots of different cultures in lots of different ways, and the different levels of conceptual development or types of conceptual development that go on is often very illuminant. For example, we know in our culture we've got amazing mathematics, and we can, as individuals, decide how much of it or not we want to learn.
- 5.10 There's a tribe, the Pirahã, that has some arithmetic, but it only goes up to 2. So 1, 2, and then anything beyond that is just many. So in fact, they have three concepts. So there's 1, there's 2, and then there's many. They have not developed further concepts. So if they were to have a number of animals or a number of people, in principle, they would not be able to tell you if there are 12 people or 13 people. They would be aware of all of the people who are there as individuals, but it's just many. And the same thing if they have a flock of goats—a herd of goats, I guess it is— the difference between 48 goats and 52 goats is just many. They have not developed those concepts.
- 5.11 Another tribe had a 16-base number system, and it stopped at 16. The way they counted was using all of their fingers all the way up to 10. That's probably why the base-10 number system is so popular. But they would then also, after getting to there, touch each elbow, and each of those would represent. So this might be 1, 2, 3, 4. This might be 11. This might be 12, and the shoulders, 13, 14, 15, 16. And then it would stop. Beyond that, it's many. The concepts and the idea of an open-ended, just add one, method procedure has not been developed by them as well.
- 5.12 Now, as we know in philosophy, everything is controversial. I'm tempted to put this out as a rule of thumb, that there's always three important positions in every issue. That's often a useful methodological check. On most philosophical issues, often people want

to give you an either-or. It's either this or that. Chances are very good, on whatever issue you're looking at, there is a third position, so don't use that as an arbitrary template all the time, but it is a very good rule of thumb.

- 5.13** Since the language is complicated, I want to read through the descriptions of the three theories of what a concept is, this mental unit is, and then we will do examples and arguments back and forth to illustrate the differences. One position is called realism. This position says, when we talk about any concept, any abstract universal grouping and the word associated with it, it is picking out a reality. There really is abstract universal realities out there that our concepts are picking up on there.
- 5.14** So, you are all individual human beings, and we give all of you individual names, and I will describe all of your particularities. If I were to take a journal of your height, your weight, your history, and so on, all of those things are particular and real. I'm aware of those, sensory perceptually, but you are also human. When I am aware of you as humans, you do have in you, or you are participating in, an abstract general category: humanity. So, your humanity is just as much a part of you as your hair color is, or the fact that you do or do not have nail polish on right now, or that you do or do not weigh a certain amount. Your humanity, as a general abstract part of your identity, is just as real as all of those other particular ones.
- 5.15** Conceptualism wants to say that the abstraction part is a function of our minds paying attention to certain information in reality, and in some ways, packaging and repackaging the information we get from reality. All that exists out in reality are particular things in their particular. There's no abstractness out there in reality. Particulars are out there, but our concepts are a result of a process that we undergo to pay attention to some features of reality and other features to ignore those, to form a new kind of mental unit that we call a concept.
- 5.16** Nominalism, as the name suggests, wants to argue that, even in our minds also, all that exists are particular entities, ideas, impressions, memories, and that all we are doing when we form a concept is applying a name, nominalism, applying a name to a grouping of things that are still particular in our minds, just as they are particular out there in reality. Now, with that abstract statement of the three things, let's get into the arguments a little more.
- 5.17** So, realism, the position that says what we're doing is, there really are out there intrinsic in reality abstract properties, as well as particular properties, and we have a mental capacity to grasp those abstract properties, just as we have mental capacities to grasp abstract particular properties. We might say that we have sense perceptual capacities, and those enable us to grasp particularities that are out there, particular sights, smells, tastes, shapes, and so on. But we have a capacity for reason, or it might be the mind's eye, whatever it is, but we have this other capacity that enables us to grasp abstract traits that are out there in reality and bring them into our minds.

- 5.18** Why might we think this is true? The most famous representative of this tradition is Plato. "But of the heaven which is above the heavens, this is the region whereof I must now speak, for thither the truly existent essence is found, the colorless, formless, intangible essence, visible only to the mind, the pilot of the soul." Now, we encountered this quotation in a metaphysical context. What we are now doing is using it for its epistemological implications. So there really are essences, abstract essences.
- 5.19** For Plato, they have to be in another dimension—we'll get to why he thinks that must be in a particular place. What happens is that all of the particular things, more or less, participate in or have some relationship to that higher abstract essence. And so, we are able to perceive particular things with our sense organs, but also our mind; the pilot of the soul, as he puts it here, is able to tap into the essential abstract essence of the person.
- 5.20** Why might we think that this is necessary? I draw again upon my art skills, and I present to you three perceptual objects. Now, we can be aware of them: one is to the left, one is a little bit to the lower right, one is a bit to the upper right. So they have different spatial positions. One is green, one is a pinkish purple, one is black, so they have different colors, and we are aware of the different particularities there.
- 5.21** But we might just say, let's focus on the spatial arrangement of each one's constituent elements. Very quickly, we would say, "Oh, those are all different from each other," and we will assign abstract labels to them: that's a square, that's a triangle, that's a circle. And it's not that this is the only square, the only triangle, or there's circularity—there's lots of circles that are out there in reality, but they're all different circles, but they're all equally circles. There's lots of squares that are out there, and I can group all of them into the category of square, and this is one example of the square. Then the same thing for the triangle.
- 5.22** So I'm aware of them as particular entities, but as also members of a more abstract general group. All of that's fine. Now, is that, though, really a circle? We all say, "Yeah, it is a circle." Is that really a square? If I adopt my Platonic hat or put my Platonic hat on for a moment, I would say, in fact, though, it's not a circle, and that's not a square. It's only sort of a circle, and it's sort of a square. My reasoning would be this. Well, what is a circle? Give me an essentialized abstract definition of a circle.
- 5.23** We would say something like, well, a circle is a locus of points equidistant from a given point. So there's a center, and then if they're all equidistant from it and it's a continuous equidistant locus of points, that's what a circle is. So it's an enclosed locus of points equidistant from a given point. That's what a circle is. That's perfect. It's beautiful. That's the abstract. I've used a whole lot of abstract concepts to describe what a circle is.
- 5.24** But now, if we look at this one, and I zoom in on it, two big glaring problems. One is, it's not enclosed. I got lazy, apparently. I almost made a circle. It doesn't fit the definition.

Also, not all of those points are equidistant from the given point in the circle. So this is only sort of a circle. It's circle-ish, we might say, but it's not a real circle. But we do have this concept of a circle that is perfect. The mathematics that we use when we do our analytical geometry is perfect and certain. There's none of this kind of, sort of circularity issues, and pretty close is good enough. No, there has to be exact precision.

- 5.25** So, Plato's answer is then to say, well, this is not a circle, but it kind of participates in a sloppy way in circularity. We have to distinguish the idea of a circle from the imperfect representations of a circle in this material world. The same thing, I would say, if we go to the square. We would have a definition of a square, we would say it's straight lines, right angles, 90 degrees and all of that. These lines are not very straight. That's not actually going to, if we put a precise protractor on it, come out to 90 degrees. That's only kind of, sort of a square, but there has to be squareness in its perfect essential trait, in some other dimension.
- 5.26** The nominalists want to argue that's ridiculous. Instead, what we're doing is, we're recognizing particulars are particulars. They're all different, but we notice that some of them are more or less the same. We don't need to get into all of this perfect abstract essence stuff. All we do is take the "imperfect"—we put that in quotation marks language because that language is somewhat prejudicial—we take the circles and squares and triangles as they are in the world.
- 5.27** The ones that are circle-ish enough, are close enough, and they're different enough from the ones that are square-ish to, for us, for our practical purposes in the real world, say we're going to consider these ones as a group, slap a label on them. We'll call these squares, we'll call these ones circles, and we will get on with life. None of this stuff about other dimensions, none of this stuff about having special cognitive faculties that are able to tap into higher dimensions.
- 5.28** We're aware of the world as it is. It's a number of particulars, and they don't have to be equal in any super mathematical precision. We take things as they are, group them as it seems to be. From their perspective, that's what we always do when we conceptualize. No one has seen a perfect circle, no one has measured a perfect circle, so abandon that quest.
- 5.29** Now, we might take another example. This is the color spectrum or the visible light spectrum as a representative. What do we do ordinarily as human beings? We divide it up into segments, and then we form these abstract categories. We will say we have red as a color category, yellow as a color category, green, blue, purple, as color categories. Then on the basis of that, we might then go on to do other abstract operations and so forth.
- 5.30** But from the nominalist perspective, reality does not come to us, that the dividing line between red and yellow is announced, that here the essence of red stops and the essence of yellow begins. Instead, what we have is something that's more or less red,

which starts to become more or less yellow. It's largely arbitrary where we are going to draw the line. That's what we do: we more or less arbitrarily draw the line.

- 5.31** We might make up this example here. This is partly my example, partly not. We might just take the whole color spectrum and, instead of dividing it into five or six, divide it into four. I like the idea better of orred and grello and bleen and blurple. That's kind of fun to say, isn't it? Blurple. All right, so let's just get rid of that old, arbitrary red, orange, yellow stuff that somebody foisted on us a long time ago. And if I said, "These are the categories that I wanted to use," you can't say, "Oh, you're wrong. That's the wrong way to divide it." All you can say is that you, subjectively, have decided to add it that way, in contrast to someone else's subjectively deciding to divide it up a certain way. Neither of them is particularly wrong; it's subjective.
- 5.32** A famous representative in this tradition: William of Ockham. "I maintain that a universal,"—what we might call a concept or an abstract statement of something essential—"is not something real." There is no yellowness; there's just things that are more or less yellow. "It's not something real that exists in the subject, but that it is a being only as a thought-object in the mind." If I'd decided I'm going to go with the standard scheme, this is the yellow scheme, now I can think about things as part of the yellow scheme, but that's just a decision that I've made as a thought-object in my mind, as a particular idea that I have there.
- 5.33** David Hume, some centuries later, another famous name in this nominalistic tradition — things in reality are particular. We have sense impressions of them that are on us. I have particular sense impressions. I remember those sense impressions. Those go into my memories, but all of my memories are particular. And then I have particular labels that I assign to groupings of particular ones that are more or less similar. So nominalism, again.
- 5.34** "All ideas are derived from impressions, and are nothing but copies and representations of them." That's the language there. It's not that we're doing something different to make them more abstract. They come to us particular, and we just make other particular copies of them. It's a photocopier, perhaps. This is a metric. "Whatever is true of the one must be acknowledged concerning the other." So, if the impressions are particular, then all of the copies of them that I make in my mind must also be particular. You can't get something from nothing. "Impressions and ideas differ only in their strength and vitality, not in their particularity and abstractness."
- 5.35** Conceptualism disagrees with both of these positions, and it argues that the concepts we have in our mind are, in fact, abstract, general, and they result from an operation that we engage in on particular sense impressions that we have initially. But what starts as an awareness of things as particulars does become an awareness of them as members of an abstract group. How does this work?

- 5.36** Rand is a representative of this tradition. "None of the traditional theories of concepts regard concepts as objective, as neither revealed nor invented." Let's pause right there. The realist position is what she is talking about as revealed, that somehow triangularity, circularity, exist as perfect forms in that alternative dimension, and those are revealed to us, to our mind's eye in our soul. Or the other position: invented. That's the nominalist position, or her language for the nominalist position. We just decide to group these people here, invent a word, slap it on that group of things, so, it's not revealed or invented.
- 5.37** The way she puts it here, "They are produced by man's consciousness in accordance with the facts of reality." So it's something that we do, but based on the facts of reality; both are important. "As mental integrations of factual data computed by man, as the products of a cognitive method of classification whose processes must be performed by man, but whose content is dictated by reality." The idea here is that our consciousness, generally speaking, is a relational phenomenon, but our consciousness is a faculty or a set of capacities that have a certain identity, so they have to engage in certain operations in processing the data.
- 5.38** We have to say, it's not just reality stamps content onto our minds. We do mental work, where that mental work is guided by a connection to reality. So, a proper account has to say something about the object side and the subject side in a connected way. Now, that's big language. Let's try some examples.
- 5.39** First example, let's try some of the arithmetical examples. We have a concept of 1, 2, 3. We also have a concept of a method, wherever you end, just add one again, and keep going. The process is open-ended. So by the time we get up to 87, 88, 89, we still know what we are talking about. When we get up to 3416, 3417, we still know what we are talking about.
- 5.40** In some context, the difference matters a lot. We're sending a rocket into space. We want to get the pressures exactly right to a certain level of precision because reality is what it is. And we've developed arithmetical tools that are enabling us to understand and control reality to that level of precision. So, while there's nothing in reality that comes with a label on it that says 3418, nonetheless, that is something that goes on in our mind that is objectively picking up something that's going on out there in reality.
- 5.41** If I owe you a debt and it's \$9,999, and you are persnickety about these things, you will count all of them. If it only comes out to \$9,997, we have a problem, because the reality here does not match what you said. You said this ends in a nine, but it actually ends in seven. The mental apparatus of the concepts that we are using, and were derived by a certain process, are picking up a reality. Now, how all of that works out, well, we have to go back to the beginning to understand the process. If I say, this is 1, well, what's the oneness? Or this is 3, what's the threeness, and how do we get to that?

- 5.42 Let's start perceptually. Now, we just look at these things and don't use any number concepts yet. Just by looking at them, we can see that the things in the upper left are sort of grouped. There are some things over toward the lower right, and they're sort of grouped, and the things in the lower left are sort of grouped. Just by looking, I can see that they're all different, but also, there are some similarities there. The upper and the one off to the right are more similar to each other than the one to the left.
- 5.43 At the same time, I am aware that the upper one and the one to the right are also different. The language we would say, the lines are not equally spaced, the lines are a little bit different. I'm aware of those differences, but I'm ignoring those differences for now. That's a mental thing that I do. I push out of my mind the difference in length and the difference in space, and I'm attending only to the thing that seems to be in feature that differentiates them from the other two.
- 5.44 Suppose I take another visual thing. These are purplish dots on some paper. I see there's some to the left, some to the bottom, some to the upper right. Again, each dot, if I look at it as a particular, is unique and particular. But the ones to the left and the ones to the bottom, there's something about them that is similar, and I can see that similarity, and it differentiates them from the ones to the upper right.
- 5.45 I might also go over here, just go, I'm exercising my foot. I experienced that, and I can remember that. Then I do it again. That was different, and I'm aware of it experientially. The first one, even though this is a leg making a sound on pavement, has something similar to those ones there, and it also has something similar to two of those there. So, suppose I take the ones that seem to be perceptually similar to me, ignoring the particular differences within them, and I do something like this: I draw a mental circle around them—that's the dotted line.
- 5.46 So I can say that I'm going to, for now, ignore the ones in the lower left and the upper right, and just focus perceptually on the ones that are inside my mental frame of reference here. They do, in fact, have something similar to each other. What I will do is then say, I'm going to consider this as a group of things and put this in the same category, ignoring the differences of their particularity, and just say, "I'm going to call that three." I've just formed the concept of three. Or I could have called it "trois" or whatever language I wanted to use.
- 5.47 Now I have a new mental unit, and anytime something new comes along, it might be very different—particularity—but also the same. That was three tosses. So three lines, three dots, three taps, three tosses, they're all the same. At the same time, they're very different. But my mind had to go through a various process of differentiating, noting similarities, and then forming the new unit. Now we've got three. We might say, this means three, or that symbol in the middle means three, or that set of letters written out means three. That all means the same thing. Those are abstract characterizations of the same thing.

- 5.48** All right, let's try another example. Suppose we want to say mathematics gets very sophisticated, very abstract, very precise. But we can be skeptical about a lot of mathematical concepts the way, say, the nominalists are. They might say things like that one, two, three, counting, we understand that that's very fine. But when we consider a lot of the measurement concepts that mathematicians use, those seem often arbitrary and subjective.
- 5.49** If we take, for example, basic measurement concepts, like foot, yard, mile, it's not the case that reality comes carved up in feet, yards, mile. Instead, feet, yards, mile is not only a cumbersome arithmetic measurement system that we now call the imperial system—the metric system is also much more efficient—but at the same time, we would say the world does not come divided up into centimeters, meters, kilometers, and so forth. So these seem to be made-up units that we arbitrarily decided to use, and we are imposing them on reality.
- 5.50** So the nominalist will then say, mathematics, for all of its vaunted precision and objectivity and its unlocking the secrets of the universe, cannot be seen as real. It's in the same category as other things. What I like to do is use these examples here about the objectivity of measurement concepts, despite the fact that we as subjects are engaged in manipulating reality in certain ways and forming mental units, in order to capture the ways we are manipulating our awareness of reality.
- 5.51** So suppose I'm just walking along and I see some sticks on the ground. The sticks are real. They're all individual. The ground is real. I pick them up, or I pick one of them up. I'm not interested in that one. I'll put this one over here, something like that. What is real is, I have a stick in my hand, and there's that thing over there and that thing over there. It really is there. It really is there. This thing really is here. I look at this, and I say, "This is an interesting thing," and I have all of its perceptual features.
- 5.52** It has a certain feel in my hand. When I look at it, it goes from here spatially over to here. If I do this, it makes certain sounds. There's all kinds of things that it really has as features, and I really am aware of all of them. Suppose, though, I say, "You know, I'm not interested in this thing, and I'm not interested in the texture thing, but I am interested in this spatial thing here." From here to here, there is a reality at work here.
- 5.53** Suppose then I'm playing around with that and say, "That's interesting," and I look at my foot, and then I do something like this, practicing my balance. That's interesting. It's about the same. Perceptually, I could know this is a stick, and this is a foot. Those are different things, but nonetheless, there's something similar about them and an object of similarity that I am now aware of. Then I say, "What I'm interested in is the [indicates distance] from there to there. I'm interested in the [noise] from there to there." I don't know, I'm going to call that a foot.
- 5.54** I just made up that label, and it's based analogically on this, but that is fine. I now have a new mental unit: a foot. What I mean by that is, this and that is characterized by a

relationship to this. This is real, that is real. The fact that they have a similar spatial dimension is also real. It's a relational word capturing real relations in reality. If I now then were to take this stick—I'm calling it a foot now—and say, "Oh, those things also have some spatial—it's a real stick. That's a real stick. It really is there. It really is there," but I'm interested in this part of it, the distance part of it, it really is the case that they are in a spatial relationship to each other.

- 5.55** And if there were another stick here, maybe I put this here, the relationship from here to here is different to the relationship from there to there. That really is a relationship. How would I capture that difference? If I were then to say, "Well, I'm going to do this," two, three, four, five, and this, oh, I counted six. So this is six feet, and this one is going to be something like three and a bit feet, three and a half feet.
- 5.56** Now, I'm using three, I'm using foot. It's true that the world does not come carved up into feet, and there's no little sign that pops up and says: six feet. I am using mental operations, but all of the mental operations, the words, the concepts I'm using, are tied to objective features of reality. The claim then is that mathematics is objective in that sense. We're not tapping into higher realities of essences, threenesses, and sticknesses, or meternesses, and so on.
- 5.57** Instead, what we are doing is taking reality as it is, noticing certain features, and using some features as ways of specifying properties, relations, dimensions of other things. So mathematics then is objective. That is to say, it is things that are going on in our mind, capturing actual relations in reality. So we have realism versus conceptualism versus nominalism. We want to go up the cognitive hierarchy further. We have all of these sensations integrated into percepts. Now we have figured out concepts as ways of organizing and retaining percepts in an abstract fashion. It enables us to do more sophisticated stuff.
- 5.58** Now we start to take concepts and put them together into sentences, and we go on to do logical operations with them. Let me say that this is a whole other fun, very complicated area, and we have a dedicated course just on logic, with some attention to language issues and grammar issues, as well as that. That is continuous with this material, but it's a deep dive into just the logic branch, and we will skip over that for this course for now. We will just make the statement that it is continuous with all of the material that we have developed so far.
- 5.59** The same arguments then need to be specified. There are those who want to argue that logic is just subjective, a bunch of rules that we have made up that we impose on reality. Some who want to argue that there are real logical operators and logical essences out there that our rational minds are able to tap into. There's a more conceptualist approach to logic that sees logic as based on objective reality, but it's an abstraction on identity, causality, and so forth. How all of that works out, if you are interested in, feel welcome to take this or, of course, some other logic course.

- 5.60 This is a famous quotation from Francis Bacon, early in the modern world, in a book that he called the *Novum Organum*, or the New Tool, the New Logic. What he was very interested in was developing an observational, rather, experimental basis for logic, philosophy, and science more generally, in contrast to what he thought was too much of a speculative top-down approach to logic and the sciences and an approach that was too reverential to old texts, just because they were old texts written by famous people in the past.
- 5.61 But I like it because it captures one of the core positions about the relationship between the subject and the object. He puts it in, first, metaphysical formulation. "Nature, to be commanded, must be obeyed." The idea here is that nature is what it is. We have to obey nature. And we, of course, as human beings, want to be able to control nature and sometimes alter nature to suit our purposes better. But we can't start from our subjective preferences and just impose them on nature.
- 5.62 We start with reality, existence, nature as it really is. It sets the terms. We learn what those terms are. Then we figure out what we need to do successfully to act in nature. Does it give us some wiggle room, so to speak, that we can manipulate nature in various ways to make it better? But it still has to be on nature's terms. The point is then that this objectivity of cognition becomes fundamental for Bacon. We have to keep the subject, that is to say, our minds, mapped onto the object, reality, and developing the tools, the new tools of logic, including very sophisticated scientific and mathematical tools that will enable us to do.
- 5.63 Now, like everything in philosophy, the concepts of objectivity and subjectivity are used variously and variably, and they are all hotly contested terms, even once people agree more or less on what the definition is. So expect anytime you're in serious discussion to have to spend some time sorting out just what these concepts mean and how they are being used in that particular context. So, as a step toward illuminating some of the controversies, it does strike me, from reading the entire tradition, that these concepts are used in two different contexts, and part of the confusion is the slipping of context. So one thing is being clear what the context is.
- 5.64 Sometimes we use objective and subjective metaphysically. What we want to do is to say, by objective reality, we mean what's out there, existence, reality, all of the objects, so to speak, that are out there. And by subjectivity, we mean the self or the I or the me that is either trying to come to be aware of the objects that are out there, or I'm a subject entering into the world of reality and trying to act and interact with it in various ways. So there's a difference between the object and the subject as a metaphysical thing.
- 5.65 Epistemologically, though, we use objective and subjective to refer to what's going on in the subject or to notice how the subject is acting cognitively in trying to come to understand reality. Here, the basic distinction is to say, I, in using my cognitive apparatus, am trying to stay connected to objective reality, in which case, I'm trying to

be objective, which is a description of my activity, being objective, in connection with a metaphysical reality of objectivity.

- 5.66** We mean subjectivity, or by subjectivity, we mean a person who is using their apparatus but not trying to be connected to reality. Or a position that says it's not possible for us to use our cognitive apparatus in a way that connects to reality. Those are subjectivist positions. So, if I say, for example, I'm observing reality, I'm making very careful notes, I'm trying to be objective, that's different from someone who says, "I'm just going to close my mind. I'm just going to make up some words and some ideas and some imaginary characters."
- 5.67** I know that those are imaginary characters that are not real. I've totally entered into a fictional space in my own mind. I'm not trying to be objective. I am being subjective. I am creating my own little world with my own little beings, with their own little rules of how the world goes. In an epistemological context, we are self-referencing and diagnosing what's going on inside the person's consciousness, trying to connect to reality or not or impossible.
- 5.68** Now, there's a third position, of course. Among those who say we should try to be objective, as I first described, stay in contact with reality, there's an important division between those we call the intrinsicists and those who are objectivists. The intrinsicists want to argue—I'll just give a headline version of this—that when we are coming to acquire knowledge, largely reality does all of the work. Truth is out there, the facts are out there, reality is out there, the essences are out there, the certain forms are out there, God maybe is out there.
- 5.69** What happens in the act of cognition is that the subject is more passive, and reality stamps itself on the subject. I might receive a revelation, and I don't have to do any work; or I might use my mind's eye, and the form of circularity is transmitted to my mind's eye, and I receive it in a kind of vision. Or perceptually, the way, say, direct realism says, you don't really have to do any work, you just open your eyes, and reality starts to flow in. Those would be closer to intrinsicist understandings.
- 5.70** The objectivists want to argue—this is right now a small-o objectivism, because it's broader than big-O objectivism, which is the name of a school of philosophy that is also objectivist. So this is another term multiply used. But the point here is to say that objectivism is a matter of us keeping our cognitive apparatus connected to reality, but it's not revelation. We need to do work, like conceptual work and mathematical work. But if we do that appropriately, then it's possible to keep our minds in connection with reality. It's not just a simple revelation.
- 5.71** Let me give you an example just to illustrate the difference between these three words and their uses. Suppose you are a professor. I am a professor, and every semester, I get student exams, and I'm supposed to grade the exams. We have big discussions among the professoriate about whether exam grading is objective or subjective, or if we

should take human beings out of the equation because we think they are necessarily subjective and just have the machines do all of the grading for us in order to be more intrinsic and accurate in our grading.

- 5.72** To illustrate it in a simple-minded way, the difference between these three, suppose I get a student's essay submitted to me, and I sit down with this student's essay on my desk, and I look at the student's essay, and the first thing I see is the name of the student. Then I realize, "Oh, I don't like that person." All kinds of things then start to happen inside me as a subject. I have certain memories, I have certain emotions that are going on in me. What I do is go to the end, write C. A is the best, B is pretty good, C, average, D, so I just do C.
- 5.73** That would be a clear example of subjective grading. That is to say, the paper is an objective reality. It has certain words that are communicating certain ideas, and according to the standards of the course, there are criteria to determine the quality of it, and I'm ignoring all of that and just basing my evaluation on subjective traits. It has nothing to do with the objective realities of the paper.
- 5.74** By contrast, suppose I said the following, another extreme example. Student's essay comes in, but there are no names on them, just some numbers. We take that possibility of subjectivity out of it, out of the equation. But I said, "I don't really want to read this whole thing, but I'm a special individual, and I should be able to intuit the value of this essay." Again, somewhat ridiculous example, but I'm going to have my hands hover over the paper, and I'm going to wait for a revelation. I can sense some be-ness about this paper.
- 5.75** That then is to say, it's not based on my feelings or my personal memories or anything here. I'm making a claim that the quality of the paper, its be-ness, is inherent in the paper, and I should be able, by a kind of revelation, to experience what that quality is. That would be an absurd example of an intrinsic approach.
- 5.76** The objective approach, or the way of objectivity, is to say that the paper has the qualities that it does. Words in a certain arrangement typically come out as a C paper. If the arrangement is really bad, it comes out as an F paper. If the arrangement comes out in a certain way, it could be an A paper, whatever the standards of the essay are. It's the right word length, it's got the right number of sources, the words are accurate descriptions of what the source has said. It's all built into the paper there. But for me to figure out if this is an A, B, or C paper, I actually have to do work. It really is what it is. But nonetheless, I have to look at the words. I have to sense.
- 5.77** There's a black type on a white page, and that arrangement of letters means this concept, and I have to remember what that concept is and connect that with the next concept, connected with the next concept. Understand what the student is saying in the whole paragraph. See the hopefully logical flow all through the 8 or 10 pages of the essay. Form an overall judgment. I'm very actively involved cognitively in the process,

but it is me using my senses, my perception, my concept, my understanding of definitions, my understanding of logic, narrative, and so on to evaluate the properties. On the basis of that, if I take all of that into account and say this is a B paper, that is objective grading. It requires an object with real properties, but a cognitive process trying to grasp those real properties.

- 5.78** The tricky thing here is, in the literature, that consciousness objectivity, as here described, is making it to be a relational phenomenon. There's objective reality, and there's a subject, and we're trying to maintain a relationship between the subject and the object, cognitively speaking.
- 5.79** In abstract formulation, what the intrinsicists are trying to do is to take the subject out of the equation as much as possible, the idea being that the subject is imperfect or as impurities to the process. So as much as possible, you want to try to get to a revelation, a mode of cognition, if it is possible to you. The other side of the argument wants to say, yes, there's the object, and there's the consciousness, and necessarily the subject has to be involved. And what that means is that it must be impossible for us ever to get to some sort of external reality. Hence, we get the three positions.
- 5.80** We'll see this when we look at perceptual illusions. We'll see this when we see arbitrary definitions or different color spectrum categorizations and spectrums. The argument there will be, yeah, there are lots of ways we can make mistakes. There are lots of options, but which of those options best fits the way reality actually is? We do make mistakes. Is it possible for us to figure out what's wrong with the mistake and use methods that avoid that particular mistake? So, yes, all of the subjective things that happen make it actually the reason why we need to have standards of logic and standards of grammar in the first place, but it's a challenge that we should be up to.
- 5.81** The other side, though, the subjectivists, will typically say there's no way to solve all of those problems, and so objectivity for them comes to be a bit of a myth. Let me give you some more name-checking, further important philosophers on this point here. The argument is that, if we were really going to be objective, then according to Friedrich Nietzsche here, we would have to be selfless. So the subject gets in the way, by its very nature, of any sort of objective process. But let's read it through.
- 5.82** "The objective man"—and he's speaking sarcastically here—"is indeed a mirror. He is accustomed to submit before whatever wants to be known." And notice the mirror metaphor. The idea here is, if we're going to be objective, reality is out there, whatever it is, and our minds would have to be like a mirror. A mirror doesn't really have much of its own identity, and the mirror doesn't really do anything. Instead, light impinges upon it, and it reflects whatever is put upon it. We're trying to be as mirror-like as possible.
- 5.83** "He is only a delicate, carefully dusted, fine, mobile pot for forms that still has to wait for some content and substance in order to 'shape' itself accordingly." Different metaphor. Maybe instead of a mirror, we will think of our mind as a lump of clay out of

which you might make a pot. But the lump of clay doesn't really have any nature or identity as possible, and it just receives impressions and manipulations from outside form. Maybe that's what we're trying to do. I'm trying to be as like a lump of clay as I possibly can be, and just let reality do with my cognitive apparatus, whatever I want to do, to be objective.

- 5.84** "For the most part, a man without substance and content, a 'selfless' man." To be without a self, a no-self, just a mirror, just a lump of clay, that's what it would require for us to be objective, to receive any sort of objective input from reality.
- 5.85** But then Nietzsche goes on to say—this is the whole sarcastic way—that's impossible. All of the various skeptical arguments show that we don't have just a mirror. Our minds are doing all kinds of stuff, engage in all kinds of process. We have a very big brain. We have all kinds of structures and so forth, so that model is not possible. And he then switches to the opposite and says, therefore, since objectivity is impossible, let's take the subjectivist bull by the horns and grapple with it.
- 5.86** "Genuine philosophers, however, are commanders and legislators." We're not obeying reality. We are commanding reality. We have our agenda, a subjective agenda, and we try to make reality conform to our agenda. Or the other analogy here is, we are legislators. What do legislators do? They make stuff up. This is what I want; this is my will, and reality will conform to my will. It will bend to my will.
- 5.87** Now, this commander's language is meant to be a direct reproach to Francis Bacon, who said, "Nature to be commanded." Yes, that's fine. That's what we want. But first, we have to obey nature. And Nietzsche is saying that no, that's not possible. We can't obey nature. We can't figure out what nature is. So we'll just go straight to the commanding part. And the commanding part starts off subjectively.
- 5.88** "Their 'knowing' is creating." That's interesting. So we're not receptive, we're not responsive—all of that language that objectivists or intrinsicists will use. Instead, what we call knowledge is a creation by me. I make stuff up, or we make stuff up. Their will to truth is will to power. So it's not the case that we figure out what's true, and then we are empowered to be able to act better in nature; instead, my will to power, I just want to impose my agenda on reality. That comes first. And whatever I call truth is just that which I think is going to help me achieve my power aims.
- 5.89** That's a strong form of subjectivism, standing at the long end of skeptical arguments against both the intrinsicists and the objectivists. Now, another piece of terminology. Sometimes you might hear the concept of inter-subjectivity. That comes next in this developmental process. There are those who will say, yes, we've read the history of philosophy. It's reached a very skeptical place, say, by the time we get to Nietzsche, we're not sure if we're dreaming or awake from Descartes, or we're not sure if to be is to be perceived as Berkeley had argued.

- 5.90 We might come to agree with Kant that we can't know objective reality as it really is, only our subjective forms and categories of the mind, structure and reality and so on. We've now read the Nietzschean thing. But the problem with the Nietzschean thing from this perspective is that it sounds too individualistic to us. Do we really like the idea that any individual can just make up whatever truth they want? People have their own power agendas, and some of that's pretty dark terms.
- 5.91 But at the same time, we don't think we can go back to traditional objectivity and say our minds are really coming to know reality, but we don't want just individual commanders and legislators of the Nietzschean strike off doing whatever they want.
- 5.92 So, maybe the best thing that we can do is to say, yes, nobody really knows reality, but we are part of a group. If, among our conversations with each other, we're not objectively knowing reality, but we are subjectively forming a kind of consensus or something close to consensus, that this is what we are going to agree in our group is objective reality, and we're going to define reality a certain way; that might be the best that we can do. So we grant the power of all of the skeptical arguments, but we get away from the individual power monster fears that we have, and we temper both of those by this concept of inter-subjectivity.
- 5.93 This is an explicit redefinition. What we will now call objective, anytime we use objective, we really mean intersubjective. "Objectivity is now conceived as inter-subjectivity. Inter-subjective norms are not agreed to by the members of a society because they are objective, but in effect, they become objective because they are jointly accepted." So, it's not the case that we all get together and we agree this is the way reality is. Rather, we make an agreement about what we are going to call reality and how we're going to think about it, and that's what objective becomes.
- 5.94 Then, a further generation, this is Richard Rorty, sometimes categorized as a postmodern philosopher, sometimes as a neo-pragmatist. We, of course, have courses and lectures on those as well. This is just cherry-picking this important quote, that, therefore, the idea of facts, truth, as in any sense intrinsic or objective out there, we abandon that. We recognize, of course, that groups are highly variable. Even in any given society, there's lots of subgroups, and you just have to choose which groups you are going to enter into and associate with and adopt that group's truths.
- 5.95 So there's not constraints from reality. The only constraints are social constraints from the groups that you subjectively have chosen to adopt. And his somewhat slangy way of putting this, an attention-grabbing way, is to say, "Truth is whatever your contemporaries let you get away with saying." Often when we use that "get away with stuff" language, we know we're doing something fast and dirty, so to speak, or something that we shouldn't be doing. But we are now at the tradition, if we're in this skeptical tradition, of saying, that's what everybody's doing all the time, anyway. There are no intrinsic or objective constraints on how we act in reality.

- 5.96** Another option is to say that there is the objectivist approach. There's all of these subjectivist approaches. Maybe the subjectivity is escapable. Now we have hope, and we go back and revive Platonism in some form. The world is the shadow, or a shadow of the true reality is the world of forms. Yes, if we're in this world, everything is subjective and messy and imprecise, right, and so forth, but we still hold out the hope that we can escape the self, we can escape this messy world and get to some reality that is perfect, true, and inspiring.
- 5.97** Here is the language, again, from Plato, Socrates as the mouthpiece, and this might actually be the historical Socrates' view. It's very hard to sort out what's Socrates, what's Plato, in the historical sense, but they're both in this same territory. "If we are ever to have pure knowledge of anything, we must get rid of the body and contemplate things by themselves with the soul by itself."
- 5.98** That's what Nietzsche had been talking about. We try to deny our bodies, make our minds as much like a mirror, and just purely receptive as we possibly can. So Plato was saying, that's exactly what we have to do, except a stronger step—get rid of the body altogether as a distraction. "No pure knowledge is possible in the company of the body. So long as we are alive, we shall continue closest to the knowledge if we avoid as much as we can all contact and association with the body."
- 5.99** So no sensory information, no perceptual information, no concepts that are empirically grounded, no attempts to take our language and make it connected to perceptual reality. All of that is a false road, and we just abandon it. "Instead of allowing ourselves to become infected with its nature, purify ourselves from it until God himself gives us deliverance." So if we can purify ourselves from the physical world, from our bodies, God will reveal himself to us. That's our only hope for true knowledge.
- 5.100** This goes all the way back to Plato, but this long tradition of mysticism, as we are now going to call it, wants to deny any hard, empirically based objectivist work, wants to avoid skepticism and subjectivism as well. The idea here is that, if you want to seek some sort of purified knowledge, what you need to do is engage in meditative practices, but meditative practices not in a California, New Age-y way, not that there's anything wrong with California or people who are trying to be New Age-y, if it's a genuine quest for them.
- 5.101** But we know what that pop version of it is, where you seriously are attempting to quiet one's body because you see the body as a source of impurity and infection. So shut down your senses, go into sensory deprivation tanks, starve the body, starve, or go to the thirst point, get past the point where those things even matter to you. So you make yourself as empty and as passive as you possibly can. The idea is that your mind or your soul will then lose its bodily physical world connections, and then it will be in a position to achieve its hopefully higher destiny, either to itself, ascend to the realm of true perfection or to receive revelations from that position.

- 5.102** Again, I want to use Kant as a transition point here. Mysticism comes in many forms, many religious traditions; some philosophical traditions coming out of Plato articulate that. Kant, as we get strongly in the modern world, is trying to split the difference between objectivist and subjectivist. And what he says here is, "Though we cannot know these objects as things in themselves"—this is a big conclusion, things in themselves, things as they really are. We cannot know them. That's the skeptical point — "we must yet be in a position at least to think them as things in themselves." So, they are possible objects of thought inside our consciousness, even though we can't actually come to know whether those objects of thought in our consciousness fit the way reality actually is out there. So we're in that middle state, and there's the possibility of a reality out there.
- 5.103** And then again, the famous quotation we'll use as a transition here, "I had, therefore, to deny knowledge, in order to make room for faith." That introduces for us another concept here. It's not sense perception, it's not scientific method, it's not just subjectivism, it's not going to be seeking mystical revelations, it's something else. So this very complicated subject, faith, that will be our next topic. See you then.
- 5.104** In the past, I've heard you speak on the subject of pain. A famous quote out there is, "Pain is as real as it gets." It encompasses the intrinsic objectivity and subjection in humans, cross-culturally. So when we talk about that in philosophy, who do we look to, and what would you teach about that?
- 5.105** From the context of this lecture, the question would be: Can we take the phenomenon of pain and give an intrinsicist, subjectivist, and objectivist analysis of it? And so, what would the place of it be? So, I do think that the direct realists would like pain as a good example, because, as you put it, that's as real as it gets. I'm in contact with reality. So hot stove, hand goes on the hot stove. So there's an experiential state that happens there. The reaction is more or less automatic. And we say very clearly, "I have learned something."
- 5.106** Or the three-year-old says, "I have learned something." Or the cat says—doesn't say, but the cat has, in fact, learned something. So is it fair to say that they have, in fact, learned something, and there was a direct connection here? And the realists, the realists will say, yes, obviously, there was a direct contact, and there's a real transformation that has occurred to the skin. In many cases, if the burn is severe enough, you can see at least the inflammation or the destruction of the tissues that has occurred there.
- 5.107** So it happened right there. And where was the pain? Well, the pain was here, but it was also very quickly and instantly up here. It was the whole system that was responding to the pain. I think the direct realists like that example. I think they would like it also because then it's time to say, "It's all in the mind." If we go to the idealist thing, that sounds a little odd at this point, because maybe it's not all in the mind.

- 5.108** At the same time, we do know that the idealists will try to double down on this. There is the phenomenon of the firewalkers, where they will have a bed of coals, and they will walk across the bed of coals, and they're not doing what you or I, I assume, would do as we were walking across the bed of coals. And the lesson there is to say, see, it is all just in the mind. The pain experience is in the mind, but I have mastered my mind such that I can ignore it or make it unreal, and so therefore the pain is unreal. I do not have any empirical studies of what is actually going on with the firewalkers, but I think that's how the debate would go.
- 5.109** My question is concerning a hot philosophy in Peterson Academy, made popular to the academy by Jordan Peterson's Maps of Meaning course, is phenomenology. Founder, I think, is Heidegger. I was trying to classify in my head, is Heidegger the father of phenomenology? Is he an objectivist, is he an inter-subjectivist, or is he a subjectivist?
- 5.110** Heidegger is the most famous representative of the phenomenological school, but he's not the founder. Husserl typically gets the designation there. I do know that Jordan Peterson, in some of his philosophical moments, is strongly in the phenomenological school, but I don't think consistently so, and that would be a further conversation.
- 5.111** Phenomenology comes out of a tradition. It's a post-Kantian tradition. Again, I have to do commercial advertisements for other courses and books, because there's a whole tradition that comes out of Kant, where Kant, as this quotation we actually coincidentally have on the board right now, says there is the world as it really is out there, and then there is this world that we have subjectively constructed in our minds. And the labels Kant gives to that is, this is the phenomenal world, the world of phenomena, and that he calls the noumenal realm, and that's the real reality.
- 5.112** The point of Kantian philosophy is to say, there's an absolutely unbridgeable gap between the noumenal and the phenomenal. That's why we are denying any possible knowledge of real reality. To put it in this language, all we are aware of is the phenomena. Phenomenology, as a school, is post-Kantian philosophy that says, "Kant is right, let's forget all of this talk about noumena and trying to reach reality in itself, and just content ourselves with, we have this phenomenal world, we will describe and categorize the phenomenal world as it seems to me, but not making any pretenses that I'm describing reality as it really is." That's one post-Kantian school.
- 5.113** Another post-Kantian school does not want to give up the hope, so to speak. Their argument is that, by the time we get to Kant, we have argued ourselves into a very skeptical and therefore subjectivist place. But that just means that we have made some mistake along the way. Something went wrong with our Baconian or Cartesian or Berkeleyan philosophy.
- 5.114** So what we need to do is to go back and look at our axioms, look at our assumptions, retrace the logic, and figure out where the mistake is, so that we can show that this gap between object and subject is broken down. Because as philosophers, we should be

about reality, not just subjective, real reality. That's the hope, that's the dream, that's part of our project. So various then post-Kantian schools had different ways of trying to bridge that gap.

5.115 To come back to the phenomenologists, now, by the time we get to Husserl in the early part of the 20th century, he's the one who takes this and turns it into a school that's recognizably so. And because he was a very sharp guy, built up a school around him.

5.116 And then in the next generation, Heidegger comes along and takes phenomenology in another direction. Heidegger is a complicated figure—we have lectures on him as well, various aspects of it. I think that he is a subjectivist who still wants to be an objectivist. So he agrees with the tradition of philosophy that comes down, that says, "We are trapped in this phenomenological context; the post-Kantian move is right. But what I want to do is still hold out hope that we can get to some sort of reality."

5.117 Now, the way he does it, though, is to say that words and logic and language and conceptualization is the biggest problem. As human beings, we start to use all these words, and we get trapped in this linguistic reality. It's not so much the perceptual, phenomenological reality that the Husserlians were much more interested in. What Heidegger wants to do is find a way—and this language might start to sound familiar if you read some later figures—to deconstruct language.

5.118 The idea here is, if we can break down language, break down words, break down logic, break down concepts, to see that they have thrown a veil up over our reality, and maybe set aside that whole way of trying to get to reality through a more logical, rational, scientific method. Instead, what he wants to do is, if we can deconstruct that approach to language and open ourselves up to a more poetic, more emotionalist language, then we might find that our emotions are more of a vehicle that will put us back in contact with reality.

5.119 I think this explains partly his rhetorical style because he tries to be inventive with language the way poets are, and he's often reflective on poets like Hölderlin in the German tradition, so that the poets are a better source, because they're not rational and objectivist; they are more emotional and creative. So something like that. So yes, Heidegger is in the phenomenological tradition, but not a linguistic one, more an emotional one. I think in our time constraints here, that's about as much as I should say at this point. Don't get me going, man.

5.120 I'm going to ask you to keep going.

5.121 Oh, OK.

5.122 The dichotomy between the phenomenological and the noumenal, using labels like analog and digital, the imposition, even in Heidegger's school, the imposition of technology, like language, and its impact on epistemological foundations for being human, humans that want to approximate themselves to reality as closely and with as

much fidelity as possible. What do you see the course of, the use of technologies, as it becomes more and more sophisticated, doing to humans in terms of humanness—and either expanding or contracting the epistemological foundations of what it means to be human in interacting in the world? What is real? What does it mean to be human, and what does it mean to be real? How do we get there?

5.123 Yes, that's right. So, huge question. Yeah, so, two quick points. I think technology is a product of excellent epistemology. The fact that we're able to do these technologies that, in fact, enable us to control and better manage the world. And then self-reflexively, in epistemology, using technological tools that enable us to know the world better. So the obvious cases being microscopes, telescopes, and so forth. So, in that sense, they are products of good epistemology.

5.124 We study how light works. We do the science and we do some experimenting with smelting and glassworking and so on. We gain real knowledge as a result of that. Then we develop technologies that in turn enable us to acquire more knowledge. Now, all of that's on the epistemological side.

5.125 The other part is that that's a part of who we are. Then it opens up issues about transhumanism. Do we use the technologies to enhance our humanness, to make us better? And of course, the other side of the argument would be that it weakens us in various ways. So the sorting, I think, of good and bad uses of technology, that would be worthy of another lecture or course, in and of itself. So, a good question.

Lecture 6: What is Faith?

At a glance — What is faith? — the epistemology of belief.

- 6.1 Our question in this session is: What is faith? We know we're immediately getting into huge controversies, and the first controversy is: What, neutrally, do we mean by this word, setting aside different versions of what faith is and the issues of endorsing or not endorsing faith?
- 6.2 The first clarification is that faith, neutrally, is used in two different ways: sometimes as a noun, sometimes as a verb. When we say it as a noun, someone might say, "This is my faith," and what they mean by that is, "Here are, say, a hundred things that people in my religion believe, and I believe them." So this collection of beliefs, as my set of beliefs, is functioning as a noun in that context. Or it might be that there's a different religion, and it believes these ninety things over here; it's a different faith.
- 6.3 We're not as interested in faith in that context in this session. What we are more interested in is faith as a verb, as a means of acquiring beliefs or a means of sustaining beliefs, such that one might also add the claim that by believing on faith, one is acquiring knowledge or one is acquiring certainty, another pair of concepts that need to be connected here.
- 6.4 Now, where faith comes into the arena is if you don't think any of the other methods of acquiring belief get you where you want to go or get you to what you think is the truth. If we stay in the domain of religions, for example, there is a distinction between the natural theological approaches and the revealed theological approaches. The natural theological approaches have no use for the concept of faith.
- 6.5 If I am a natural theologian, I will say that I believe in the existence of God on the basis of looking at nature. We gather some evidence. We notice that things are in cause-and-effect relationships, and there must be a beginning cause. Or things exist— that's a natural fact. We can observe that, and we reason to the idea that there must be a being that created the universe. Or we would observe that the universe is complexly ordered and do all of the science and agree with all of the science and then make an argument that there must be a divine being that is behind all of the ordering.
- 6.6 We don't need faith in the sense that we have observation, we have logic, we have reasoning, we have scientific method, and we have proved the existence of God if we think all of those proofs. So whatever faith means, there's no additional stuff that we need.
- 6.7 But if we think that we cannot prove, or make reasonable, or scientifically justify, or whatever other epistemology we want to use, belief in the existence of God, then we are going to say things like, "It requires a leap of faith," or "It is a matter of faith." The point of the leap metaphor is to say that, in some sense, we're going beyond the

evidence or that our reasoning has taken us to this point, but to get there, we have to go beyond reason or do something non-reasonable in order to get there. What exactly that is and whether it's legitimate, that's the first descriptive part of the study: faith as a verb and then the normative part of faith as a verb.

- 6.8** Now, in the middle part of the 20th century, Oxford University had a conference exactly on this issue, and it invited three eminent philosophers of religion— there were other participants, but these were the featured speakers to take up this issue of exactly what faith is. What is going on in human psychology if one is making a faith move, as opposed to "I observe God," or "I have a mystical revelation," or "I have proved the existence of God." What exactly is this? These were collected and published shortly after the conference, and they have become staples and foundational texts in the discussions of faith by philosophers of religion.
- 6.9** Now, Professor Antony Flew introduced his understanding of faith by means of a parable. Let's imagine two explorers. They're way out, far away from civilization. They're going through the jungle. Days and days later, the jungle opens up before them, and they see this lovely, lovely vista in front of them. They see there's a stream that runs through, and there's some collecting pools over here, and there's some lovely shade trees over there. There's some tall vegetation here, and there's some plants over here that love more sun but lots of water. There's some other plants over here that are flourishing in the shade of the tall trees, and they don't get as much water as well.
- 6.10** One of the explorers says, "Wow, this is so beautiful. Who would have thought that there is a gardener out here in the middle of the jungle?" The other guy, his companion, says, "There's no gardener. This is just a product of nature." So they have a debate between them over whether this arrangement, that is very beautiful and seems well-structured, is natural or whether it is a garden.
- 6.11** The one guy wants to say, "It can't be just natural because over here we have all of the plants that need a lot of sun and they need a lot of water. It can't be an accident that they're all here, and they're not in other places in this vista. Over here, we have all and only plants that need a lot of shade, but not quite as much water. They're all over here. None of them are over there. That can't be an accident. Someone intentionally must have planted them here. There must be a gardener."
- 6.12** The other guy says, "No, there's no need for there to be a gardener for that, because what happens is, at the end of the season, the plants go to seed, their seeds scatter. There's plants here that need a lot of water and a lot of sun. They scatter their seeds, but the seeds, when they get picked up by the wind, if they go over into the shade, they land in the soil, but they don't flourish; they don't survive. If they go over here, where it's very dry, they don't get enough water, so they don't survive. The plants that need both a lot of sun and a lot of water result from natural processes, not a gardener."

- 6.13** They go back and forth on this debate, and in effect, we are having the argument from design or the teleological argument all over again, focused on this garden. Then, he says, "Let's put this to the test. If there really is a gardener, as you suggest, then we should just camp here for a while, and sooner or later the gardener will show up to tend his garden, right?" Yeah, that's a legitimate test. Gardens require maintenance, and if there's a gardener, he has to show up at some point.
- 6.14** So they camp. They wait a day, they wait a couple of days, they wait a whole week, they wait two weeks. Nobody ever shows up, but the garden carries on just as it is. And so the one guy, the skeptic, says, "There's no gardener. See, I told you there is no gardener. This is all just a result of natural processes. End of story." The other guy says, "No, no, no. That's not the end of the story. Of course, there's a gardener. It's just that it's an invisible gardener. And that's why we didn't see him any of the time when we were keeping watch for him. He did come, he tended the garden, but we just didn't see him."
- 6.15** The guy says, "Oh, so now it's an invisible gardener. Okay, all right, let's put it to the test. If there's an invisible gardener, he still has to come and tend his garden. Maybe what we can do is,"—these are very well-equipped jungle explorers—"we'll take some wire, and we'll do a trip wire thing all around this area here at knee height or whatever. Maybe we'll attach some empty food cans, so they will make a clattering noise if anyone disturbs them. We'll wait and see for your invisible gardener to show up."
- 6.16** They set this all up. They wait a night, they wait two nights, a week. Nothing ever happens. A couple of animals will disturb the wires, but nobody ever shows up. At the end of that, the guy says, "See, there's no gardener, invisible gardener." The other guy says, "Come on, there's got to be a gardener. It just must mean that it's an immaterial invisible gardener. That's why, as an immaterial being, he doesn't trip the wire, but he still must exist."
- 6.17** Flew stops the story and says that this tells us something about the nature of faith. Faith is the psychology of the believer in this case, the believer that there is a gardener. If we put it in quasi-scientific epistemological language, he has a hypothesis, and his hypothesis is that the gardener exists, and as part of his epistemological commitments, he is willing to put that to the experimental test. He is willing to say there are experiments that we can do that will determine whether the hypothesis is true or not, and those tests will terminate in observational data. So the believer, to a significant extent, is functioning as a small objectivist/scientific believer in this case.
- 6.18** But the faith part comes in that every time evidence comes along that contradicts the hypothesis, he changes the hypothesis. Flew then generalizes this as a defining feature of faith, that the person of faith is always going to believe what they believe, no matter what; that's the commitment. And often they will play the game, sometimes genuinely, of evidence and experiment, but they're going to modify their hypothesis to avoid any unpleasant evidence that comes along.

- 6.19** Flew is himself not a believer, so he wanted to see this as a dismissive point, to say this says something about faith. And the normative point is that this should be illegitimate, because if you're just going to keep changing your hypothesis to avoid the evidence as it comes along, that's not a legitimate epistemological move. But that normative claim can be separated from his descriptive claim that what faith amounts to is having a hypothesis that you will modify in the light of new evidence as it comes along. Keep that one in mind. That's our first story.
- 6.20** Now, the second important panelist here was a guy, R. M. Hare, who is more friendly to faith, although he gives an anecdote that sounds like the faith believer is not someone particularly admirable, but we'll set that aside. His story is, imagine that you are a younger person and you've just gone off to college, you've gone off to Oxford University or Cambridge University, and you are put in a dorm room with one other first-year student. Immediately, you meet this person, and you can see that he's a little bit paranoid. He's looking at you very suspiciously, wondering if you're going to steal his stuff. He seems very protective about the stuff. But you try to be friendly with him, and he warms up to you, and you can have a relationship.
- 6.21** But the very first day that he meets you, he tells you, "I don't think I should be here. I think the dons are trying to kill me. They want to kill me. The dons here are like tutors or professors one works with more closely than the professors, who are more exalted, and you see them only at a distance once a week or so. My parents sent me here because they know that the dons have it in for me, so my parents want me to get killed, and this is their way of getting rid of, and having an alibi and so forth."
- 6.22** That sounds a little bit crazy. But then it's the beginning of your semester, and you've all been invited to a garden tea party social mixer. So you and your roommate go to this garden tea party social mixer, and everybody's nice and very pleasant. You also see your new roommate having what seems like a very nice conversation with a couple of dons over there in the corner. And then, you get back to your rooms at the end of the evening, and you say, "See? People like you here; they're not trying to kill you." He says, "No, no, no. They are trying to kill me. They're just pretending to be nice to me so that I will let my guard down and they will be able to do their dastardly deeds on me at some point when I'm not expecting it."
- 6.23** Then, time goes on, and you can't talk him out of this. You write your first papers, and you submit your papers to the dons. The dons grade your papers. Your roommate gets back his first assignment and is full of glowing praise: "very well-researched, brilliantly written," a top-grade mark, and so on. You look at him and say, "See, they like you. They admire you. They want you to succeed." And he says, "No, they hate me. They're going to kill me. They're just setting me up for the biggest possible fall," and so on.
- 6.24** To stop the story at that point, Hare wants to say, set aside the fact that this guy is crazy, or just focus on the psychological mechanism of what is going on here. He has a belief, and we might call it a hypothesis, but it's not really a hypothesis in any scientific

sense. Somehow, subjectively, he's come up with this idea that people are out to get him and the dons are out, and this is a big world conspiracy to get rid of him. We don't know where that idea came from.

- 6.25** You present new evidence to him. You say, "Why would your parents pay so much money to send you to an expensive university? Why would they talk to you so nicely at the garden party? Why would they give you really good grades on your paper if all of that conspiracy stuff was true?" In each case, what the guy does is find a way to dismiss that evidence and reinterpret the evidence.
- 6.26** Hare's point then is that this is different to the account that we got from Flew. Flew said what happens in the gardener case is that the guy keeps changing his hypothesis every time there's new evidence that comes along. It starts off with a gardener, then it's an invisible gardener, then it's an invisible immaterial gardener, and so on and so forth. Hypothesis changes in the light of the evidence, but that's not what happens in this case. What happens in this case is that the guy's hypothesis stays exactly the same: they're trying to kill me. The evidence, whatever evidence you present to that person, is cleverly reinterpreted in a way that fits the hypothesis. It's not that the hypothesis changes to avoid the evidence; it's that the evidence is reinterpreted to fit the hypothesis. So it's a different psychological mechanism that's being proposed here.
- 6.27** Now, the third panelist was Basil Mitchell, also more sympathetic to the cause of faith. He also told a story. He says, imagine yourself back in the time of World War II. You are in France, and you are a member of the resistance, but the Nazi Germans have taken over France, and you are in a Nazi-controlled part of France, but you are part of an underground cell that is resisting the Germans, trying to fight back through various undetectable ways.
- 6.28** One night, your cell is meeting in a secret farm, barn house or something like that, and a stranger shows up and knocks on, say, the barn door. You open the door, and this guy is there, and there's something about this guy that you like. You feel the way it is with some people. You feel a connection to them. Other people, you feel, there's something off here. But you feel this strong connection. So you ask him what his business is. He says, "I heard there's a meeting. We're going up against the Nazis. I would like to join." You have a good feeling about the guy, so you let him in.
- 6.29** The other people in your cell say, "We don't know this guy. Why are you letting him in?" You say, "Well, I just have a good feeling about him. Let's give him a chance." So, you integrate him into the cell, and you're making plans to blow up, I don't know, some German supply train at some point in the future. This guy is helping out with the plans. He seems to want very detailed information about what the plan is, and that raises a question in the minds of some other people, but you are quite convinced that this guy is trustworthy.

- 6.30** The night of the raid, when you are going at 2:00 in the morning, this German supply train is going through, and you're going to blow it up, you and your team show up, and the stranger also shows up, but this time he's with a whole bunch of other Germans, and he's wearing a German Nazi officer uniform. Your friends say, "See, you are wrong to trust him. Your feelings were wrong in that case." You, at that point, say, "No, let's still give him a chance. Maybe he is secretly working on our side but pretending to be a Nazi in order to set himself up for greater resistance efforts in the future. I think I can still trust this guy and give him a chance."
- 6.31** Now, Mitchell ends the story there and then asks us to reflect on this. What he wants to say is that faith is not like the other two stories. Those two stories were too much about the language of hypothesis and evidence and running experiments and trying to figure out what was going on inside the person's mind as they're processing their evidence and relating it to their hypothesis and deciding what they're going to believe or not. All of that is too rational, logical, cognitive in its understanding of faith. Faith is not like that.
- 6.32** Faith is much more like a personal, feeling-based relationship. People of faith have a certain experiential state that they have that is unique to them, that is not going to be communicable by means of words put into the form of evidence and logical arguments that support or do not support a hypothesis. If you have that feeling of direct personal connection, then you are not going to be able to be talked out of it, no matter what the evidence is.
- 6.33** So we have three different accounts of faith on the table so far. I want us to pause on trying to figure out what faith is and now bring in some important related terminology in explicating the landscape of faith. While we are in the domain of religion or not religion, specifically, we know there's the language of atheist versus theist and the language agnostic versus gnostic. As with all abstract, complicated, and controversial topics, these are used variably. One is always obliged to define one's terms. So let's go ahead and do those. I am going to do this in a strict, historical, philosophical fashion as these words were first used and continue to be used by those who pay attention to that tradition.
- 6.34** Originally, these are Greek roots: theos, "God," and agnostic, a word for "knowledge." The "a" prefix is just a negation; it just says not. The two issues here— if we are a theist or atheist is to say: Do I believe in God or do I not believe in God? Theists believe in God; atheists do not believe in God. The agnostic point is the word "know." Do I know what I'm believing in, or do I not know what I am believing in? Belief and knowledge are related, but different and importantly different concepts in the human psychological apparatus. So the two questions: Do I believe in a God, and do I know God?
- 6.35** Now, if you take those as defined, then we end up with four possibilities. We have people who believe and people who don't believe, but there's also people who know or

claim to know and people who claim not to know or that one cannot know. So there are going to be people who believe and claim they know, people who don't believe and claim they know that that's the right proposition, people who believe, but they don't know, and then people who say that you can't know this, so you don't believe it. We want to fill in all four of those cells and keep them straight in our thinking. Let's spend a little time doing that.

- 6.36** In many cases, the first conceptual hurdle is for people talking about religion to recognize that most, but not all, mainstream religions are agnostic. That goes against much of the social media use of the language of agnostic. That is to say, most mainstream religions, orthodox positions, will say, "We believe in X, Y, Z. Here's the list of things that we believe in, but we are not making knowledge claims." I will give you examples from the three major Western religions on exactly this point.
- 6.37** Here is a Christian example from one of the early fathers in church history, stating fundamentally what Christianity is. "We'll start with a conception too sublime and overwhelming for man to grasp, the dispensation and divinity of our Savior, Christ." So we believe in the dispensation and divinity of Jesus Christ. We believe that, but it's beyond our capacity to understand, to wrap our minds around, to know. It's too sublime. It's too overwhelming. We don't know what that means. We can't make logical sense of it. We can't give you arguments and evidence for it. We are agnostic.
- 6.38** The contrast then would be to forms of Christianity and other religions that will say, "We believe in God, but we know this about God, and we know this about God. We know lots and lots of things about God. We're really sure that we know all of these things about God. We comprehend and understand God." That would be gnostic theism. One form of it might be mystics. If it's a mystical version of religion, "I am one of those individuals who has a direct personal experience with God. So I believe that God exists, but I also know directly in my own experience. That mystical tradition is in Christian traditions, but it's a minority tradition. This is much closer to the Orthodox mainstream version of Christianity.
- 6.39** Same thing in Judaism, from a rabbi writing a prayer book for Shabbat. We believe in Yahweh, we believe in God, but what do we know about God? "The sum total of what we know of You is that we do not know You. The meaning of having been created in the image of God is a mystery." We believe that we're created in the image of God. We have no idea what that means. It's a mystery. "It is impossible to say exactly what it means to have been created in the image of God." So theist and agnostic.
- 6.40** An Islamic version of the exact same point from one of the Surahs. "Limitless in his glory is God above anything that men may devise by way of definition." So we can't define it. We can't say exactly what this is. It's beyond man, and man's comprehension. We believe in God and in his limitlessness, even though we can't say that we know what that means.

- 6.41 Our important set of distinctions. Now, here is where one of the big controversies comes in. Suppose we agree with the definition of terms, we categorize different approaches to religion in different ways. Suppose we say believing is one thing and knowing is the other thing. What should one do when one finds they are pointing in different directions? In our most optimistic approaches to epistemology, we want to say that we should believe in things that we know.
- 6.42 But what if we need to believe in certain things or find ourselves believing in certain things, but we don't have knowledge? It also seems to be that if you come to know that something is true, you should also then believe in it. Those two should match. But what happens, in sometimes frequent cases, where there's a mismatch between our beliefs and our knowledge.
- 6.43 The two standard answers are to say that if you don't know, don't believe. Put your belief on hold. Keep an open mind, maybe, until maybe more evidence comes in and your knowledge increases to the point where it sustains your belief. The answer says that is not what we want to do. That's where the faith concept, once again, is going to come in. We don't know, but we are still going to believe, so we need something extra. We need to take a leap of faith or will ourselves to believe.
- 6.44 The person who, in contemporary times, coined the term agnosticism, drawing on the long history, but quite accurately, was Huxley. Going through all of the definition and category work, he then reaches the normative conclusion about what you should or shouldn't do if you don't have knowledge. "Agnosticism is not a creed, but a method. How do you use your mind in cases where you don't have good evidence or even close to proof? A method, the essence of which lies in the rigorous application of a single principle. Positively, in matters of the intellect, follow your reason as far as it will take you without regard to any other consideration."
- 6.45 So you follow the evidence, form the concepts, do the experiments, use your reason, do the logic as far as that will take you, and only to that point. If your reasoning tells you that you only have a little bit of evidence, maybe you don't believe it yet, but maintain it as a hypothesis, keep an open mind. If you got a lot of evidence, maybe lean into it, believe it, but be willing to change your mind. The degree of your commitment to the belief depends on and should match and scale up and down according with the degree of evidence that you have.
- 6.46 "Negatively, in matters of the intellect, do not pretend"—that's strong language there—"do not pretend that conclusions are certain which are not demonstrated or demonstrable." The difference here then would be between someone who says, I got a certain amount of evidence, say, and I think my religious belief overall is reasonable, and therefore I'm going to continue to believe my religion in a reasonable way, but keep an open mind that maybe I need to change my mind on certain things or back away. Huxley says that's fine, but if you are in that same religious tradition, you've got the same amount of evidence and the same psychological state that my reasoning, and

the evidence takes me up to this level of reasonableness, but I say, "No, I'm absolutely certain, and I'm gung-ho committed entirely to this belief system." Something else has happened—that's the faith move. Huxley wants to argue that's not legitimate. You should not do that.

- 6.47** But the other side of the argument wants to say that it is legitimate sometimes, and so we need to turn to what the considerations are that lead in that direction. The language that is often used here is the language of willing to believe, the will to believe. You will yourself. Maybe you're having some doubts, but you find a way to override those doubts and continue to make yourself believe a certain thing. That's one of the psychology, or we use, again, this leap language.
- 6.48** One example of this is the pragmatist philosopher William James, who wanted to argue that faith is fine. Faith is fine in some cases, but not in other cases. There are cases where we have evidence for or against a given thing; we should believe based on the evidence for or against, if evidence is available. But if there is something that is important and we don't have evidence one way or the other with respect to it, then, because it is important to you—it is what he calls a live option, psychologically, it gets your juices flowing, so to speak; it really matters to you whether you believe this thing or not, but the evidence for or against is not forthcoming—he argues that in those cases, it is legitimate to will yourself to believe or to let yourself to believe in that case.
- 6.49** His language is partly negative. In this case here, he says that it's one thing to say we should strive for objectivity, and I get that argument, but as he puts it here, "Objective evidence and certitude are doubtless very fine ideals to play with, but where on this moonlit and dream-visited planet are they found?" The argument here is that we can understand the argument about objectivity and evidence and so on. But he is now inheriting a certain amount of the skeptical traditions: "We don't think objectivity is possible." So, insisting that everybody should be following the canons of objectivity is not realistic, and it certainly is not realistic with respect to human psychology, because we need to believe things. We want to believe things. And if the beliefs are not going to come from objective evidence, it has to be something that we will ourselves to believe. And that is a faith.
- 6.50** So he draws that conclusion, "Faith based on desire is certainly a lawful and possibly an indispensable thing." So once again, we are not attending to words, and logic, and evidence, and language. He says, "Faith based on desire." It's based on our emotions. So if we reflect on the earlier stories—the gardener, the guy who thought everybody wanted to kill him, the Nazi resistance group—James is much more on the side of the person who says faith is not a logic-hypothesis-evidence issue. It's feeling, it's trust, it's commitment, it's a will to believe, not a reasoning to believe. It's a subjective commitment.
- 6.51** Now, why one should put this subjective commitment? Sometimes this is reinforced socially by fear. If you are in a social commitment, "You shall not put the Lord God to

the test." Sometimes that is with finger-wagging and implied pressure of some sort. But it is also in neutral epistemological points to say that your belief should not be subject to test, which is to say subject to evidence, subject to experiment, subject to hypothesis. So there's biblical support for this approach.

- 6.52** Kierkegaard, name-checking, some other important ones from a different religious tradition, eschewing entirely the language of logic, evidence, and proof, again, somewhat sarcastically, "For whose sake is it that the proof [of Scriptures' truth] is sought?" Why would you be trying to prove the Scriptures? "Faith does not need it; aye, it must even regard the proof as its enemy." Two interesting points captured in that last sentence. If you have faith, you don't need proof. You don't need evidence. Your faith is something different, something strong, something independent of that entire psychological framework.
- 6.53** Not only that, he is in the tradition of saying that we have looked at the various arguments for and against the existence of God, and we find, unfortunately, if we are on the believer side, that the arguments don't work. In fact, the arguments seem to lead in the opposite direction. So, that then is to say that proof is the enemy of faith. The whole idea of seeking evidence, trying to be logical, rational about faith and religious belief, is the wrong approach entirely. It has to be a willed commitment by the subject.
- 6.54** Martin Luther, a couple of centuries earlier, the first of the major Protestant reformers, agreeing—not agreeing with Kierkegaard because he was centuries earlier—but agreeing across time, so to speak, with that exact position here. Luther making an absolute dichotomy between reason and faith. "Reason is the greatest enemy that faith has. It struggles against the divine word, treating with contempt all that emanates from God."
- 6.55** So, if you're a rational person, Luther's position is that you have to disbelieve and actually be a little bit contemptuous of the things that Christians are supposed to believe. The virgin birth was unreasonable. So is the resurrection. So are the Gospels, the sacraments, the pontifical prerogatives, the problem of life everlasting. Luther is quite forthrightly saying that Christianity in all of its core doctrines is an unreasonable position. There's no way to wrap our minds around or get any evidence to support the idea that miracles occurred, that a young woman can somehow just find herself pregnant, and she didn't have sex with a man. Oh, come on. That's unreasonable.
- 6.56** But rather than that being an insult against Christianity, from Luther's perspective, what you have to do is to say the insult is people who think that evidence, proof, argumentation matter. Reason is the enemy. Faith is a completely different commitment, and it should embrace irrationality, embrace the contradictory, embrace things that are mysterious, embrace things that one cannot explain.

- 6.57 Both Kierkegaard and Luther reflected on the story of Abraham. And Abraham comes down in this Judeo-Christian and Islamic tradition as the hero of faith, and they are very concerned that we get the account of faith correct in the story of Abraham. The basic outlines of the story are that Abraham had a covenant with God that through his seed, the future generations of Israel were going to be born. Initially, he was disbelieving because he was married with Sarah. They were getting very old. No children had been born to them. It was thought that Sarah was barren because Abraham had had a child with his slave woman, Hagar. So he was not the problem, so to speak. So it must be something to do with Sarah. But nonetheless, he was disbelieving.
- 6.58 Then Sarah got pregnant. Isaac is born. Isaac becomes a youth. Angel of the Lord comes back and says, "Take this son of yours up to Mount Moriah and offer him as a sacrifice." What does Abraham do? Now, if we think of Abraham as a man of reason, then he's going to start saying things like, "God maybe has been good to me in the past. He's always followed through, so I don't understand it right now, but I'm going to be calm and cool and collected, and the next morning I'll take Isaac up to Mount Moriah, and we'll go through the motions. But I know I can rely on God, and he's going to save things at the last minute. I'm going to try to kill him, but then the angel interferes. I knew it, no problem."
- 6.59 All right, so if we try to characterize Abraham in those rationalist cool ways, both Luther and Kierkegaard and many others will say that we are missing the point. Abraham was not a man of reason. This did not make sense to him because God had had this covenant with him. It seems like he only has one kid: Isaac. "If Isaac doesn't have children, there's not going to be a mighty nation of Israel. That would mean that God is breaking the covenant, but God's not supposed to be the being who breaks a covenant with me. He's asking me to kill a kid. What's this kid done to deserve death? Nothing. God's supposed to be just. This is an injustice. I don't understand this. He's asking me, the father, to kill his own son. Can you imagine anything worse than asking a parent to kill their own child? That's sick. That's sadistic. What am I going to tell Sarah if she asks before or after, "What are you up to?" "Oh, I'm going to take our kid, and blah, blah, blah. She's going to look at me—
- 6.60 This is completely incomprehensible. So to the extent that he is a man of reason, all these questions are going to come up in him, and he's going to be torn up inside. He's also going to know he doesn't have any answer to any of these questions. But nonetheless, according to the story, what he did the next morning was get up, take Isaac up to Mount Moriah, and he was willing to go through with it. He was willing to kill his own son, even though it didn't make sense to his reason. That's faith, according to Luther and Kierkegaard.
- 6.61 The way they put it is very dramatic: "It is a quality of faith that it wrings the neck of reason. But how? It holds to God's Word: let it be right and true, no matter how foolish and impossible it sounds. So did Abraham take his reason captive and slay it." So

interestingly, what Abraham is sacrificing is not his son; he's sacrificing his reason. That's the true lesson of faith. "Inasmuch as he believed God's Word, wherein was promised him from his unfruitful and as it were dead wife, Sarah, God would give him seed. There is no doubt, faith and reason mightily fell out in Abraham's heart about this matter."

- 6.62 So Luther and Kierkegaard both have very harsh words for those who want to take the story of Abraham and massage it and explain it away and say that Abraham didn't really have to worry. No, Abraham did have to worry. This was the mightiest of all mighty battles inside one's soul. Are you a man of faith or are you a man of reason? And the wrong choice is the man of reason. "Yet at last did faith get the better and overcame and strangled reason, that all-cruellest and most fatal enemy of God." So we have a fundamental, antirational, even irrational epistemology that is being valorized here.
- 6.63 Generalized point, "So, too, do all other faithful men who enter with Abraham the gloom and hidden darkness of faith." I want to pause there right now. God will ask you to do some pretty gloomy things by your lights. And the hidden darkness is a metaphor for you won't understand. That's what faith involves: being willing to do things that seem gloomy, dark, bad to you and not understanding them. They "thereby offer to God the all-acceptablest sacrifice and service that can ever be brought to Him." Sacrifice your reason, that's the essence of being a man of faith.
- 6.64 Kierkegaard, carrying on in the rhetoric, is equally strong. "Faith requires a crucifixion of reason." The Romans put Jesus on the cross. That's a crucifixion sacrifice. What you have to do yourself is put your reason on the cross and kill it the same way. "One must learn to relinquish his understanding and his thinking, and to keep his soul fixed upon the absurd." In his judgment, Christianity is an absurd doctrine to your reason, but you have to believe it anyway. Keep your soul fixed on that and do whatever psychologically necessary to keep yourself oriented towards it.
- 6.65 This is a project that might be done by the individual. You might take Abraham as the model, but you, the individual, work, and some people might be up to it. Other faith traditions will argue that by and large, this is too much to accept for individuals. Many of the protestants were more individualistic. It's an individual responsibility and quest that one goes on.
- 6.66 More in the Catholic tradition, this process tends to be socialized. The way one becomes a person of faith despite whatever doubts one has, is you try to set those aside, but you join a social group and you do what the social group does. As a result of that, you become more socially conditioned and it starts to feel like your faith in that case is a part of who you are and your doubts, your rational doubts will melt away.
- 6.67 The way Blaise Pascal puts it is this way: "Learn of your inability to believe since reason brings you to this." We have a rational mind, and it raises problems for our faith, and maybe we get ourselves to the point where we have doubts: I just can't make myself

believe it because I've got all these rational doubts and so forth. "Endeavor then to convince yourself, not by increase of proofs of God, but by the abatement of your passions. [There] are people who are cured of an ill of which you would be cured. Follow the way by which they began by acting as if they believed,"—so act the way a believer will act—"taking the holy water,"—this would be in a social context, in an established institution—"having masses said, etc. Even this will naturally make you believe and deaden your acuteness."

- 6.68** So the problem is you have a sharp mind, you have an active brain, you're very rational, you're trained to some extent to think about evidence and argument and so on. That needs to be dulled and deadened, and the way you do that is by immersing yourself in a social organization, social institutions, and let it become habitual and be reconditioned, and then your rational acuteness will go away. That is the essence of faith, according to Blaise Pascal. He seems to accept the path and follow the path as set by other people.
- 6.69** We are deep into this now, and by my count, we have six different or six distinctive descriptions of what faith is. This is part of what makes this topic so controversial and why the debates on them are long-lived and seemingly never-ending. Let's run through them now, taking all of the stories that we've paid attention to, the lessons that are drawn from them.
- 6.70** The first one: faith means avoiding evidence by modifying your hypothesis. You're willing to acknowledge the evidence, but you're going to change your hypothesis to avoid that evidence so you can maintain belief in some variation of that hypothesis. That's the gardener.
- 6.71** Faith means reinterpreting evidence to fit the hypothesis. You're going to stick with your hypothesis no matter what, whatever it is that you're going to believe, and you're going to need to be clever enough to take whatever counter-evidence apparently comes along and a way to make that counter-evidence not seem like counter-evidence. Fit your hypothesis. That's what faith is.
- 6.72** Faith is not at all cognitive in that sense. Instead, it's about a feeling. You have a personal feeling, a connection to the divine, to God, to whoever it is, and that grounds, justifies it for you. As long as that feeling persists, that is your faith.
- 6.73** Faith means something where the evidence does not help us one way or the other, but it's an important, desirable thing, and in that circumstance, I just make a will. I will myself to believe this various thing.
- 6.74** In stronger form, it's recognizing that what you want to believe is irrational, non-rational, going on the offensive against the whole idea of reason, logic, science, rejecting them in total and leaping into something that you know does not make any rational sense to you.

- 6.75 Then, the sixth one was faith is a matter of habituating yourself in action. It doesn't start in one's feelings. It doesn't even start in one's cognition. Instead, it starts by action. You act in certain ways, and with those actions comes certain feeling, cognitive states that enable you to believe whatever the faith is.
- 6.76 Now, I think those are six distinctive accounts of faith. I commend them to your attention. I don't mean for them even to be an exhaustive list, because the literature is big in this area, but at least it's a very promising start and staking out much of the territory of the faith and faith debates. All of those are subject to criticism, just as everything is subject to criticism. Different versions of faith, of course, criticize each other. But people who are outside of any faith tradition will also criticize all of those. So, I want us to walk through the six descriptions of faith with the usual appropriates.
- 6.77 Now, if we say, as with Flew or with Hare, the gardener or the guy who thinks the dons wanted to kill him, that faith is something like either avoiding evidence by modifying your hypothesis or reinterpreting evidence to fit the hypothesis. The people who are on the pro-reason side will say both of those are illegitimate psychological moves. Just don't do those. Whatever the evidence is, you always face the evidence. You don't just reinterpret it away. You should always be willing to, if there is evidence, change your hypothesis to fit the evidence. Don't explain away the evidence. Also, don't be pig-headed about your hypothesis. You need to be open in both respects.
- 6.78 Here's Carl Sagan, for example, a philosophical scientist of the late 20th century: "My view is that if there is evidence for it, then forget about it. An agnostic is somebody who doesn't believe in something until there is evidence for it, so I'm agnostic." So belief fits the evidence all the way through. Sometimes, of course, we have to go back and forth, but we're always cognizant of the evidence and tying ourselves fundamentally to the evidence.
- 6.79 If we switch away from those more cognitive understandings of faith, faith is a feeling of trust, connection, or a willing yourself to believe something that you think is desirable. In both cases, we're talking about desires, we're talking about feelings, we're talking about subjective connections and so forth. The opposite argument always is to say that feelings and desires are not cognitive tools. They have an important place in our psychological life and our broader life as a lived experience, but if we're interested in grounding our beliefs, believing things that are true, believing things that are justifiable, desires and feelings don't help us there. We need a commitment to objectivity, not these subjective states.
- 6.80 Rand makes the claim, for example, on this point here, "Emotions are not tools of cognition." It's just a blanket. Whatever we do with emotions, it's not consult our emotions and therefore decide what we are going to believe. They might be a byproduct of thinking processes. It might be a byproduct of various physical states and so forth. They might value-orient us once we have decided to believe certain things as true, but nonetheless, they can't substitute for the evidence and for the thinking. "One

must differentiate between one's thoughts and one's emotions with full clarity and precision."

- 6.81** Suppose we then more robustly take these accounts of faith. Faith is going on the offensive against reason: sacrificing reason, diving into things that you know to be absurd. The pro-reason position always is to say that anytime you have an absurdity, anytime you have a contradiction, that is the wrong thing to believe. Contradictions don't exist. Reaching an absurdity is a sign that you've made a mistake somewhere. So don't believe the contradiction; see the contradiction in your thinking as a sign that something has gone wrong. Go back and start the process over. Find the mistake.
- 6.82** This one goes all the way back to Aristotle in his metaphysics with the principle of non-contradiction as an axiom. It is fundamental. "Such a principle is the most certain of all; which principle this is, let us proceed to say it is that the same attribute cannot at the same time belong and not belong to the same subject and in the same respect." Things are what they are. That means in our thinking, we have to have things being what they are. If we find ourselves saying something contradictory: "It is this, and it's also not that," or "It's this and this opposite thing," or I start off believing this, and through a process of reasoning, I end up reaching the opposite of that. That tells me I've made a mistake somewhere in my logic. Contradiction is to be avoided, not embraced.
- 6.83** Faith—if faith is a matter of following the authority of others, Pascal: just go to church, just do what other people do, recondition yourself, dull it in your mind. The opposite policy is to say never do that. Always use your own judgment, not the judgment of others. You're not an animal to be conditioned and reconditioned according to various social mores. You are a thoughtful being.
- 6.84** Long quotation here, John Locke, I think, expressed this view strongly in the early modern world. His important *Essay Concerning Human Understanding*: "Men must think and know for themselves. Not that I want a due respect to other men's opinions, but, after all, the greatest reverence is due to truth," not other people's opinions. I revere other people, but my commitment is to the truth. "We should make greater progress in the discovery of knowledge if we made use rather of our own thoughts than other men's to find it. For I think we may as rationally hope to see with other men's eyes as to know by other men's understandings." That's meant to be a little bit sarcastic. You can't see the world through someone else's eyes. Use your own eyes. The same thing holds on other issues. You can't know the world if you are filtering it through someone else's mind and just parroting what that person says. You have to know it by your own mind for yourself.
- 6.85** "The floating of other men's opinions in our brains,"—getting even more sarcastic here; they're just floating around in their other people's opinions—"makes us not one jot more knowing, though they happen to be true. Aristotle was certainly a knowing man, but nobody ever thought him so because he blindly embraced and confidently vented the opinions of another." Aristotle became a knowing person, not because he could

quote Xenophon and Plato and Socrates and Parmenides very confidently, parroting all of their opinions. He thought through, decided some things are true, some things are not true, developed his own mind, and that was his comprehension.

- 6.86** "In the sciences, everyone has so much as he really knows and comprehends. What he believes only, and takes upon trust, are but shreds." This may or may not lead us in the direction of believing in a religion or not believing in a particular religion. Of course, many people in religious traditions will say that if you are trying to be too rational about it, you're just going to argue yourself into atheism. Of course, lots of atheists want to say that if you are rational about religion, you will end up in our camp because we are the camp of pro-reason and so forth. But for this course, let's leave that as an open question.
- 6.87** Here's an example of a giant on the landscape, Galileo Galilei, who is a man of science, but also a man of reason. He argued that those two should be perfectly compatible. No faith, but if you really believe that God created the world and created us as we are, He gave us our senses. He gave us our reason. God doesn't give people capacities without any reason. If He gives you a capacity, He wants you to use it. So if you use your senses and you use your reason, you should be able to get to a reasonable religion, whatever that pops out. That's his judgment call there.
- 6.88** The implication here does seem to be critical, though, of any non-rational approach to faith where you say that no, God gave me my senses and God gave me my reason, but I'm going to turn them completely off and believe things that I think are crazy. Galileo is not endorsing that approach to religion or faith. We'll stop there. Thank you for your attention.
- 6.89** My question has to do—I might stumble through this question a little bit—the aspirational aspect of faith and its ability to, it's going to sound a little woo, but to create a world, or your world or your experience of the world as more positive. I think in the second or third lecture, you gave a definition of religion that described rituals to gain supernatural reality, a vision of such, and the shaman would play with that and imbue that vision onto the initiate. And so you have this vision from a future or potential reality, and then the initiate aligns his life and acts in accordance with that. I see that there's a power aspect of faith.
- 6.90** It doesn't have to be—there's almost a William James utility in it, that you have to act in accordance to something. Whether you're an atheist, you act in accordance to some philosopher echoing in your head that you aspire to. Whether you're a child, you are aspiring to be like your dad. Even right now with you, you're the professor. I'm trusting in your knowledge. I'm going to seek it out for myself, but I'm aspiring to somewhat gain what you have. So I see that there is a reconciliation, perhaps, between argument two of—can there be a reconciliation between argument two of the updating your beliefs and argument three about the trust aspect? Can they be reconciled?

- 6.91** That's a very interesting psychological delineation of the territory. I think if you start off in a social context, then partly the way to describe that would be to say if the novice, the novice, recognizes someone, say, as the shaman or a leader in a certain way of life, and comes to that person and says— this is, I think, where the first distinction has to come—is it that I am making a commitment to this shaman, and he's going to lay out some possibilities for me, and I'm going to stay through it, but I don't have anything to base it on. That, I think, would be one commitment.
- 6.92** Another commitment would be to say that this guy seems to be knowledgeable and has some powers that I don't yet know because I'm still a novice, and other people speak well of that person, so I'm willing to give him a shot, so to speak. So then, I would become a novice, but in a provisional sense. If my reasoning then is that I will give it a try experimentally, and we'll see if it works out, if it does seem to be leading in the direction of greater knowledge or greater aspiration, fulfill, then I will continue on that path. But I'm at the same time maintaining a critical stance, that if it doesn't seem like it's getting anywhere, I'm going to back off and say that maybe this is just wrong. That's to put it all in a social context.
- 6.93** I think if we scale that out to ordinary or contemporary examples, it would be the difference between someone who says, "I am born in a faith tradition, and I have already decided I'm going to be in that faith tradition, and I know that I don't know very much about it, but there are leaders in this faith tradition. I'm going to follow them as far as I possibly can." That would be one aspirational approach that I think is closer to faith.
- 6.94** Versus, say, someone who is a young person and says, "I recognize that there are several different faith traditions and religious traditions, and what I'm going to do is shop around, say. All of them are aspirational, but I'm going to go to this church for six months, then I'm going to go to this synagogue for six months, then I'm going to go to this mosque for six months, and then this temple for six months, because I'm a teenager and I still have a certain amount of time. All the way through, I'm doing a lot of careful assessing about strengths and weaknesses and exercising my own judgment." In that case, I think also I'm still fitting with your aspirational model and following the lead of other people who have gone before, but that is compatible with a critical reflection model.
- 6.95** I think in both cases, though, we would also need to drill down and say, when you're inside that process, what epistemological methods are you using or not using? Where we want to maybe draw on some other examples might be, I'm thinking of examples of the lives of some Buddhist monks. Apparently, there is a Buddhist monk who reaches enlightenment, not every generation, but it's very long and infrequent.
- 6.96** So you see a young person who then makes the novice commitment, say in their late teens. And then, 20 years later, they've been really dedicated, but enlightenment has not happened. And then 20 years later, they've been serious about it and

enlightenment has not happened. And then at the end of their life, enlightenment has not happened, and this seems to be the truth. Was then, we say, the initial commitment that they made an inappropriate commitment, even though it was aspirational?

6.97 In contrast, say, someone who said, "I'm going to make the wholehearted commitment, but in the back of my mind, it'll be 10 years, and if enlightenment has not happened in 10 years, then that's going to suggest to me this is not the right path. There has to be some termination clause.

6.98 Now, I know that doesn't speak to everything that you are involved. But I think in other domains outside of religion, this aspirational component and the trust component—we think of romances. We gather evidence about each other. When we're dating each other, there's a huge information flow, so to speak. People are drawing on evidence and looking for signs and red flags and green flags and all sorts of stuff. It's a very rational as well as emotional and aspirational process.

6.99 Do we want to say, though, that one should not make the commitment or the leap into the relationship until the evidence is sufficient? Or does that spoil the process? The way that love, say, really works is that you don't have very much evidence. You just make the leap into the relationship, and you hope it works. Then, if it does work great. If not, then you have to bail at some point. I think the same psychological, aspirational, emotional, passionate evidence dynamic works in that domain as well.

6.100 When you transfer it to the business realm, I think of someone who's an entrepreneur. Almost always, entrepreneurs at the beginning have a kind of an idea of where they want their business to go. Sometimes, actually, they have a more definite idea, but they have to rethink it and abandon it and go in a different direction. But at the same time, it is aspirational. But for that aspiration to work out, I have to make a total commitment to that aspiration without knowing that that is the right way to do it or even that it is possible. Again, all of those dynamics could transfer over into the business and entrepreneurial space.

6.101 That's a very rich question. I think, once again, we'd have to have lectures in the whole course devoted just to that. Nice job.

Lecture 7: The Human Being

At a glance — The human being — turning the inquiry toward ourselves.

- 7.1** We now want to turn to the human being. In this course so far, in Metaphysics and Epistemology, we have scaled up from the small to the big galaxies, the cosmos, the gods. And now we want to zoom in on something that is equally significant and important, especially to us, since we are human beings, and ask the question: Who are we? Where did we come from? What is our place in the big picture, the grand scheme, the whole shebang, as we variously say?
- 7.2** One way to approach this, particularly in a course on metaphysics, is to say we can think big picture, and we can also focus on the smaller picture, the macro and the micro. Most theories of what the human being is are a microcosm of the macrocosm, that what you think the biggest explanations are of the universe and how it works, that the human being is a unique but micro-representative of those larger features of the universe as a whole.
- 7.3** Three important, broad approaches to the human being have developed in schools of philosophy across time. They are still with us in our contemporary debates. The labels we will be using here—there's some slippage in philosophical use, religious use, common parlance, so we'll define the terms—but one is to understand the human being as a dual creature, and as the name suggests, made up of two very different kinds of things.
- 7.4** One, as the name suggests, sees the human being as an integrate, and the third that wants to reduce or explain away or eliminate some aspects of the human being as merely epiphenomenal, slightly technical term to say that things that might seem to exist but really are apparent or just byproducts of more fundamental realities.
- 7.5** Now, let's take these three labels and work through them, initially as a description of what each of these theories says about the nature of the human being. We'll also try to name-check some of the important representatives in philosophical history and key quotations from them on that position. Let's start with dualism and take as our representative Martin Luther, one of the giant names, obviously, in religious and philosophical history.
- 7.6** Here's what he says fundamentally about man. "Man has a twofold nature, a spiritual one and a bodily one." So we are two different kinds of things that have come together. We are a dualistic entity. "According to the spiritual nature, which men refer to as the soul, he is called a spiritual, inner, or new man. According to the bodily nature, which men refer to as flesh, he is called a carnal, outward, or old man, of whom the Apostle writes in II Corinthians 4, 'Though our outer nature is wasting away, our inner nature is being renewed every day.'"

- 7.7 We are dual in the sense that we are made up of two different kinds of stuff, if we can use that: material stuff and soul stuff. These are not only different; they are opposite in all of their features, fundamentally. One is wasting away, one does not waste away. One is made of material stuff, and the soul is made of non-material stuff, whatever that is, spiritual stuff and so forth. He goes on to make this a stronger claim. Not only are the two elements within a human being different from each other or different to each other, they are also opposed to each other.
- 7.8 "Because of this diversity of nature, the Scriptures assert contradictory things concerning the same man, since these two men in the same man contradict each other, 'for the desires of the flesh are against the Spirit, and the desires of the Spirit are against the flesh.'" So they are different in their makeup, the ontological stuff, and what they want, so to speak, is completely different and contradictory things. And so the human being is a battleground of contradictory desires based on different kinds of stuff.
- 7.9 Now, approximately at the same time, Leonardo is a slightly older contemporary of Martin Luther. We know that in Europe, particularly in southern Europe, there's a development of a more humanistic, more worldly, more nature-friendly understanding of the world. This Renaissance humanism, of which Leonardo was an artistic and engineering expression of it, starts to look at the human being not through the eyes of dualism. It starts to ask, can we understand the human being as more continuous with nature and perhaps as a part of nature? This is just the beginning of a story.
- 7.10 We see in this sketch, or what Leonardo is doing, partly the label of the sketch, the Vitruvian Man, as it is called, is that what Leonardo is looking to is not older Judeo-Christian scriptural references, but going back to the Greeks, or in this case, the Roman models, and seeing if we can see human beings more as of a piece with nature.
- 7.11 When we are designing buildings and architectural structures, we're very concerned with material forces and material shapes and how they all fit together. And that same geometry that applies to describing natural phenomenon also seems to apply to describing the human being. So, we can see the human, as we do in this case, as a natural being of certain proportions that we can describe by means of circles and squares. How far can we push this more naturalistic investigation of the human being?
- 7.12 Within a generation, there are serious anatomical studies starting to occur. In most cases, anatomical studies had been banned or forbidden. But as things start to open up in the early modern world, giants of the anatomical profession, in this case, Andreas Vesalius, do serious numbers of dissections, and then start to write up anatomical studies in textbooks, accompanied by beautiful, detailed drawings. One of them includes this one of Vesalius himself, showing himself working on a human arm, what's under the skin, all of the tendons, all of the veins, all of the bone, and so forth. This is published to great acclaim in 1543. It's a landmark in the study of anatomy.

- 7.13** A little bit earlier, an English physician named William Harvey had also done some anatomical studies on human beings and some other animals and learned something that seemed shocking at the time, which was that the heart is at the center of a circulatory system of the blood and that the blood goes all through the body and ends up back in the heart where it's renewed and rejuvenated, so to speak, and so forth all the way through.
- 7.14** This was a great anatomical discovery, but it was also shocking in the following sense that, if we think more traditionally about how we think about the heart, and we still have this language to some extent now today, where we think about matters of the head and matters of the heart, and so we're starting to think of the head as more rational, logical, that's where we do the thinking, but the heart is more where we do our feeling, and the heart is the seat of the soul, as we used to think about it.
- 7.15** What Harvey and some of the other anatomists are doing is, they're saying, "We've looked at the human heart, and it's just a pump. We don't see a soul there. Instead, it's just a mechanical or natural device that exists to pump blood around the body. We've looked also at the hearts of pigs and dogs and various other animals, and they're basically the same."
- 7.16** So, there's nothing special about the human heart. We start to have a deflation of many of the connotations and more poetic, romantic understandings that are associated with the heart. It's a more materialistic understanding of the human being. Again, how far can we go with just describing the human being in purely physicalistic, naturalistic terms and not talking about spirits and souls and the other apparatus in the traditional dualist language?
- 7.17** By the time this works into serious philosophical discussion, what the scientists are discovering, we have a British philosopher—or he was an English philosopher at this time—Thomas Hobbes, writing in the 1650s, saying that basically, science has come in. We've studied everything with respect to the human being. What we realize is that the human being is not a dual creature. The human being is a naturalist creature, a physical creature, a materialist creature, and all of the things that the human being does can be described in such terms.
- 7.18** The way he writes it here in the *Leviathan* is, this is what human life is. "For seeing life is but a motion of limbs, the beginning whereof is in some principal part within; why may we not say that all automata, engines that move themselves by springs and wheels, as doth a watch," so we consider a watch, and it moves in various ways. It's a kind of artificial life. And then the analogy to human beings. "For what is the heart in a human being but a spring? And the nerves, but so many strings? And the joints, but so many wheels, giving motion to the whole body?"
- 7.19** So, we have a natural life described in material mechanistic terms. Then we can look at a watch, very complicated, and it does the same kinds of motions, by the same kinds of

mechanical devices. And therefore, it's an artificial kind of life. This is controversial and, to some, distressing. In many cases, when we start to think of the human being, we try to ask the question, how far can we go in this naturalistic direction? How much do we need to go in a more dualistic direction?

- 7.20** In many cases, the fact question, or the open philosophical investigation question, gets buried underneath some value prejudgements because, at the same time, we want to think of ourselves a certain way. We are special. We have values. We want to have a destiny, and maybe we want to have an ultimate destiny. Maybe that ultimate destiny requires that we think of the human being as outliving its mortal shell, so to speak. So, even the question that that might not be the true account of what the human being raises, lots of important value questions.
- 7.21** One of my math and logic professors in graduate school was this gentleman, Raymond Smullyan, and he said something I thought was quite perceptive, just on this fact and value issue, particularly with respect to bracketing the value issue for now. Let's read this through. "Recently, I was with a group of mathematicians and philosophers. One philosopher asked me whether I believed man was a machine. I replied, 'You really think it makes any difference?' He most earnestly replied, 'Of course! To me, it's the most important question in philosophy.'" Is man a machine?
- 7.22** "I had the following afterthoughts"—the 'I' now being Smullyan—"I imagine that, if my friend had finally come to the conclusion that he were a machine, he would be infinitely crestfallen."—Oh, man, I'm just a machine—"Oh, my God, how horrible! I am only a machine!" But I should find out, or if I should find out that I were a machine, my attitude would be totally different. I would say, 'How amazing! I never before realized that machines could be so marvelous!'"
- 7.23** This is interesting because there are then two sets of issues here. One is, we come into these issues, almost all of us, with a prior assessment of the dignity and worth and value of human beings. We might come in as very pessimistic—human beings are terrible and awful in various ways; or we might come in more optimistic—human beings are great and wonderful; or we might have a mixed version. But whatever our view of the status of human beings is, morally speaking, we should not let that prejudice our question of the underlying ontology, so to speak, or the underlying metaphysics of what the human being is.
- 7.24** Because it might very well be that we have an optimistic view of human beings, but that is because we are material creatures who can do amazing things. Or it might be that we are coming with an optimistic view of human beings, and that is because we believe we have an immortal soul that can transcend our underlying physicality with all of its darkness. So all of those possibilities are still open questions, but what we are not interested in is letting the value question prejudice or close our minds to the underlying questions.

- 7.25 Now, what is the relationship between the mind and the body, the physical and the psychological? This is the classic mind-body problem. When we talk of the body, sometimes we talk of the bodily, sometimes we talk of the material, sometimes we talk of the physical. We've got some terminological issues there. What I want to do is use those more or less as synonyms right now. We can tease out some differences later when necessary. Typically, for abbreviation, I like to use the Greek symbol phi for the physical, just to refer to the physical, whatever that is.
- 7.26 On the other side, we've got the mind, the body and the mind, or the physical and the psychological. And this I am characterizing initially as a very broad label, category of things, the mind, the soul, the spirit, consciousness, awareness, whatever we think is going on on the psychological side. Our thoughts, our emotions, our desires, our memories, everything going on on that side of the equation, so to speak, let's just put that in a broad category—the psychological—and then raise the question, what then is the relationship between the physical and the psychological? I like to use the Greek symbol psi for the psychological.
- 7.27 If necessary, then we can define elements on either side of the divide more precisely as we're developing one of the things. As we have seen by an initial indication, the dualist approach is to say that the physical and the psychological are two different kinds of stuff, and so we should understand the human being as an aggregate. The body came from one place, the soul came from another place, and then they have been aggregated.
- 7.28 That aggregation often is seen as a temporary aggregation for as long as one's natural life exists. But that's not to prejudge or make impossible the idea that there is, after one's physical life, at least for part of our being, that a life continues, and that would be on the psychological side. So, just as the psychological side of us and the physical side of us were initially aggregated, they can be disaggregated.
- 7.29 Perhaps at the moment of death, what happens is that the physical side wastes away, starts to decompose, but the psychological side separates, retains its identity, and carries on. It goes to Valhalla, it goes to Hades, it goes to heaven, it goes to wherever. Or it might, in fact, be reincarnated, according to some traditions, where it separates from this particular body, and then after a certain period of time, there's another carne, another body, another physicality, and it gets plugged into or reaggregated with that body, and there's an ongoing cyclical thing.
- 7.30 Now, there is a catchy metaphor here. To take this abstract label, to say that what we understand then is the human being is partly physical, but we can understand the physical in materialistic and mechanical terms, all that is fine. But we have to understand the psychological side as different kinds of stuff, and it is a kind of ghost.
- 7.31 In English, that comes out of the German "geist," that is meant to be a spirit or supernatural creature. But we all apparently have one that is currently conjoined with

our body. In fun form, the best way to think of the human being is as a ghost in a machine. We also have the phrase, when you die, you "give up the ghost," so to speak. That is to say, your ghost, your body rather gives up the ghost; the geist, the spirit, whatever it is, leaves the body and carries on with whatever its destiny is.

- 7.32** The reductive materialists, as we want to call them, want to then say, everything is ultimately material. The reductive part then is to say, all of the psychological things, thoughts and feelings and desires and memories, we're going to reduce them, or rather, that is, to explain them totally in terms of underlying physical processes or material processes, things that happen in the brain. So a thought is a byproduct of certain brain activity, say, in the neocortex. A memory experience is a byproduct of certain neural activities happening in the hippocampus. We might broaden that to the neural system, to talk about sensations and perceptions and so forth.
- 7.33** The emphasis here is on the words "only" or "byproduct." So the real things, the real activity is happening at the level of physicality. When we do our science, when we do our understanding of what it is to human beings, the total story is going to be written in the language of material, physical science.
- 7.34** We don't need to talk about anything on the psychological side because it is causally inefficacious. It doesn't do anything. It's just along for the ride. It is a phenomenon, but it's what we call an epiphenomenon. It's just a phenomenon that is along for the ride, just to continue with that metaphor, with the real phenomenon, which is the underlying physical material processes. The catchy label, or the catchy phrase I like to use in this case, is to say, we don't then think of the human being as a ghost in the machine, but rather, smoke and fire.
- 7.35** When we think about the psychological or the mind, the mind is to matter what smoke is to fire. The real action is happening in the wood, say, that we are burning and the gases that are in the wood. We add an ignition agent, and there's oxygen, there are gases, and there's a physical combustive process that occurs, and we can describe that in purely physicalistic terms. Then, of course, light is emitted, heat is emitted, smoke is emitted, but the smoke isn't doing anything. It's more or less a useless byproduct of the real underlying physical processes that are going on.
- 7.36** Integrationism is a big multisyllabic word. There are other closely related labels. We'll introduce some of those a little bit later. But I like the integrationist label because what it wants to do is to say that both the dualists and the reductive materialists are wrong, that what we need to do is find a way to integrate the two. We understand the psychological and the physical, or the mind and the body, as two aspects of an integrated being. It's not that they are different kinds of things. There's this one unified being, the human being, and we can focus on its physical activities and its psychological activities, both stemming from this underlying entity.

- 7.37 The important concepts are going to be to say that we should understand the mind, not as a dual, different kind of thing, a spirit or a soul or so forth, but as something that emerges from underlying physiological brain processes. Once those get complicated enough, there's a new kind of causality, consciousness, including self-reflective consciousnesses and so on. And those emerge out of the underlying reality.
- 7.38 So that is different from or different to the dualist approach. That is to say, the mind is dependent upon the underlying physicality, but at the same time, it disagrees with the reductive materialist position because it wants to say this is a real phenomenon and it has real agency; it has real causal efficacy. So my body, so to speak, we might start there and say it goes through various processes and gives rise to psychological phenomena. And those psychological are emergent realities. They really exist. We're not just going to explain them away.
- 7.39 In some cases, what can happen in the psychological side of things is, since it's a real phenomena, it can do what it does. And one of the things it does is it has thoughts, it makes decisions, and then it, in turn, can causally determine the physicality. So it's not just smoke to fire. It is a real phenomena that depends on the underlying physicality, has its own identity, that in turn can be causally influential and even efficacious with respect to the physical.
- 7.40 Sometimes the label functionalism is used, also. This point of emergentism we'll come back to. It's important. This concept of function. What we have to do is understand the mind or the psychological as one way of describing intimately physical activities. Often very popular in this tradition is to make analogies between human beings and computers, and to say that the important thing is not the underlying hardware or materiality. The important thing is: What is that materiality doing? What functions are being performed?
- 7.41 When we're talking about thinking, when we're talking about feeling, when we're talking about decision-making, what we're talking about are functions that are going on in this complicated system that in turn depend on the hardware, but in turn, rather, influence the underlying physicality. A catchy phrase to initially try to capture the aspiration of this account of the human being is to say, the relationship is not ghost in a machine, it's not smoke and fire. Rather, mind is to body what software is to hardware.
- 7.42 If we understand that software is causally efficacious with respect to the underlying hardware, but you need to have some hardware to run your software. We work out that whole computer analogy; that's a good analogy. How far that can go? Open question, but we transpose that back to our understanding of the human being. We know that an operating computer system is an integrate. You need the software and the hardware integrated and running together in order for anything to happen.
- 7.43 Now, let's try to make some arguments for and against these positions. We've described them, but why might we think that one or the other is true? The most

common arguments for dualism come out of a broader metaphysical understanding. In many cases, there's a general metaphysical position, how we understand the cosmos as a whole, that there's a natural world and there's a supernatural world above that. On the basis of that kind of metaphysics, we elaborate an entire understanding of the direction of the universe, and then we position human beings in that, and then we say that the human being is a microcosm of the macrocosm.

7.44 So if the universe as a whole is dual, nature and supernature, then the human being has a foot in each camp, so to speak. We must be partly natural and partly supernatural. And so, in many cases, people will adopt a dualistic understanding of human nature because it comes bundled with this broader understanding of the cosmos.

7.45 But supposing we isolate the question of human nature from those broader frameworks and just say, if we look at human beings, or I look introspectively and I analyze myself, why might I think that my mind and my body are two different kinds of things?

7.46 We might say, for example, look at the language that we have to use in order to describe these two things. I don't know my north, south, and so on, but I'm just going to arbitrarily say, suppose that that is west, and I can say my arm is pointing to the west, and now I have turned my body to the west, and I am moving my body to the west. All of that makes sense physically. But if we think about thoughts, so right now I am thinking about how old was Aristotle when he went to study at Plato's Academy. So I've got this thought going on in my mind.

7.47 That's a real phenomenon, but it makes absolutely no sense to say that that thought is moving west or that it's moving east. That language does not apply to thoughts. It's not a spatially oriented thing, but physical things are spatially oriented things, and then we do develop coordinations and so on. I might say my jacket is a tan color and I'm wearing a blue shirt. Then, I think, think about my thoughts. I'm thinking $2 * 7 = 14$. I'm having a certain thought, and that's a real phenomenon. If I ask myself: What color is that thought? Again, it doesn't make any sense even to try to describe a thought in terms of colors.

7.48 We develop this elaborate language to describe physical things, but it seems like we have to develop this entirely different language to describe all kinds of psychological phenomena. If we have these two vocabularies, two different languages, it must be because they, or at least the reasonable inference, is that they are two very different things. If we can't find a unified language to describe all of them, then we're left with the conclusion that they must be ontologically or metaphysically different. So it's one kind of argument that points us in a more dualistic direction.

7.49 Dualism also is criticized by the other two approaches. So one standard criticism that's raised of dualism is called the problem of interaction. We'll run through this one as well. When we're trying to explain human beings, it does seem to be the case that our

mind and our bodies interact with each other. So I do certain things with my body, or certain things happen to my body, and that has causal implications for what goes on in my mind.

7.50 It might be a loud explosion occurs over there. That's a physical thing. It impinges upon my ear. It goes into my brain. But psychologically, what is happening in my mind? Suddenly, my attention is focused over here. I'm saying, "Oh, my goodness, what has happened over there?" I'm very curious. Maybe I've heard explosions like that before in my life, and something bad had happened. So, I'm suddenly feeling anxious and even fearful. I'm getting into a certain emotional state. All of these physical things are causing certain psychological states.

7.51 It also seems to go the other way. I can have various thoughts. I can be standing here right now, saying, "I really feel like moving toward the west a little bit, and maybe in a couple of seconds, I should give an instruction to my body to turn and face the west and go to the west. And so, how about now?" So I made a decision, I turned my body, and I go walking over here toward the west. So that then is to say, my psychological has caused certain physical things to happen in the world. And I might hit the wall and make some sounds, but that's because of psychological phenomenon.

7.52 So there's this two-way interaction always going on between mind and body, psychology and physicality. The criticism then is to say that dualism has a hard time explaining this duality. If we go to the metaphor that we floated a little bit earlier, or the analogy that we floated, that the human being should be understood as a ghost in a machine, then what we have is a metaphysics that says the physical side of us is this mechanical, physical thing, but the psychological side of us is this completely different psychological thing, like a ghost.

7.53 And what we do know, and this might be a pitch to Hollywood, but let's stick with the Hollywood movie understanding of it, ghosts cannot be affected by physical things because they are not physical beings. So I'm remembering a horror story that I only watched the first 15 minutes of, but the story is a very stereotypical story. There's two young teenagers, and they'd heard about this abandoned house at the edge of town, and there was some sort of ghost story attached to it. Then, like young teenagers, they said, "Let's go investigate this house at midnight on a dark and stormy night."

7.54 The next scene has them showing up at the front door of this scary house, and the music is getting foreboding. Then they cautiously open the door, and it's very dark inside. One of them says to the other, "You mean, you didn't bring a flashlight either?" This is not a great movie. So yes, they don't have a flashlight, and they've heard all these scary stories. They said, "OK, let's investigate." So they go inside anyways, and then they get to the landing, and then they do the other stupid thing. They say, "Let's split up. You explore the lower level, and I'm going to go upstairs and explore the upper level."

- 7.55 So it was the girl who went upstairs. And apparently, the story was that some murder had occurred in the bedroom at the far end of the hall, and everything is creaking and so on. She goes down the hallway and she opens the door to the bedroom and she looks in and she sees the ghost. Her eyes go big. She gives a shriek. The ghost sees her, and then she turns around and starts running down the hallway, running down the stairways, and then we go back.
- 7.56 She had remembered to slam the door behind her, the bedroom door. And of course, that's completely pointless because the ghost just sails right through the door. The door is merely a physical barrier, and ghosts are not affected by physical barriers. So she goes down, she ends up in the kitchen, the ghost sails down, and ends up in the kitchen as well. She gets into the kitchen, she starts grabbing all kinds of kitchen things and throwing them at the ghost. Some of them go wild, but some of them, if they are on target, just go right through the ghost, because physical things can't affect the ghost.
- 7.57 Then, eventually, the ghost gets her trapped in one corner of the kitchen. The ghost reaches out its ghostly hand and picks up a big butcher knife and stabs her to death. That's cheating because it's a ghost. It doesn't have a hand, and the big butcher knife is a physical object that's got mass and its weight. It can't pick it up. It might try to pick it up, but its hand is just going to go right through the knife. So if we are—that's impossible.
- 7.58 Then we generalize to that. If we really think that the human being is a ghost in the machine, then it seems like it's impossible for the ghost that is inside us, the geist, the spirit, the soul, or whatever it is, if it's purely immaterial, to cause things to happen in the physical body. It also seems impossible for things in the physical body to affect the ghost. How do we explain this? If there's a ghost inside me, the ghost of Stephen Hicks, I'm standing here, and I weigh a certain number of kilos or a certain number of pounds, I'm going to make myself jump up and down.
- 7.59 In three seconds, I'm going to do a little hop. Three, two, one. My ghost says now, and so I do a little hop like that. Well, how did a ghost lift that many kilos up off the ground? It's impossible. It's like the ghost in the ghost story trying to pick up a knife; it can't do so. It should go the other way. Whatever is happening in the physical body should not be affecting what's going on psychologically.
- 7.60 Now, to flip it around and do an example of that, if we think what happens to us when we drink a little too much wine or we imbibe various other substances, all of those things, wine is just a set of chemicals, and it goes into our biochemical bodily system. We might understand that the wine will interact with our biochemical system and make my body do various things. In my case, it makes me want to just relax and go to sleep if I drink too much. But other people, it might make them do crazy party kinds of things or get very angry and go out and do violent things.

- 7.61 So, fine. We understand all of that at the physical level. But if it really is the case that my mind or my soul is a different kind of thing, it's not chemical, it should not be affected by whatever level of alcohol is in my system. But we do know, as a matter of fact, that when one drinks too much wine— I'm just going to stick with that example—one's mind is also affected, one's emotions, one's psychology is also affected.
- 7.62 You start thinking—getting silly now—that you're a lot smarter than you really are, or that you are a lot wittier than you really are. Jokes are much more hilarious when you tell them than, in fact, they are, or that you're much more attractive to people than really you are in life. That is to say, not only does your body get messed up, your mind and your entire psychological side gets messed up. But how can merely some chemicals cause such effects? That's a big problem if we go strongly in the dualistic direction.
- 7.63 That points people then, typically, in the direction of a more physicalist understanding of the human being. If it's the case that when we drink too much wine, it affects our minds as well as our bodies, then it must be that what's going on in the mind is a function of underlying bodily processes.
- 7.64 Let's switch over to that one. Maybe we can explain everything in terms of underlying physical problems. And so we end up going strongly in a reductive materialist direction. A standard problem in the reductive materialist is that it tends to say, things that we want to say are important and real, like the thoughts that we have, the memories we have, the emotions and desires that we have, that it wants to explain those away as merely epiphenomena. And so, really, the only description of the universe we need to make will be in terms of physical equations, and those physical equations won't make any mention of psychological phenomena.
- 7.65 Or not just merely having thoughts, having feelings, and so on, on the psychological thing, denying any agency to those things. So, whatever is going on on the psychological side is just a byproduct of underlying physical processes. They don't have any independent action on their own, and they certainly, from this perspective, don't have any causal downward direction, that just by thinking, we cause certain physical states to come into existence.
- 7.66 All of that seems to fly in the face of the initial phenomenological data that thoughts are real. Emotions are real, my sense of agency is real, and my sense of causal efficacy when I am acting in the physical world is also real. If we want to take that as foundational data that is real, the problem of reduction of materialism is that it seems committed to just explaining away the data, rather than trying to give an explanation of the data. That, methodologically, is a tricky bit.
- 7.67 I want to introduce the third alternative by means of a comparison of the two. Often, when one is doing philosophy, and you find two positions that seem opposed to each other, sometimes they are opposed in some respects, but not in others. In many cases, they have a common premise or a common assumption. Often, that assumption is left

somewhat unstated, but once it is brought explicitly to the fore, you then see what one might need to think about and perhaps challenge in order to see what other alternatives are available to one.

- 7.68** Let's try that here. Suppose we take the dualist position. What dualism is saying is that the psychological cannot be explained in terms of physical models. It's just too different. But we do believe that the psychological exists. It's real. And so, logically, then it follows that the psychological must not be physical. So if we can't explain the psychological in physical terms, but the psychological is real, then the psychological must not be physical. Good logic.
- 7.69** Now, from the reductive materialist perspective, we say something like this, or the eliminative materialist position. They will also say, we cannot explain the psychological in terms of physical models. It doesn't fit into our equations. It doesn't fit into our scientific understanding of the world. What they then say is that science is the best thing that's going, and whatever doesn't fit into our physical models must therefore not exist. And so, then it follows that the psychological doesn't exist.
- 7.70** What they disagree on is whether the psychological is real or not. That's their ultimate conclusion. They seem opposite each other, but at the same time, they do agree on some fundamental premise. If we look at the first premise in both cases, it's identical. We cannot explain the psychological in terms of physical models.
- 7.71** Is that true? Is it that we haven't been able to explain in terms of physical models yet, or is it impossible by the nature of the phenomena to be able to do so? What the integrationists want to do is say it's that former. Traditionally, up until, say, modern science, modern psychology, modern philosophy, we've not been able to do so, but let's not give up the quest so far. Let's try again and see what we can do.
- 7.72** An important concept here is one that came up earlier when we were talking about perceptual issues. From this perspective, we're starting naturalistically, and we're going to understand sense perception as an embodied phenomenon, and that the psychological state that we call sense perception or awareness of particular things, it's not a ghostly thing that happens inside the human being, but rather it is a different phenomenon, a base-level cognitive phenomenon, that depends upon an embodiment.
- 7.73** It depends on you having an eye and an optic nerve, and that you are able to then interact with physical reality in a certain way. And then what we then want to do is scale up to that when we start talking about memory, when we talk about concepts, when we talk about emotions, and see if we can nonetheless see all of these cognitive phenomena, broader psychological phenomena, nonetheless as real emergent phenomena from these underlying complicated physical processes.
- 7.74** Let's try a thought experiment in order to explore this a little bit. Where is my self or my "I"? Because I tend to use this language, and we all tend to use this language of saying, "I am aware of the world. I have a pain in my right shoulder. I am hearing a

symphony. I am remembering my second-grade karate teacher." All of these things that are going on, I am projecting the future. We want to say that this is, in some sense, different psychological phenomena, but nonetheless, they're all integrated in some way. They make up my I, myself, what makes me me.

7.75 We might say, well, this spatial question is not quite the right question. Just as we don't want to say slots are moving in northwest or east directions, but nonetheless, we're going to say that I am somewhere around here. I am currently in Phoenix, Arizona, but my I is not in Portland, Oregon, or floating around orbiting Mars, whatever it is. It's got to be also somewhere in the Phoenix area and pretty closely associated with my body. Where exactly is that?

7.76 I want to draw on a very enjoyable, deeply philosophical article from some years ago, now written by a philosopher of mind, Daniel Dennett, called "Where Am I?" I commend this to your attention for further reflection. I'm just going to glancingly dive into it and borrow here and there things that I want to borrow for this lecture here. Let's start to tell the story. Suppose we say, "Here I am. I'm a human being. I'm a particular human being. I have my identity, and you will recognize me as having a certain identity. My wife will recognize me, my parents will recognize me, my friends will recognize me as the same entity."

7.77 We're increasingly comfortable with the idea that we can take various parts of my body and do transplants. So, for example, suppose I was drinking too much of that wine and I started to destroy my kidneys and/or my liver and I needed to have a kidney transplant. We're comfortable with the idea of saying, "You might be a kidney donor. Give me one of your kidneys," so we will take his physical stuff and put it into my body. And when I wake up from the surgery, it's still me, even though it's 99-point something percent me, plus some other human being's kidneys.

7.78 Now, suppose my other kidney started to fail, and I needed another one. If I got a kidney from you and I put it in, it would still be me. If I got a slice of your liver, and typically liver transplants only take a slice, we put that in, when I woke up, it would still be me. Suppose my heart failed, and someone had a serious motorcycle accident, died, but was an organ donor, and I got that guy's heart, and that's plugged into me. But it's still me, still Stephen Hicks, even though I don't have my heart, my kidneys, even my liver, and so on. We can do the same thing with corneas. If my corneas start to have problems, I can get a cornea transplant.

7.79 The question is: How far can we push all of this? We know we can get skin grafts from other people, as long as there's a basic compatibility. I can get blood transfusions from various people. If my veins start to fail, I can have vein transplants or possibly artery transplants. We just go through all of the organs of the human body and say, by and large, that's fine. We're comfortable with transplanting, but I'm still going to retain my identity, myself. It's still Stephen Hicks. If I show up at my parents' house for a holiday,

they're going to still say I'm their kid. Not going to say, "Sorry, too many transplants, not you anymore. Bye, you're out."

7.80 Now, we haven't talked about the brain yet. The brain is also an organ. It's a physical organ. You can weigh it. It's got certain chemical structures, neurons, electricity running through them, and so on. Suppose we change the experiment a little bit to the following. I'm going to step out of this one because I don't want to be a subject in this particular experiment.

7.81 But suppose I chose you and I chose you, and I'm an evil scientist, so to speak, and I kidnap both of you and make you unconscious, and I lay you on various slab tables in a sterile environment. What I do is, I very carefully go inside your cranium, you're lying here, and I take out your brain. I go over here, and I, very carefully, take out your brain. Initially, I plop them out into a little bowl with some brain-sustaining fluids. So I've got these two brains and these two bodies.

7.82 Then I say these brains are both connected to spinal cords. But what's the spinal cord? The spinal cord is a system of electrochemical neural connectors, and we know that we can disconnect electro-neural connectors and reattach them in various ways. It's just sending signals back and forth. We're very good now at doing that. So, I'm going to very carefully detach this brain from the spinal cord, keeping it in the spinal cord fluid, whatever it is. Same thing over here. But I'm going to take this brain, put it in this head, reattach it to the spinal cord, and I took that brain, and I put it in this head and reattach it to the spinal cord.

7.83 OK, science fiction, yes, but not unimaginable science fiction, because now we're just talking about brain transplants instead of heart transplants or kidney transplants. It's just a more complicated organ.

7.84 The question then is: When these two individuals, you and you, wake up from the anesthesia, and you sit up, where are you? Or where is she? And where is he? It seems like what we would say is, she is now over here in a different body, and he is now over here in a different body. Before, we were willing to say, if we ask where I am or what makes me me, well, my kidneys are a part of me, but they're not really essentially me. I can have different kidneys and still be me. But my body can't take a different brain and still be me, because we said, she goes with her brain over into this body, and he goes with his brain over into this body.

7.85 Why is that? We say there's something uniquely important about the brain, which is that's where your thoughts happen. That's where your memories are stored. That's where, as a result of various decisions and experiences you've had, certain parts of the brain can fire or not, and you have certain emotions that might then infuse the rest of your body.

7.86 But you then are close to your brain. The tricky bit is that it seems like all I did over here was take a material organ out of this body and move it into this physical body. I didn't

say, "I'm taking the brain, and oh, don't forget the soul," and I grab the soul, and I bring the soul over, too. All I did was move the brain. So am I, fundamentally, my brain? That's the question.

- 7.87** Another way to slice all of this. As well as transplants, we are increasingly comfortable with the idea of artificial transplants. We might say, for example, suppose my kidneys fail, but there is no one around who can do a kidney transplant. Instead, what I do is, when I'm in the hospital with my kidneys failing, is, they give me a machine, a kidney dialysis machine. That's a machine that cleans my blood and various other things. There's a tube that comes out of me, goes to the machine, the stuff gets cleaned, and it gets through another tube back into my kidneys, and then my body carries on.
- 7.88** Now, the dialysis machine might be clunky. I need to have it on a trolley of some sort. But maybe I've got my kidney dialysis machine, and I'm going about my daily business. I have my kidney machine here, but it's still me. The important thing here is not that I have my natural biological kidneys, but that I have something that can perform the function of the kidneys, even if it's made up of artificial parts, like the kidney dialysis machine.
- 7.89** We also know there are many artificial hearts. They are basically plastic and titanium of various sorts, one of the famous early ones developed by Jarvik and successive versions of that. So my heart fails, and they take out my biological heart. There's not a heart donor available, or maybe I'm open to the experiment. I will take one of those plastic and titanium hearts with little motors in it that pump. It's just a pump. Put that in. So now I've got a chunk of plastic and titanium in here. It's not biological, it's not original to me, but it doesn't matter because it's performing the function appropriately.
- 7.90** Then we start to get more systematic. Suppose I lose my legs in a terrible accident, and you give me prosthetic legs. I lose my arms in another terrible accident. I'm an accident-prone kind of guy, and you give me—we're starting to turn me into a, still me, but largely robotic being. We might say my eyes fail. So we take out my eyes, and we put in a little camera here, and we find a way to hook that up to the optic nerve. My ears fail. I get artificial hearing aids. Plugged in, and my hearing is sustained.
- 7.91** Can we systematically go through every organ that performs a function for the human being and replace it with artificial parts, and still have the human being? If we can, then we are closing in on a more physicalistic, integrationist understanding of the human being that says all these functions have to be performed, but not necessarily biologically.
- 7.92** Now, we come back to the brain. Suppose we just take one brain. I'll only pick on one person this time in my evil scientist mode. I'll pick on this gentleman this time. I sneak up on him, chloroform him, abduct him off to my evil scientist laboratory, and there he is for whatever it is that I want to do. Here we pick up with Daniel Dennett's "Where Am I?" story and a variation on that.

- 7.93 Suppose we say what we want to do, as Dennett tells us in the story, there is some underground thermonuclear device that was developed during the Cold War, and it's starting to leak radiation. We don't want that to continue; we want it to be diffused in some sense. But we can't send a normal flesh-and-blood human being down there because the radiation would just destroy him or her right now.
- 7.94 What I decide to do, me channeling Daniel Dennett's story, is, I'm going to take this guy, and I'm going to take his body, lay it out here. I'm going to make the opening at the back of his cranium. I'm going to plop his brain out into a little dish that's got nice brain juice fluids to keep it going, whatever it needs. Then I'm going to say, what if this connection between spinal cord and brain, we separate the two by means of some radio transmitter devices? Because, really, all the spinal cord is doing is just sending communication signals back and forth between the brain and the rest of the body.
- 7.95 We know that we can send communication signals back and forth all sorts of ways. We can use radio, we can use microwave, we can use digital versions of this, that, and the other thing. We're fine with translating them all back and forth. We take a cell phone, and it's just sound energy that goes into the microphone and it gets translated into digital and it gets translated, say, into, I don't know, some sort of radio wave that gets bounced up to a satellite and then down to someone else's phone, then it gets translated back into sound energy and it goes into their ear, and they have a certain psychological experience in their brain. So all this multiple translation of communication signals, we're very fine with it.
- 7.96 And yes, the spinal cord is kind of tricky, very dense communication. But suppose we then say, let's separate those two. But what I'm going to do is just have a little radio transmitter device of some sophisticated form at the back of his brain, where the spinal cord ordinarily would attach, and we'll have another one at the end of the spinal cord. So, now what we can do is keep his body on vegetative life support over here and his brain over here, and they can communicate with each other. Science fiction, but again, imaginable science fiction. We don't need to say, no, we need to add some soul transmission stuff going on there. That doesn't seem to be a part of the equation right now.
- 7.97 So, now we've got a brain in a vat—that's the phrase we like to use in this description, and it's fine, and we have a body. Suppose we then take that body and we send it down, and then we very quickly learn that, sorry, it can't handle the radiation. The body is gone, disintegrated. But his brain is still here. And earlier, we were comfortable with the idea that that's where his thoughts and feelings and memories and emotions and so forth are.
- 7.98 The problem that we now have for him is, he's only a brain and he probably would like to have a body too, so he can get on with his life. Suppose we then say, well, we've also been very comfortable with this idea of prosthetic legs, prosthetic arms, artificial hearts, kidney dialysis machines. Suppose we just make a kind of robot that's got all of

those parts, performing all of those functions, and we just have it communicate with his brain.

- 7.99** So now he's got his original biological brain, but his body is transformed into a robot. If he were sending his robot around various— initially, we would not recognize him. Maybe you knew him from third grade and he starts walking up to you and says, "Hey, how are you? Do you remember in third grade when we did this thing at recess?" You have no idea who he is, but it might take him a while, if he has the right number of memories, to be able to convince you that this is just a new robot body replacement and it's still him communicating with her. He's inside there somewhere or other.
- 7.100** If this thought experiment is still legitimate, then this opens up a whole lot of possibilities. One, of course, might be that we could build two robots. We've all had that feeling that we would like to get more stuff done in a day, so now we would have two robot bodies doing different things but controlled by the same mind. It might also be a benefit to him because his old biological body got tired at various points, and now he can go 24 hours a day, possibly, if he wants. So that could all be very much an improvement.
- 7.101** But what about this brain that we now have in the vat here? What if we extend this idea of saying that when we were talking about the other bodily parts, it wasn't important that my heart is made of flesh and blood; it could be made out of plastic or titanium, as long as it gets the job done. It doesn't matter if my kidneys are the biological ones. So, as we developed a very compact kidney dialysis machine—it's basically a fanny pack you can carry around with you—the important thing is that it's getting the job done of cleaning the fluid, pumping the blood around. Or we had the ear implants, the hearing aid or the corneas. They could also be artificial corneas. It's just a lens for bending the light, protecting the eye. It doesn't matter if it's the original biological material. Does that apply to the brain?
- 7.102** Then we start to look at the brain in terms of its functions. What does the brain do? Then we start to say things like this. Well, the brain takes the information from the senses. The senses, all the energy structures that are out there in the environment, they impinge upon the senses, and the senses send it back to the brain.
- 7.103** Some of them then get put into memory in our brain. The brain has reception of sense perception. It has retention of memories. The brain also does information processing in the neocortex: thinking, decision, mathematics, and so on. Then the brain sends signals to the body to do various things: walk in this direction, close your eyes, lift your leg like that, say these words from your mouth. And so what we then do is, we break the brain down into, initially, these four operations, and all of them are done by biological materials, so to speak, in the brain.
- 7.104** So then we introduce the function language explicitly. What the brain does is like every other organ in the body: perform a set of functions. Takes some inputs, it stores them

in memory, it does this thing we call thinking or information processing, and then it outputs various results to the body. Can we duplicate those functions in non-organic or non-biological materials?

- 7.105** Where the analogy to a computer starts to have a bite is when we start having sophisticated technologies enough to say that the keyboard is an input device. It works by touch, just like we have touch input devices. A computer will have a camera. It's receiving visual inputs. We can have the camera in front of us, store this picture of me in the computer memory, or take this picture of me, retain it as video and transmit it to someone else so that person can see me. And of course, my sound as well. It has a microphone for auditory signals. That's at least three, and they're just functionally similar to the three sensory organs that we have here.
- 7.106** I imagine we could be able to devise some device that's chemically sensitive and also taste, whatever that is, some sensors that would be unique to the different chemical structures that make up taste, and then that would input into the computer. A computer stores stuff in memory it has a central processing unit, sometimes more than one, but it just processes information, manipulates it according to programs, and then, as a result of that, it sends it out to whatever bodily things it wants: print this, display this on the screen, or control this robot as it goes through various mechanical motions.
- 7.107** Structurally, this is identical. Functionally, this is identical. Now we're then on the cusp of re-asking that same question that Thomas Hobbes asked some centuries ago, where he said it seems like a watch is performing all the same actions—springs and wheels and levers and so forth—so can we say that it has artificial life? And now we have machines that are biologically, and they're performing the exact functions, once they are sophisticated enough and integrated enough, can we say we have artificial intelligence or artificial thinking?
- 7.108** At that point, what we are doing is thinking of the human being as a set of functions. Just as the computer, we quite well now understand it because we have built them from the ground up, that it is a set of underlying physical components that have a program, and that once that program is operational—as they say, the software is operational in an integrated way with the hardware—all of these functions are performed. It becomes aware of the environment. It's able to remember things about the environment. It's able to think and come up with solutions and decisions based on its thinking, and to make decisions that cause the robot body to act with purpose in the world.
- 7.109** So, if we have then a sophisticated—and again, we're still at science fiction levels, but we are arguing about being on the cusp of all of this, that if we have a sophisticated enough robot performing all the functions, a sophisticated enough computer performing all of the cognitive, psychological side functions—are we comfortable with saying, well, that would be artificial life or at least artificial intelligence?

- 7.110** Another important concept in this literature, I want to, before closing, lay it out, is this is concept of emergence. We have to then start saying things like, if we're thinking software and hardware, the hardware can't just go of itself. It needs the software. Otherwise, we just have an inert computing system. But the software also all by itself can't do anything. It needs the hardware. We can't think in dualistic terms, that the software has its own life, so to speak, and the hardware has its own life. Instead, it's an integrated being. Really, what we're doing is we're teasing out two aspects of it, the hardware side and the software side. But when the system is running, they're not separable. They're only separable in thought, not in reality.
- 7.111** The point here is that we seem to then say, we have another level of causal reality that has come into existence. We've taken these parts, material parts, put them together in various complicated ways, and out of that emerged a new level of phenomena, the system running and doing things that the underlying system could not do on its own.
- 7.112** So this concept of emergent properties, emergent states, and the idea that what we call our psychological, our capacity for thinking, might be an emergent state out of a complicated enough brain. Our emotions are an emergent state out of a complicated enough neural system, physical system, and so on.
- 7.113** The reductive materialists disagree with this move—this is where the controversy is—because they want to say, at the lower physicalist level, we can explain everything. All these apparent emergent phenomena are byproducts of the underlying physical states. This position wants to say, no, something new comes into existence, and it has its own causal identity and its own causal efficacy, and we have to take that seriously.
- 7.114** To give a very simple example of what is meant by the idea of emergence, we take a straight line, and that's a certain amount of, say, physical material. It's a real stick. What we can do with that real stick of that length, it has a certain mass, certain length, and so on, is, we could take the exact same stuff and bend it into the shape of the square. It's the same physical stuff, but the shape is only different. Or we could have taken that exact same stuff and arranged it in a circle. So materially, these two things are not different in any particular way, except for the shape of them. But the fact that they are in a different shape, which is a new property state to be in, different kinds of phenomenon are going to result when they interact in the world.
- 7.115** So if we take, for example, the square and put it on a slope, what's going to happen is, from this position, the square will, the right side, fall down a little bit, and then it will stop. But on the right side, if we have the circle on an identical slope, it will, as a result of gravity, the right side, get pulled down. But then it will continue to roll, and that's something different. So even though we have the same physical stuff in each case, we have a different emergent property—rolling comes into existence, and rolling is something real and phenomenal. But it can't be explained simply in terms of, we had a certain amount of physical stuff made out of a certain amount of physical material.

- 7.116** So if we put that in a chart, in both cases, we've got a line that's of a certain length. One is formed in a square, the other is formed in a circle. Both of them are put in the same kind of slope, subject to the same kind of gravity. But in one case, a new phenomenon emerges or comes into existence. It's a real phenomenon, and so we go on. That was a very simple example, but if we increase the complexity many, many fold, then we start to get to possibly explaining psychological phenomena.
- 7.117** Let me end with reiterating, in motto form, the three competing theories, all of which are still with us, the arguments going back and forth about the sufficiency and problems of each. Do we think of the human being as a ghost in a machine? Do we say that the mind is to matter like smoke is to fire? Or do we think that the mind is to the body what software is to hardware?
- 7.118** In our next session, we'll turn to another possible emergent phenomenon, volition, or free will, and ask the question: Is that thing real? If it is real, can we understand it in emergent terms? Or if emergentism is a failure, are we left with certain other alternatives? See you then.

Lecture 8: The Meaning of It All

At a glance — The meaning of it all, opening with Buridan's Ass.

- 8.1 Let's start with Buridan's Ass, or perhaps more delicately, Buridan's Donkey, so named for a medieval philosopher named Jean Buridan, who left us a very interesting thought experiment about volition and determinism or causality, as we understand it, applied not only to animals, but most specifically to human beings.
- 8.2 So the thought experiment is to imagine a donkey positioned between two equally attractive bales of hay or bundles of hay. Now, the donkey is hungry, the donkey is a biological being, and so it wants to eat, and it eats hay. But in its environment, it has two equidistant and equally attractive bundles of hay. If we imagine positioning the donkey exactly in the middle of them, so he can see both of them, is it possible for the donkey to eat from one of the bundles of hay, or will it be, so to speak, paralyzed in inaction and not be able to act and thereby starve to death?
- 8.3 The question is initially puzzling because we know the donkey is going to be moving around, and the wind might be blowing a certain way, but abstract away from all of those. If we simplify it to the biology of the donkey means it needs to eat, and so it's ready to act in a particular way, but it needs a stimulus from the environment.
- 8.4 There's two causal factors that have to operate. It's a biological need and imperative to act, and then something in the environment that attracts its action. But if we have two equally attractive things in the environment, one that would pull it in this direction, another that would pull it in this direction, do not those two environmental forces cancel each other out, thus leaving the donkey paralyzed and unable to act?
- 8.5 Then we want to say, "Well, that's very interesting, Jean Buridan. Thank you." We're interested in human beings because we know in part we are like a donkey. We have lots of biological imperatives. We have lots of enticing and repelling things in the environment, and we might act with respect to stimulus and response from the environment, or just automatically act with respect to biological imperatives.
- 8.6 But often when we think about ourselves in the donkey's situation, we say it's easily resolved because a human being would just choose this one or just choose that one. We have this capacity to will, that is an additional capacity to our stimulus-response capacity and our biological imperatives. Actually, the human story would be more complicated. We are biological creatures, we are environmental response creatures, but we also are willing creatures, and so we have free will or volition.
- 8.7 The question, of course, is: Is that true? Or is it the case that really we are just like Buridan's Donkey? We're partly stimulus-response machines, and we are partly biological machine; or we are stimulus-response organisms, or biological imperative organisms. All of those are subject to principles or laws of cause and effect. The entire

network adds up to there being no room for will, for choice, or breaking the laws of causality in order to act one way or the other.

- 8.8** Now, this is a thesis then about cause and effect, and we've looked at cause and effect in the big picture. Now we are zooming in on seeing whether cause and effect applies entirely to the human being, whether we need to expand our understanding of cause and effect or not.
- 8.9** A prominent position is determinism, which says that cause and effect dominates the entire universe, it rules the entire universe. Everything is subject to cause and effect laws. That leaves no room for options. There's no wiggle room, so to speak, in the universe, and that leaves, therefore, no room for choices among options. The universe is always at a certain overall complex state of affairs. Given that complex state of affairs and all of the laws of cause and effect that are operating at all levels of analysis, there's only one thing that can happen. The universe moves into another state of affairs, but then it moves into another state of affairs, and so the entire universe goes along what might then be seen as a preset, determined path.
- 8.10** Now, strictly speaking, we have to talk about determinisms because then you say, what is the nature of cause and effect? There are different causalities, and some versions of determinism will say, these are the most important causal forces, and they explain everything, particularly about the human being.
- 8.11** There are biological forms of determinism that say—typically these are very compatible with reductive materialist understandings of human nature more generally—the human being is a biological system or a complex of biological subsystems, all of which work together. But all of those biological systems are physical and chemical systems. There are strict laws of physics we know, strict laws of chemistry. Those all come together in complex form. It's just a complicated biological story. We're just a more complicated biological story than the other animals. We don't think that worms and robins and chipmunks and so on are out there making free will choices and so forth. They are just acting according to their biological nature's cause and effect rules, including the human being.
- 8.12** Other versions of determinism will emphasize the environmental condition—sometimes as a subset—the social environmental condition, seeing the human being as, yes, a set of biological capacities, but those are by and large, indeterminate, plastic, malleable. You take the same human being and put it in different environmental circumstances, it will be conditioned according to those environmental circumstances. And according to that condition, we can predict exactly how it is going to act. Same causal circumstances, same action effects.
- 8.13** And there's a lively debate back and forth between those who are saying biology is dominant versus those who say environment is dominant. And of course, one can combine them in various ways and say it might be 75% biology, 25% environment, or

80% environment, 20% biology, so you can mix and match, but they still are deterministic theories that rule out the possibility of choice.

- 8.14** Another major category of determinism is a divine determinism, or supernatural determinism, as we've used it in this course. We're familiar with the idea that God created the universe; God created the laws, and is the lawgiver who imposed the rules upon the universe; God is the architect of the universe—all of these metaphors, and what happens in the universe is totally a function of God's decisions. If we think of God as omnipotent or omniscient, then he has all of the power to make things happen and we are relatively powerless, and so we are determined to follow along God's plan for the universe. That would be a divine version of determinism.
- 8.15** Drilling down on these two here, it seems to be a matter of logic: if God is omnipotent, that is to say God has all of the power—and we stick to that point as a strong theological, metaphysical commitment—then it follows that human beings do not have any power. If God has it all, we can have none. But this idea of free will or volition, it's a power, "It's up to me, I am the one in charge to make certain choices." That seems to then be ruled out.
- 8.16** Another variation of this, if we emphasize God's omniscience, that God is all knowing—that is to say, he knows everything in the past, he knows everything in the present, he knows everything in the future. That sounds like, logically, the past has preconditioned the present, and the present is preconditioning the future. God is not doing guesswork about what is going to happen in the future. This is what is going to happen. That would then rule out the possibility of any of us now making a choice that might set the universe off on a different path. We have to make the choices that are going to lead to whatever God's foreknowledge is. So some divine deterministic models then rule out free will as well.
- 8.17** In Eastern form, and sometimes in Middle Eastern form, particularly in those understandings of the divine that are less personalistic—if you don't think in terms of Yahweh, or God, or Allah, or Krishna—if you think of the universe as governed by more impersonal divine forces. You call it fate, or sometimes providence is the language that is used, we can get the same result.
- 8.18** Here's a famous [quote] from a poet, we'll just read it because it's delightful. "A chessboard made of Night and Day is used by Fate on which his game to play. With men, put up and push around and then Return each one to where before he lay." So the giant universe is analogized to a chessboard, and fate is playing chess, and we are the chess pieces, and we're just pushed around by fatalistic forces beyond our control. And eventually we are laid down to rest.
- 8.19** Astrology—long-lived system, there are Eastern versions, Western versions—often sees the stars or the heavens as a middleman or a mid-level of causality between higher divine forces and what happens down here on Earth. A famous saying in this tradition is

to say that not only our actions and our character are preset and governed by these astrological forces, the stars govern mankind, but God governs the stars. So we are here, and the astrological charts will determine, or give us the information about how we have been determined in terms of our character and action, but behind the stars is the heavens and the ultimate forces as well.

- 8.20** Now, let's go into the arguments for each of these versions of determinism. The arguments for divine determinism presuppose that we make arguments for the existence of God. We've spent some time on those. If we think those arguments are successful, then the very generic arguments that we have developed so far would need to be particularized in a particular theological direction, and then to specify that God must be an omnipotent being, and therefore, the way the world works is strictly determined according to that divine plan. So there's some work to do, and we've only seen in this course some initial steps down that particular road.
- 8.21** Environmental determinists will, when they're making arguments for their version of determinism, emphasize the plasticity of human nature so that we can take human beings who have not yet been conditioned and make of them pretty much whatever we want of that. The data they will emphasize is, if we think about cognitive power, language being an enormously distinctively human, powerful cognitive capacity— but it seems, relatively speaking, determined.
- 8.22** When we are born, we're born into an environment where there's a predominant language, typically that of our parents. They start stimulating us from the environment with certain kinds of sounds. The human mind of the infant is plastic; it just soaks it up and absorbs it and becomes conditioned in the vocabulary, the grammar, and in some cases, the content that is built into whatever that language is. It's not a choice, and particularly, we find it very difficult as we get older not to think in terms of whatever language it is that we are exposed to.
- 8.23** If we're exposed to multiple languages when we are young, we can, of course, soak those up like a sponge. But again, it requires that environmental stimuli conditioning. If you were to take the exact same infants and instead of having them placed into, say, a family that speaks this language and place them in a family that spoke this other language, it's not that, "No, as an infant, I'm going to resist learning this language. I want to learn that other." You have no cognitive capacity or power to do so. You are totally environmentally constructed.
- 8.24** Same thing for food preferences. It's very common. We eat certain things, we get to like them, by the time we are old enough to be aware that there are other cultures with certain food preferences. Often, it's impossible for us even to entertain the possibility of eating some of the things that people from other cultures eat. Our gorge rises when we consider the option. I remember watching a documentary of some Mongolians who, to get through the long, harsh winter, eat every part of the yak—this great beast, lots of meat and proteins and fats and so forth—including the eyeball of the yak. It's

considered a great delicacy within that culture to be the person who gets to eat the eyeball of the yak.

- 8.25** I imagine you're maybe not too far from my reaction to that. If I even try to imagine myself eating an eyeball, it doesn't work. I don't know if I could force myself to eat an eyeball. It's outside the range, so to speak. But if I had been born in Mongolia, then I would be one of those people clamoring, saying, "No, pick me, give me the eyeball. I love it. It's so tasty," and so on. Again, much of that seems to be environmentally determined.
- 8.26** More provocatively, we might say, what about people's political opinions? Or is it not the case that in many cases, young people are merely reflective of the political opinions of their parents or the religions of their parents? If we imagine—this would be an experiment you are not allowed to do—but imagine you were in charge of an orphanage and you on a regular basis have babies coming in just day-old or less, and it's up to you to try to place them with families. But suppose you were to collect, I don't know, 20 newborn infants and ship them off to strict Hindu families somewhere in Southern India and place all of those infants with the strict Hindu family.
- 8.27** Then you fast forward 20 years and you go back and you check those 20 infants who are now adults. What religion are they? We say, well, 19 out of the 20 of them are going to be Hindu, maybe 20 out of the 20. Then we say, what about that one that didn't? Well, that one was just aware of some other religious influence, read some different stimulating books from some other tradition and just came conditioned in a slightly different way.
- 8.28** But if we had taken those 20 kids and placed them with strict Baptists in the Southern part of the United States, fast forward 20 years, they're all going to be Baptists. Or if we had placed them with strict Islamic families in the middle of Saudi Arabia, they all would grow up to be Muslim. So that data will then seem to indicate from this perspective of the environmental theorists that we're not autonomous in making our own religious choices. Instead, we are conditioned by the predominant environment in which we grow up.
- 8.29** Now, we might say there is some religious diversity. Again, out of the 20, one or two might change his or her mind about religion. But the environmentalist determinants will just say, well, they saw a movie, or they had a teacher in seventh grade, or they read something that grabbed their attention and reconditioned them or made some wiggle room, so to speak, in their conditioning set that other environmental determining forces could enter.
- 8.30** The biologists will counter all of that and say either they have explanations for those things, or that there's countervailing data that says we should take much more seriously the biological data. The studies that are fascinating in this case is twin studies, which have now been going on for over a century. You take two twins who are born,

and if you can find ones who have been separated at birth and raised by different families. You do all of your research to identify those. Now, in those cases, we know the biology is identical or near identical, but because they were separated, the environment in which they are raised is different. We have the perfect controlled and uncontrolled variables in this case.

- 8.31** Then we can go and see if the twins, having grown up, they're now 20 years old, 30 years old, 40 years old. If we find, as it seems like we often do find, that the twins, despite different environmental upbringings, are very similar—not only in their physique, but often in their temperaments, in their personalities, sometimes in their choices of careers—then that would be evidence for the human being being, by and large, biologically determined. That what we are born with is a set of biological capacities, and they are just going to express themselves as they develop through childhood, into youth, into young adulthood and so on.
- 8.32** Another version of this is not to focus on twins, but say, to focus on sex differences. Common patterns, you can do the studies of the crime statistics all over the world. It seems to be the case that if you have your taxonomy of crimes—violent crimes, and non-violent crimes. And inside the category of violent crimes, murders, and rapes, and assault, and so forth, then it does seem to be the case that in the violent crimes category, the vast majority of those crimes are committed by males. Other kinds of crimes are, the majority of them, committed by females.
- 8.33** Why is it the case that that is so? We can imagine doing some statistical— or some experiments that might yield some statistics. If we imagine, for example, all male environments, and a lot of these have been studied, say a mining camp, where some remote part of the world, there's something that is being mined, and all of the guys go out there to seek their fortunes, or they're hired and so on, but you end up with an environment that is 99.9% male. The social dynamic then seems to be fairly consistent, whether it's a mining camp populated by people from North America or Africa or Asia or whatever, they all get into fights. They all like to drink a lot, and there's a certain amount of violence of various sorts and so on.
- 8.34** Then you would contrast that with taking places where you have females. You have a large group of females who are more or less isolated. You don't get the same crime behavioral patterns emerging in those 99.9% female. You get different patterns. Maybe the case is that we're all born with the same basic human template, but the important biological difference is your maleness or your femaleness, and that is the thing that really predicts what your thoughts and values and the actions are going to be.
- 8.35** Now, that's an initial run-through of the kinds of arguments, and there are empirical as well as strictly philosophical arguments that can be raised. One critical question that is often raised for all versions of determinism—a very interesting one to think through—is that all determinists, of course, argue that on epistemological grounds, we are being scientific, we are using empirical methods, we are proving this, that, and the other

thing. And so we know that, say, everything is biologically driven or everything is environmentally driven or whatever the proportions are that we can do so. So that determinism is a proven or at least highly reasonable theory to engage in, given the data.

- 8.36** There's an interesting question that we can then put to such people is that: Do you know that to be true? Again, we have to reflect on what do we mean by something being known as opposed to merely being believed? Or is it the case that, according to your own theories lights, you have only been conditioned or biologically or divinely ordained to believe in the deterministic theory?
- 8.37** If we back up and try to put it in the form of a contrast, we have among human beings some people who believe in determinism and some people who believe in free will or volition. The determinists over here say, "We know that our theory is true. It's knowledge. This theory over here, people who believe in free will, that theory is false. They are wrong." But according to determinism's own account of the things, is it not the case that they have to say, "We were, as a result of whatever combination of divine, biological and environmental forces determined to believe what we believe, and it was a different combination of biological, environmental, and divine forces that determine that these people were going to believe what they believe."
- 8.38** So all we can then say is people have different determining influences or different causal factors, and we can't say which one is right or wrong. If I, now as a determinist, had been subject to the same determining forces as these guys, I would be over in this camp. If any one of those had been subject to the same determining forces I was subject to, that person would be over there. So it's not that I know, it's just that I am forced to believe, and that's not the same thing.
- 8.39** So the question then is whether versions of determinism become eliminativist with respect to knowledge. They can't claim knowledge. All they can do is claim that people believe what they believe and try to give an explanation for that, which seems to indicate that if we're going to use the concept of knowledge, that when we unpack that concept, part of it has to do with being able to say that people are able to become aware of the evidence and say, "I can sort the evidence this way, or I could sort the evidence this way, I could make this inferential move, or I could make that inferential move. This one doesn't work, but this one does work because I'm able to make all of the proper moves."
- 8.40** Therefore, since I am in control of the process and I've done this sorting, my judgment is that this is true. But if I'm just automatically going through a certain process, I can't say that I've gone through that process, and so I can't claim knowledge. That is a very interesting issue about the epistemology of determinism. I just leave it for your consideration.

- 8.41** All of these versions of determinism, in order to round out a little bit more the vocabulary, there's also an important position that we call indeterminism. And what indeterminism does is deny the fundamental premise of causality. It says it is not the case that all of the universe is governed by strict cause and effect. There is some chaos built into the universe. God plays at dice, or the universe itself plays at dice in some sense.
- 8.42** In contemporary forms, some versions of quantum mechanics want to say that we can rescue human beings from a strict, deeply deterministic analysis by saying that once we get down to atoms and subatomic systemic particles and quarks, their cause and effect break down—some versions. Some of them want to say it's not just that there are more complicated versions of cause and effect that we haven't figured out as we are breaking things down—this subversion of quantum mechanics or this interpretation says causality goes out the window at this level. So, there are some randomness at the lowest level that is operative. Then every once in a while, that randomness just filters all up the chain, and human beings do something that counts as free will.
- 8.43** Now, that is part of the intellectual landscape. People who are in favor of free will or volition, we're going to turn to them now. Obviously, they don't like all of the deterministic theses. They also don't like indeterminism, because on this account, if we're going to have free will or volition, as we standardly characterize, it's not just that weird stuff happens at the quantum level, and there I'm doing whatever it is that I'm doing. It's I am making choices. I am making decisions. I am aware of real options before me, and which one is taken is a function of me, not only my biology, not only the environment, not only God's forces, but me.
- 8.44** A poetic expression of this from Henley's *Invictus*: "It matters not how strait the gate, how charged with punishments the scroll. I am the master of my fate, I am the captain of my soul." That point of being the master and being the captain in positions of power, positions of authority, is that I have that authority power, and that free will or volition is a real power that exists in the world in addition to all of the environmental powers, the biological powers, the divine powers, and so on. So the understanding of cause and effect needs to be enriched from this position.
- 8.45** Now, the two phrases that are commonly used here are free will and volition. I'm mostly going to use the volition formulation here. But there are some dualistic understandings of free will. We go back to a notion of God, and part of the religion's responsibility on this issue is to determine how powerful or not the gods are. If you're a polytheist, you take all of the power, you divide it among a bunch of gods and goddesses. Monotheistic versions typically concentrate all of that power into one being, and some of them will take it to the extreme and say that that amount of power is infinite.
- 8.46** But if one is a dualist, one is very often associated with a religious tradition. If one wants to argue that one's dualistic power on the psychological side does include a

measure of power, free will, and thereby moral responsibility, that I am a moral being, maybe I get my moral Commandments from a God being, but it's still up to me to choose whether I'm going to follow those Commandments or not. If I'm going to go seriously down that route, take morality and moral responsibility seriously, it must be that I am a moral agent, and as a moral agent, I have to have a certain amount of power.

- 8.47** If I have a certain amount of power, it has to be that God, at least right now, does not have all of the power. Maybe God is a limited being, very powerful, but not all-powerful, and some power is distributed among human beings. Or I might say God, in principle, has all of the power, but as part of his design, he has delegated some of his power temporarily to us, and that would enable us to become actual moral agents in the here and now. So there are dualistic accounts of free will, but it does require some adjustment to one's broader metaphysics.
- 8.48** I want to do a similar thing, say that the idea of saying that our free will has to be a part of a soul that comes from a supernatural source, share something not necessarily obvious in common with all of the versions of determinism that want to eliminate volition from our vocabulary and our understanding of the world.
- 8.49** If we look, for example, at this reconstruction of a deterministic argument, it would say that we can understand the world in terms of biological forces. We understand the world in terms of environmental forces. We can't make sense of free will as somehow an exception to all of this. It seems like free will would have to be some magical thing that breaks the causal chain of events that are going on.
- 8.50** The way of formulating that more strictly would be to say, if volition exists, it's an exception to natural cause and effect because natural cause and effect is whatever comes out of physics, whatever comes out of chemistry, whatever comes out of biology, and anything more complicated than that is just a very complicated physics, biology, chemistry story. So there's no room left over, so to speak, for volition. Even if we're going to try to think about it, we have to be stepping outside of nature to imagine it as some magical special force. But we are naturalists. We don't think there is such a place. No exceptions to natural cause and effect exist. That's going to be our second premise. Therefore, we have to deny the possibility of volition.
- 8.51** What is interesting is if you go to the dualist understanding of human being, you get a similar premise, but a different conclusion. It also agrees that if we look at the natural world, it's a world of biology and environment and so forth. If volition exists, it is an exception to the natural cause and effect. We're perfectly fine to say at the physical level—biology, chemistry, physics, and so on—we are giving the account of what happens in the natural world.
- 8.52** But we really believe there is free will. Maybe it's because we think people are morally responsible agents, but then that means that morality can't be coming from nature—

our capacity for being moral agents can't be coming from nature. If we were just part of nature, we would be like the other animals, not moral agents, not morally responsible. Therefore, it is not part of natural cause and effect; it doesn't come from nature. We have to believe that there is something called a soul or a spirit that is beyond nature. It has certain supernatural powers that are inhering in us as a dualistic being while we are in this particular world.

- 8.53** The important point is that both of them are agreeing with this initial claim that volition cannot be made sense of with respect to natural cause and effect. It would have to be an exception. But we don't think there are any exceptions, or we're willing to go to another metaphysical dimension to find that exception.
- 8.54** It is the contemporary advocates, some of the classical advocates, who want to challenge that common premise and say that it is a false premise. And what we then need to do is try to find a way of accounting for volition while adhering to the idea that the natural world is cause and effect. Very often, the story, as we get further along, is that our understanding of how complicated the natural world can be, all of these different causal forces, including emergent causal forces as our science gets better and better, we're going to then be able to understand the possibility of volition.
- 8.55** We have already challenged this enough by saying we start off with a bunch of non-living things, and out of that, life emerges. Then out of that, eventually, with greater complexity, emerges awareness and sentience. Then out of that emerges complex conscious self-awareness and perhaps conceptual consciousness. So, we're getting better and better at the science as it gets more and more complicated and all of these emergent causal forces. Eventually, maybe we can figure out what volition is: some self-regulative conceptual consciousness. Since for most of human history, we've not been at that level of cognitive development or scientific development. It has seemed like a mystery.
- 8.56** But the most important assertion is to say that volition or free will must be understood not as an exception to cause and effect, but as a kind of cause and effect. There are lots of different causal forces out there, and volition has emerged as one kind of causal force in the world. So, we have to give a naturalistic understanding of that. What would that mean? This is an ongoing project, but this is where the debate stands right now.
- 8.57** One way to approach it is to say what we need to do is try to see volition as a causal force or as a set of causal forces. But let's break down exactly what we mean by volition, start off doing some of the phenomenology, so to speak. We can do this introspectively, and it seems like it has to start introspectively because the core power that we're talking about is an agency power that each of us has as an individual. But I can't see in your head to see your agency, and you can't see in my head to see my agency, so we'll start with some introspective data.

- 8.58** We are aware then of this first phenomenon, the phenomenon of attention. We all are aware directly that when I am in a situation—I might be reading a book—I can pay attention or I can let my attention wander. Or I'm listening to a professor drone on and on, saying, "When is this going to end?" and I just let my attention wander off to something else. As opposed to—I also know I have this power—I can say, "My attention is wandering—focus!" and make myself pay attention to the professor, make myself pay attention to the book. That, then, is to say it is a self-regulation of my attention. Attention is a psychological phenomenon. What we mean is self-regulation with respect to attention, so to speak—focus, not focus. Pay attention, sort of pay attention, don't pay attention at all. That's under my control.
- 8.59** Another aspect of it is where to focus. I can make a decision—there's lots of things in my visual field. I'm going to look over here, I'm going to look over there, I'm going to look up, or I'm going to look down. There's nothing driving me, holding all of the environmental conditions the same, that makes me do so. I can look over here twice, I can look over three times, I can look down. Again, introspectively, I am the one in charge of directing that focus. I can also change the level of my focus. I can say I am going to focus in my visual field very specifically on that particular thing. And what happens as I focus on that particular thing is that everything else goes out of focus in the rest of my visual field.
- 8.60** At the same time, I also seem to be able to let my focus go in a way where I'm not focusing on anything in particular; I'm just trying to take the whole thing in and see if there's anything that emerges. A common example of this is asking people to imagine walking through a rough neighborhood at night. You're a little bit unfamiliar with the neighborhood, and you're a little bit apprehensive. What you're trying to do is not focus on any particular thing, but you're trying to keep tabs on everything possible in your visual field in an equally diffused in a collective way.
- 8.61** Then, if, of course, there was a sound that suddenly came over there, your entire attention would collapse and focus in the direction of that sound, "Oh, that was nothing." Then you say, "I need to pay attention as well." So diffused or focused in that way as well seems to be a power. So attention up or attention down, focus in, focus out: two phenomenological data to do so.
- 8.62** Then another phenomenological datum is that I seem to be able to think about various things, feel various things, attend to various things perceptually in my environment, and at a certain point, decide I'm going to act in this way or think about this thing, say this particular and not this one over here. I have various options that are in my mind, and I'm entertaining them. I could use this word, I could use that word. I could take a step in this direction, I could take a step in that direction, but I'm not doing any of those yet. Those are just candidates in my mind. Then I select this one for focus, and then decision. Then, having made the decision, the corresponding action occurs.

- 8.63** Those then are the phenomena. We want to say those are the things that comprise our sense of agency, that I'm in control of what I'm paying attention to, how carefully I'm paying attention to it, whether I'm acting on it or not. That's the agency that underlies my moral responsibility. So if I choose to say this word when I could have said some other word, or I choose to act in this way when I could have refrained from acting in that particular way, I am then morally responsible for doing so.
- 8.64** That is fine. That's what we're doing at the moral responsibility and the agency level. What we have to pay attention to is these things that we say constitute free will phenomenologically. Can we understand them as real natural phenomena? What I want to make as a suggestion here, obviously, despite the great advancements we have made in neuroscience and brain science more generally, and seeing the brain as part of a physical system, I think we still are in our infancy on lots and lots of those issues—just some speculative biology or some speculative brain science here.
- 8.65** If we are interested in our attention going up or our attention going down— and it seems to go in an analog way: I can be hyperattending to something, sort of attending to something, or blanking out altogether—that up and down does map onto the functions that a rheostat does. We do have lots of physical models of rheostats, where it operates on a dial, and whatever it is, the phenomenon, it gets adjusted in an up or down way. So, is there in the brain or among the functions of the brain, something that can be physically modeled to function with respect to our attention and our self-awareness in a rheostat way? Is there some brain mechanism that would do so?
- 8.66** The idea of focusing or not—we are familiar with lots of physical analogs. The one I'm going to use here is a spotlight. We know that I can have a light, and there are various ways of taking the light and focusing it very intensely on a particular spot, or taking that same spotlight and directing its light in a much more diffuse, generalized fashion. So my capacity to take my attention and focus it on something or let it be more diffuse—whatever that direction of the attention, analogically to the spotlight—is there then again something analogous to a neuroscience spotlight in the brain function that can be developed?
- 8.67** The idea is that whatever it is that we're doing, it's not magic. We're aware of similar phenomena already in nature and similar devices in nature. So, why can't there be then brain neuroscience versions of the same thing?
- 8.68** The same thing with respect to decision. If you're old enough to remember driving a standard transmission, we know that we can have our brains functioning in imaginary mode, or in fictional mode or hypothetical mode. Then the idea of decision is that I'm going to actually act on that. The analogy would be to an old car where you can turn the motor on, and the motor is running, but as long as you keep the clutch depressed, the motor just spins. We can rev it up or rev it down, and we can do this and turn that way. But all of that is still considering my options here.

- 8.69** Then at a certain point, I make a decision, and then the body acts. The analogy here is that's releasing the clutch, and there's an engagement between the motor and the drivetrain in that case. In this case, the motor would be analogized to the brain. The clutch would be your free will decision making. So again, it's not magic. We can understand there are physical functions that are performed by devices of various sorts. Just as in our early days of neuroscience, maybe look in that direction for some analogies.
- 8.70** So what does it all mean? We're now at the tail end of our Introduction to Metaphysics and Epistemology course as a part of substantial and foundational part of philosophy as a discipline as a whole. We know that one of the classical and deservedly so questions of all of philosophy is what is the meaning of life?
- 8.71** So what I want to do is turn, in conclusion to this course, go through all of our issues. We've spent some time talking about being and beings and identity and cause and effect, and our awareness of these as phenomenon, and whether part of the awareness is constructing space and time or whether space and time are somehow out there, and we come to be aware of it.
- 8.72** What does it mean for—we thought about ourselves as cognizing being. Do we need to use our senses and reason? Do we have mystical power? Should we resort to faith or follow the authority of smarter people who have gone before us? What are our responsibilities in all of these areas? Taking those macrocosmic considerations and then focusing on what it is to be a human being ourselves. Can we make sense of the physical and the psychological and the relationship between the two of them? Do we have choices in the matter or is the entire meaning or the purpose of the universe mapped out independently of any choices that we might make?
- 8.73** About 50 years ago—this is going back to the 1970s—there was a wonderful video series that came out also as a book called The Powers of Ten that I thought was a wonderful rendition of the cognitive move we do when we scale in and out as we start to think about metaphysics as well.
- 8.74** This one started with the middle right one. There was a man lying on his back and you see his hand on his chest. Apparently, it's he and his girlfriend are having a picnic and they're relaxing in the park. Then you can go in two ways. One is you can go focusing in and then you see his hand very closely. You go in another power of 10, and then you see the skin at a high level of resolution. Another power of 10, you start to see the individual pores. Another power of 10, smaller, you see the cells and then the subcellular components.
- 8.75** Or you can go back to the man's hand resting on his chest and scale out by the powers of 10, and then you see the two of them on a picnic blanket surrounded by some green. Then you scale out another power of 10. It's like you're standing toward the top of a tall building, looking down at the park that they are in. You see the whole neighborhood

that they are in. You see the whole city that they are in—that's actually the western shore of Lake Michigan. He's in Chicago. Then you scale out to a significant portion of the Earth, another 10, you see all of the Earth. Then you see the Earth as a small sphere surrounded by blackness. You scale out to the solar system. You scale out to galaxies.

8.76 It's wonderful to be able to do that and to recognize that we can scale out and scale down all the way. When we start to formalize this, typically, we put this along a vertical axis. We plunge into this scaling somewhere between planets and cells. That's where we spend most of our time and most of our being. We're right in the middle there. If we're interested, we can scale down all the way down to currently string theory, or we can scale all the way out to the universe as a whole. We want to make sense of all of it as a whole.

8.77 Now, this is a vertical axis, and this is a spatial array from very big all the way down to very small. We can also take the whole shebang and layer it more on a temporal axis, and typically, we'll put it on a horizontal framework. We might go back to something like the Big Bang, formation of the earliest elements, the Dark Era, the proto-galaxies, then our galaxy, the Earth, and so on. Then maybe, behind the beginning of it, is the gods or God, or surrounding all of it is some supernatural dimension as well. We love this thing as well, scaling back and forth across the billions of years of time, as well as the unimaginable amounts of space, from tiniest to largest.

8.78 What does it all mean? This has been a course on metaphysics and epistemology. When we answer the question of the meaning of life robustly, philosophy is complicated. There are other questions that need to be addressed, and there are non-philosophical questions, I think, that also need to be addressed. We've bumped up against some moral responsibility and some normative issues here and there, but we stopped. We've not talked about morality and ethics in any serious way. We've not talked about politics in any serious way. We've not talked about the beautiful and the ugly and the aesthetic in any serious way, and so on.

8.79 What I want to do in our closing here is focus on the contributions of metaphysics and epistemology, what the options are as they've emerged out of the material we have covered so far. But leave open that there are other elements of this big question that would need to be incorporated.

8.80 There are four broad options. I'll run through them quickly, and then we'll come back and explore them in fuller detail. One is to say that the ultimate meaning of the story, the ultimate purpose, is that it is God's drama for the universe. Another, I'm going to call the eternal recurrence. Another, I'm going to call entropy. And another, I'm going to call the not-yet-known or the unknown.

8.81 Now, if it's the case, and I'm starting each of these with an if, that there is a God, God ultimately powerful, all-powerful, all-knowing, and God has created the universe and made the structures within which we exist, and he made us for particular purposes, and

he has something in mind with respect to doing all of this, then the standard analogy is to think of the universe as a drama, a script that God has written and is now being played out, and we are part of that ultimate drama. Then one answer to the meaning of life question would be to say that my purpose is to accept my role in God's drama. That's the implication.

8.82 I might think that we start from a Big Bang account of the universe—I think that one is the best worked out one, that all of the universe then has gone through 13 plus billion years of development, that it's all going to continue for a certain amount of time, but eventually the universe is going to slow down its rate of expansion, and then it will all start to fall back into the initial singularity point, at which point there will be a Big Crunch. But then, it will bang again, another however many billions of years—we don't know how many cycles the universe has already been through or how many it will go, but it's potentially indefinite, that the universe is a constant cycle of Big Crunches and Big Bangs.

8.83 If that's the case, then, since we are small and the universe is big, we again accept our current space-time localization as part of something beyond our comprehension that's just going to carry on. But it also might mean that, and probably could mean, as the universe a million times, a billion times, a trillion times cycles through the same eternal recurrence, we will be here again in exactly the same formulation. And maybe we already have been here before in an earlier recurrence of the same Big Bang, Big Crunch cycling.

8.84 Another possibility so far is that we start with a Big Bang, expansion, cooling, the universe spreads out, and it keeps spreading out, and it continues to spread out until it spreads out into an entropic state. For things to happen in the universe, there have to be energy differentials: high-temperature, low-temperature differentials. In that gap, something can happen. High space in a gravitational field, lower, something can happen. High temperature, low temperature, high voltage state, low voltage state. But if it's the case that the destiny of the universe is to keep spreading out, keep spreading out, eventually there will be no energy differentials, and nothing will happen.

8.85 If that is the lesson of the universe, then our ultimate destiny is rather pointless, and one's job is to recognize that and accept the ultimate pointlessness of it all. Or we might say that's a bit depressing. At least while we're here in the here and now, I don't want to think about that too much. And so what I will do is I will avoid that by plunging into what might be consoling or fantasy worlds that help me get, at least through my life. I might escape into my art. I might escape into the laboratory, feel safe and comfortable. I might escape into a religion because it provides comfort for me.

8.86 Another possible answer, drawing down a little more, is to say that we know a lot of things. We've figured out a lot. But human beings, a couple of hundred thousand years at most, and really only half a millennium of serious science and investigation, we're still in the early days, and there's a lot that we don't know, and perhaps it's the case

that we don't know yet. We have provisional data. We can form some speculations: maybe there's God behind it all, maybe eternal recurrence, maybe Big Crunch. But our job right now, our meaning, our purpose is to continue to do what we've been doing and keep searching for the purpose of life, the meaning of life, and to get the data and the best arguments that will enable us to answer that question.

- 8.87** Maybe we will be good enough at it that we will say that yes, we seem to be stuck in the here and now, this particular chunk of space, this rather short lifespan, but the more we know, we're going to be able to escape, say, our earthly bounds, explore the entire universe, extend our longevity, our lifespan. So there are still some possibilities for optimism out there, and enjoy the process.
- 8.88** Part of this course has been to lay out, not an exhaustive, but a substantial number of the fundamental issues in metaphysics and epistemology, the debates and controversies over each of them, and then to name-check, as I have been saying, with quotations from some of the most famous names associated with the positions for our further research. Let's do the same with respect to these positions that we have developed here.
- 8.89** The God's drama approach—in a Hindu formulation from the Bhagavata Puraṇa, the whole world is the play of God. We start with Vishnu. What Vishnu does is say, "Everything comes from Me." So it's an emanation of him, or he has created the spiritual world and this material world in each of them, because this expression of his power and his plan is a play. Then we are, as human beings, having roles in that play.
- 8.90** It might be as in the Ramayana story, that the ultimate explanation is to say that really everything that happens is directed by Vishnu in various avatar forms. Sometimes, what Vishnu does is he assumes a human form as he does. He becomes Rama, and he then has various adventures, and it's all written up in this grand Hindu epic called the Ramayana. We should understand then that Vishnu takes various avatar forms, and we should nonetheless see ourselves as relating to divine forces in various ways, but it's ultimately a drama.
- 8.91** In a Western tradition, that same notion of drama is very prominent. In some sense, God is watching the world He has created, partly as script writer, but also partly as audience. And those people who have before us died, some of them have gone to heaven, some of them have gone to hell. Part of the spectacle is that those who have gone to heaven get to see the drama, the terrible things that are happening to various people in hell.
- 8.92** But ultimately, the entire universe is going to realize God's script as a beginning, a middle, and the end, and that will be the greatest spectacle of all. The greatest spectacle is the coming of the Lord when this world, so long the stage of iniquity, shall pass away. So it came into existence. Good things happen, bad things happen, but by

and large, the bad dominated and ultimately, for its sins, the entire world will pass away. And that's a morality play built out of a certain metaphysics.

- 8.93** Slightly secularizing this, as we get into the modern world, some words from Shakespeare. He has one of his characters say explicitly this formulation, "All the world's a stage. And all the men and women, merely players, they have their exits and their entrances." So, the sense that divine or not or divine-assisted, it's a theatrical production. Our parts are written for us, and our job is to perform the functions that have been written for us. Plays have beginnings, middles, and ends, and at some point we will exit and go out of it.
- 8.94** Eternal recurrence—the idea in purely secular terms would be Big Bang, however many billions or trillions of years later, returns in the form of a Big Crunch, out of which comes the next Big Bang back and forth. Two famous philosophical advocates, I'll draw some quotations of them. One from Schopenhauer, German philosopher: "What is born must die, and what dies must be born again." So a cyclical nature of the universe is being articulated here.
- 8.95** The eternal recurrence phrase, most explicitly and obviously, is associated with Nietzsche: "Behold, we know what you teach out of this philosophical system that all things recur eternally, and we ourselves with them, and that we have already existed an infinite number of times before and all things with us." If we have a strictly deterministic causal understanding of the universe, then that makes sense. Whatever the initial conditions were causally in the Big Bang, the universe works out according to scientific script until the Big Crunch occurs, but that also is going to be governed by physics. That means to the ultimate point of detail, whatever happened in each cycle is going to be the same thing over and over again. We've been here before. You've heard this lecture before. You will hear it a trillion times again in the future.
- 8.96** Your reaction to that? "Uh-oh." Or—not Nietzsche's solution—"My formula for the greatness in a human being is amor fati"—we are all fated to go through these cycles to be here again, rather than, "Oh, no," embrace it, love it; love your fate—"that one wants nothing to be different, not forward, not backward, not in all eternity, not merely to tolerate necessity, still less to conceal it, but to love it." There's a hint of a suggestion here that at least your attitudinal state with respect to realizing your fate is up to you. You can either fight it—pointless, or embrace it and love it.
- 8.97** The entropy option—maybe Big Bang goes out, but there's not enough gravitational or whatever force is at the center to slow down the expansion and draw everything back in. So it's going to go on in all directions to the end until there are no energy differentials and we have the heat death of the universe as it is sometimes described.
- 8.98** An early suggestion of this is in Ecclesiastes— we'll read this one through. A weariness that comes with this realization. "Vanity of vanities, says the Preacher. Vanity of vanities. All is vanity. What does man gain by all the toil at which he toils under the

sun?" We pause, we reflect, we realize: I'm working hard, I'm working hard, I'm trying to get somewhere. But when I step back, I'm like: What am I working so hard for? It all seems pointless in some sense.

- 8.99** "A generation goes, and a generation comes, but the earth remains forever. All things are full of weariness. A man cannot utter it. The eye is not satisfied with seeing or the ear filled with hearing. What has been is what will be, and what has been done is what will be done, and there is nothing new under the sun...There is no remembrance of former things, nor will there be any remembrance of later things, yet to be among those who will come after." Whatever it is that we're doing—pointless. We will die. We will be forgotten. It's all vanity.
- 8.100** Bertrand Russell, 20th-century philosopher, perhaps the most famous philosopher in the world, in the early middle part of the 20th century. Not religious, but reflecting the same sentiments: "All the labors of the ages, all the devotion, all the inspiration, all the noonday brightness of human genius are destined to extinction in the vast death of the solar system; and the whole temple of Man's achievement must inevitably be buried beneath the debris of a universe in ruins." It's not just your achievement; the whole universe will go to ruin that entropic state. That's the philosophical perspective on our ultimate destiny.
- 8.101** That's hard to absorb. Why not just commit suicide right now, so to speak? There are philosophers who, reflecting on this, will say that is the most important question for human beings to face. The existentialist Albert Camus, "Why not commit suicide right now?" You have to answer that question. You have to find a reason for living. The philosophy is about unpacking what possibly that reason might be for you.
- 8.102** Now, Freud took a different tack, arguing that, yes, the universe seems ultimately pointless, scientific, medical man. The reality is that for most human beings, that's too hard a thought to bear. They don't like that direction. They won't let themselves go in that direction. They avoid it. The way they get through their lives from day to day, they sense that it's ultimately pointless, is to create fantasy worlds for themselves in order to get through the day.
- 8.103** Illusions—not truth, not fact, not reality, not knowledge—illusions are indispensable to the human race. They are, in fact, indispensable to human happiness. So it's not the idea that knowledge brings happiness. Knowledge brings pessimism, futility. It's illusions that are going to save you. They are the one thing that enables men to tolerate life which is otherwise intolerable. Among these illusions, we find religion, art, and culture, which help restrain the brutal impulses of mankind and make life bearable.
- 8.104** So we just make a commitment to our art. We make a commitment to our religion. We're not really too concerned with figuring out whether they're true or not. What we want to know is: Does this bring some consolation to me? Does this give me some sense of purpose, however illusory in the here of now? We, wise philosopher types,

need to recommend art, religion, and so forth to the vast majority of human beings, even if we don't think those things are true, because out of a sense of benevolence, we think that's going to make life tolerable for the vast majority of human beings.

- 8.105** The fourth option is to say that we've made progress, but there's a lot yet that we don't know, and maybe we shouldn't be committing to any of the three options before us right now. Charles Darwin, for example, very impressed with the developments in biology, including his own contributions to it, but he pushed as far as he could go entering philosophy mode. He says that we've pushed things back evolutionarily to a certain point, and beyond that, we don't know yet. "The mystery of the beginning of all things is insoluble by us; and I, for one, must be content to remain an agnostic."
- 8.106** His point is that maybe we don't have all of the answers yet. We have a lot of answers, but not all of them, and we should be content with what we have accomplished so far, and leave open the idea that subsequent generations will extend further. It's not a pessimism, but a contentment and a certain measure of pride.
- 8.107** Late 20th-century astrological—sorry, astronomical scientist, a big difference there—Carl Sagan making a very interesting point that if the naturalistic Big Bang story and evolutionary stories are true, then we are pretty special as far as we know in the evolution of the universe. "The cosmos is within us. We are made of star stuff." So it's not that stars are alien things that are out there, but we are a complex, an impressive complex that has emerged into existence out of that star stuff. And then this arresting point: "We are a way for the universe to know itself." Much of the universe is not conscious, but suddenly it has acquired consciousness and then self-consciousness and then the ability to do all this science to reflect on the universe itself. That's a fascinating process to think about.
- 8.108** We can take the knowledge that we acquire and the things that make us pessimistic: the shortness of life, the limitations of physicality. We seem trapped in our bodies, trapped on Earth in various ways. An early Nobel scientist speculating that science is going to provide grounds for optimism. Death is not inevitable. The prolongation of life is a problem for medicine, and the possibility exists to greatly increase human longevity through hygiene, diet, and medical progress.
- 8.109** It is interesting to reflect on the fact that life expectancy around the globe was around 40 at the time of the beginning of the 20th century, and by the time we get into our 21st century, it's now about 80. That's a doubling over the course of one century. And so then a projection, if we then go another century forward, can that trend line continue? It's a positive trend line, supposing that we think human life is a good thing. Again, we've not gotten into the moral and value branches of philosophy very systematically. But we do have a philosophical scientist saying that with the right metaphysics and epistemology, maybe we can shed some light and provide some ammunition for certain moral approaches.

- 8.110** By the time we get into the middle third of the 20th century, a stronger claim, not just longevity and extending it: man can be immortal if he can find the means to renew himself continuously. Yeah, I ordinarily would have died of heart failure, but then I get a Jarvik-7 heart. And then, 85 years later, that one wears out, and then we replace it with a Jarvik-28 heart, and then that one wears out, and we do that with all the various parts that fail. Interesting. Science has only begun to open the door to the prolongation of life.
- 8.111** This optimistic vein continues on into the early 21st century, where we will end. "By the 2030s"—we're getting close—"the nonbiological portion of our intelligence will predominate. We will gradually merge with our technology, extending human lifespans indefinitely." If we choose to, and if we have the right epistemology and the right understanding of metaphysics. Of course, if we enjoy that process, continue to do that as well.
- 8.112** Now, this has been an Introduction to Metaphysics and Epistemology. We have covered a lot of territory, a lot of issues, and a lot of controversies. My hope is that you see this as an invitation to further exploration of metaphysics and epistemology. We have name-checked important people, but not all of them. I also would recommend you take those quotations, read them in a fuller context, read the controversies and the criticisms of all of those. In my view, philosophy really is a do-it-yourself enterprise if it is going to work. This course can help. I hope it has. The great minds around us and the past can help as well. But take it up, do the metaphysics, do the epistemology, and work out the meaning of life for yourself. Thank you.
- 8.113** Great. All right, this question is pertaining to our last lecture, and you went, obviously, you could give a whole course on the philosophy of medicine. Let me just interrupt at that point. My PhD is in philosophy, but my university required that we do an outside minor in another department. Mine ended up being in the history of science and history of—as it happened, the history of science department was heavily populated by philosophers of biology and people specializing in the history of medicine. So that was great. There was just wonderful, wonderful stuff there. So thank you for the suggestion.
- 8.114** You're welcome. Are we fundamentally our brains? I mean, with the modern phenomena, such as you went into heart transplants, you went into other organ transplants. Today, we possess the evidence that a donor can actually receive a heart and then have characteristics of that person they received the heart from, the most famous example being classical music or the ability to play the music from that. So given that, people like philosophers like Hobbes and Luther had no idea about these things, how does that change dualism, reductionism, and how we think about those things in philosophy, given these truths that we only know recently?
- 8.115** Yes. I think that's a fascinating set of issues. I have read journalistic accounts of exactly those things, but I will say that I don't have any expert perspective on those. But

supposing the reports are accurate, then what that would suggest is that, if we say the person is their brain, and so we do the thought experiment of the brain transplant, that would be too simplistic. We might then have to back off from that and say the person is 98% their brain, but there is lots of cellular exchange, including memory cellular exchange or conditioning behaviors because it's the whole brain and nervous system interacting with all of the organs.

- 8.116** So, it could very well be that certain traits—and we have to get into personality traits and character traits and knowledge traits and so on—are stored or inherited, or the action component that expresses that trait is not localized only in the brain, but also can be, as you suggest, in the heart. That might then be the explanation, so the story then gets more complicated, but also a lot more interesting. So if you want to become a classical pianist, maybe there would be a shorter cut, right? Yeah, that's right. You've got to get the right kidneys and the right—
- 8.117** But there is also an interesting issue there. I had piano lessons when I was a kid, and we do talk about finger memory, and all musicians talk about finger memory, and that's a little bit analogical. But if we think of memory just as something stored in the hippocampus and various other parts of the brain, maybe that's not the right way to talk about it. There are automated things, and there might be memory cells of a sort in other parts of the neurosystem, and that's going to be the fuller explanation. Just an ongoing scientific and, I think, also philosophical project as we go. Good question.
- 8.118** All right. Thanks a lot. The attention was great.